PEARSON EDEXCEL INTERNATIONAL A LEVEL CHEMISTRY · WCH11

Unit 1: Structure, Bonding and Introduction to Organic Chemistry

Revision notes, spec point by spec point (Topics 1 to 5)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

Contents

Topic 1 – Formulae, Equations and Amount of Substance

1.1 Atoms, elements, ions, molecules, compounds and formulae

1.2 The mole and the Avogadro constant

1.3 Balanced full and ionic equations with state symbols

1.4 Relative masses, molar mass and parts per million

1.5 Concentration in mol dm^{-3} and g dm^{-3}

1.6 Empirical and molecular formulae from experimental data

1.7 Reacting masses

1.8 Gas volumes, molar volume and pV = nRT

1.9 Percentage yield and atom economy

1.10 Determining a formula or confirming an equation by experiment

1.11 Core Practical 1: measuring the molar volume of a gas

1.12 Equations from simple test-tube reactions

Topic 2 – Atomic Structure and the Periodic Table

2.1 The structure of an atom

2.2 Relative mass and charge of protons, neutrons and electrons

2.3 Atomic (proton) number and mass number

2.4 Working out the particles in an atom or ion

2.5 Isotopes

2.6 The mass spectrometer and interpreting mass spectra

2.7 Predicting the mass spectra of diatomic molecules

2.8 First, second and third ionisation energies

2.9 Orbitals

2.10 What affects the size of an ionisation energy

2.11 Evidence for shells and sub-shells from ionisation energies

2.12 Shapes of s and p orbitals

2.13 How electrons fill orbitals

2.14 Electronic configurations from hydrogen to krypton

2.15 Electronic configuration determines chemical properties

2.16 Blocks of the Periodic Table and sub-shell capacities

2.17 Graphs of ionisation energy and periodic properties

2.18 Trends across Periods 2 and 3 and down a group

Topic 3 – Bonding and Structure

3.1 Evidence for the existence of ions

3.2 Forming ions by losing or gaining electrons

3.3 Dot-and-cross diagrams for ionic compounds

3.4 Ionic crystals are giant lattices

3.5 Ionic bonding is a net electrostatic attraction

3.6 How ionic charge and radius affect the strength of ionic bonding

3.7 Trends in ionic radius

3.8 Polarisation of ions

3.9 Polarising power and polarisability

3.10 Covalent bonding and the evidence for it

3.11 Dot-and-cross diagrams for covalent substances, including dative bonds

3.12 Giant lattices of carbon: graphite, diamond and graphene

3.13 Electronegativity

3.14 The bonding continuum: from covalent to ionic

3.15 Polar bonds and polar molecules

3.16 Electron-pair repulsion theory

3.17 Bond length and bond angle

3.18 Shapes of named molecules and ions

3.19 Predicting the shapes of other molecules and ions

3.20 The structure of metals

3.21 Metallic bonding

3.22 Explaining the properties of metals

Topic 4 – Introductory Organic Chemistry and Alkanes

4.1 Hazard and risk

4.2 Hazards of organic compounds and risk assessments

4.3 Reducing risk

4.4 Homologous series and functional groups

4.5 IUPAC naming and drawing organic compounds

4.6 Classifying reactions

4.7 Homolytic and heterolytic bond breaking

4.8 Free radicals and electrophiles

4.9 Alkanes and cycloalkanes

4.10 Structural isomerism

4.11 Isomers of alkanes and cycloalkanes up to C6

4.12 Alkanes as fuels: fractional distillation, cracking and reforming

4.13 Pollutants from burning alkane fuels

4.14 Why CO, NO_x and SO_x are a problem

4.15 Alternative fuels: sustainability and climate change

4.16 Carbon neutrality

4.17 Reactions of alkanes: combustion and halogenation

4.18 Free-radical substitution: the mechanism

Topic 5 – Alkenes

5.1 Alkenes: unsaturated hydrocarbons with σ and π bonds

5.2 Geometric (E/Z) isomerism

5.3 The E–Z naming system

5.4 Reactions of alkenes

5.5 Testing for C=C

5.6 Electrophilic addition: mechanisms

5.7 Addition polymerisation

5.8 Dealing with polymer waste

Topic 1 – Formulae, Equations and Amount of Substance

SPEC 1.1

Atoms, elements, ions, molecules, compounds and formulae

WHAT THIS SPEC POINT ASKS

Know what is meant by atom, element, ion, molecule, compound, empirical formula and molecular formula.

1.1 The words you must keep apart

Term	Meaning	Examples
Atom	the smallest particle of an element that keeps its chemical identity; neutral (protons = electrons)	Na, C, Ar
Element	a substance made of only one type of atom (all atoms have the same number of protons)	sodium, oxygen (O_2), ozone (O_3), diamond
Ion	an atom or group of atoms carrying an overall charge, because electrons have been lost or gained	Na^+, Cl^-, NH_4^+, SO_4^{2-}
Molecule	two or more atoms held together by covalent bonds, as a discrete particle	O_2, H_2O, CH_4
Compound	a substance made of two or more elements chemically bonded together	H_2O, NaCl, CO_2
Empirical formula	the **simplest whole-number ratio** of atoms of each element in a compound	CH_2O for glucose
Molecular formula	the **actual number** of atoms of each element in one molecule	$C_6H_{12}O_6$ for glucose

1.1 Quick sort: which words fit?

Species	Element or compound?	Particle type
Ne	element	atom
O_2 and O_3	element (two different molecules of the same element)	molecule
H_2O	compound	molecule
NaCl	compound	**not** a molecule: a giant ionic lattice; NaCl is its formula unit
SiO_2	compound	**not** a molecule: a giant covalent lattice
NH_4^+	part of a compound	ion (a polyatomic ion)

1.1 Empirical vs molecular formula

Substance	Molecular formula	Empirical formula
ethane	C_2H_6	CH_3
butane	C_4H_{10}	C_2H_5
benzene	C_6H_6	CH
glucose	$C_6H_{12}O_6$	CH_2O
water	H_2O	H_2O (same)

- Divide the subscripts by their highest common factor to get the empirical formula.
- Giant structures (ionic or giant covalent) have **only** an empirical formula: NaCl, SiO_2, MgO. There is no "molecule" to count.

EXAM TIP

Counting hydrogens in a skeletal formula one by one is slow and error-prone. Count the carbons, start from the alkane formula C_nH_{2n+2}, then **subtract 2 H for every C=C and for every ring**. Don't forget the carbons in methyl branches.

MARK-SCHEME LANGUAGE

- Molecular formula = "the actual number of atoms of each element"; empirical formula = "the simplest whole-number ratio".
- For a giant structure say "relative **formula** mass", not relative molecular mass.

Often asked as: MCQs on formulae and particles; give both the empirical and the molecular formula of an alkene or alkane; work out a molecular formula from a skeletal formula.

COMMON MISTAKE

Giving the molecular formula when the **empirical** formula is asked for (or the other way round). Calling NaCl or SiO_2 a "molecule". Mixing up "atom" and "ion" when counting electrons or charges.

BEYOND THE SPEC, BUT EXAMINED

- Working out the molecular formula of a big natural molecule (rings plus several C=C bonds) from its skeletal formula (WCH11 Jun 2022).
- Using the alkane general formula to get the formula of a large alkane such as squalane (WCH11 Jan 2021).

SPEC 1.2 The mole and the Avogadro constant

WHAT THIS SPEC POINT ASKS

Know that the mole is the unit of amount of substance, and use the Avogadro constant (6.02×10^{23} mol^{-1}) in calculations.

1.2 What a mole is

- A **mole** is the amount of substance that contains **6.02×10^{23} particles**. This number is the **Avogadro constant**, L (or N_A), units **mol^{-1}**.

- It is the number of atoms in exactly 12 g of carbon-12. So 1 mol of any substance has a mass in grams equal to its relative mass (molar mass, 1.4).
- The "particles" can be atoms, molecules, ions, electrons or formula units. **Always say which.**

1.2 The three links you use all the time

Want	Use	Units
moles from mass	$n = m \div M$	m in g, M in g mol^{-1}
number of particles from moles	$N = n \times L$	$L = 6.02 \times 10^{23}$ mol^{-1}
moles from particles	$n = N \div L$	

The route is always **mass ⇄ moles ⇄ particles**. Go through moles.

WORKED EXAMPLE: ATOMS IN A MASS OF A GAS

How many **atoms** are in 5.60 g of nitrogen gas, N_2?
Moles of N_2 = 5.60 ÷ 28.0 = 0.200 mol
Each N_2 molecule has 2 atoms, so moles of N atoms = 0.400 mol
Atoms = $0.400 \times 6.02 \times 10^{23}$ = **2.41×10^{23} atoms**
(If the question asked for molecules, the answer would be 1.20×10^{23}.)

WORKED EXAMPLE: MOLECULES FROM TONNES

How many molecules are in 1.00 tonne of ethene, C_2H_4 (M = 28.0 g mol^{-1})?
1.00 tonne = 1.00×10^6 g
Moles = $1.00 \times 10^6 \div 28.0 = 3.57 \times 10^4$ mol
Molecules = $3.57 \times 10^4 \times 6.02 \times 10^{23}$ = **2.15×10^{28} molecules**
Don't multiply by 6 here: 6 would give atoms, and the question asked for molecules.

WORKED EXAMPLE: COUNTING IONS AND ELECTRONS

(a) 17.1 g of aluminium sulfate, $Al_2(SO_4)_3$ (M = 342.3 g mol^{-1}). Moles = 17.1 ÷ 342.3 = 0.0500 mol.
Each formula unit holds 2 Al^{3+} + 3 SO_4^{2-} = 5 ions, so ions = $5 \times 0.0500 \times 6.02 \times 10^{23}$ = **1.50×10^{23} ions**.
(b) 1.00 g of Mg^{2+} ions. Moles = 1.00 ÷ 24.3 = 0.0412 mol. Each Mg^{2+} has 12 − 2 = 10 electrons.
Electrons = $10 \times 0.0412 \times 6.02 \times 10^{23}$ = **2.48×10^{23} electrons**.
(c) Mass of one carbon-12 atom = $12.0 \div 6.02 \times 10^{23}$ = **1.99×10^{-23} g**.

EXAM TIP

Huge and tiny numbers are normal here. Use the EXP or ×10ˣ key, keep the full calculator value between steps, and round only the final answer (usually 3 s.f.).

MARK-SCHEME LANGUAGE

- Show moles first, then multiply by 6.02×10^{23}.
- Count **atoms** when the question says atoms (×2 for N_2, ×6 for C_2H_4 and so on), and **molecules** when it says molecules.

Often asked as: MCQs on numbers of particles or formula units; atoms in a mass of gas; molecules in a mass given in tonnes or kg; molecules allowed in a room (with ppm).

COMMON MISTAKE

Mixing up atoms and molecules, which halves or doubles the answer. An extra factor of 6 when counting molecules of ethene. Dividing by *L* when you should multiply.

BEYOND THE SPEC, BUT EXAMINED

- Unit conversions given in the stem: tonnes → kg → g (WCH11 Jan 2020), hectares → km^2 (WCH11 Jan 2021).
- Choosing between formula units and individual ions in an MCQ (WCH11 Jan 2026).

SPEC 1.3 Balanced full and ionic equations with state symbols

WHAT THIS SPEC POINT ASKS

Write balanced full equations and ionic equations, with state symbols, for chemical reactions.

1.3 Balancing a full equation

1. Write the correct formula for every reactant and product (use the ion charges below for ionic compounds).
2. Count the atoms of each element on both sides.
3. Balance by changing the **numbers in front** (coefficients). **Never change a subscript**: that changes the substance.
4. For combustion, balance **C first, then H, then O** last. A fraction such as 11/2 O_2 is allowed (or double everything).
5. Add state symbols: **(s)** solid, **(l)** liquid, **(g)** gas, **(aq)** dissolved in water.

WORKED EXAMPLE: COMBUSTION

Complete combustion of heptane (a liquid):

C: 7 C → $7CO_2$. H: 16 H → $8H_2O$. O on the right = 14 + 8 = 22 atoms = 11 O_2.

$C_7H_{16}(l) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(l)$

Dodecane: $C_{12}H_{26}(l) + 18½O_2(g) \rightarrow 12CO_2(g) + 13H_2O(l)$ (or double it: $2C_{12}H_{26} + 37O_2 \rightarrow 24CO_2 + 26H_2O$).

1.3 Ions you need to know

Positive ions	Negative ions
H^+, Na^+, K^+, Li^+, **Ag^+**, NH_4^+	OH^-, Cl^-, Br^-, I^-, NO_3^-, HCO_3^-
Mg^{2+}, Ca^{2+}, Ba^{2+}, Zn^{2+}, Cu^{2+}, Fe^{2+}, Pb^{2+}	O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}
Al^{3+}, Fe^{3+}, Cr^{3+}	PO_4^{3-}, N^{3-}

To write an ionic formula, balance the charges: Al^{3+} and SO_4^{2-} need 2 × (+3) = 3 × (−2), so **$Al_2(SO_4)_3$**. Use brackets around a polyatomic ion when you need more than one.

1.3 Writing an ionic equation

1. Write the balanced full equation with state symbols.
2. Split every **aqueous ionic compound** (and strong acid) into its ions. Leave solids, liquids (including water), gases and covalent molecules as they are.
3. Cancel the ions that appear unchanged on both sides (**spectator ions**).

4. Check that atoms **and charges** balance.

WORKED EXAMPLE: IONIC EQUATIONS

Precipitation. $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
Split: $Ag^+ + NO_3^- + Na^+ + Cl^- \rightarrow AgCl(s) + Na^+ + NO_3^-$. Na^+ and NO_3^- are spectators.
$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
Acid + carbonate. $BaCO_3(s) + 2HCl(aq) \rightarrow BaCl_2(aq) + CO_2(g) + H_2O(l)$
$BaCO_3$ is a solid, so it stays whole. Cl^- is the spectator.
$BaCO_3(s) + 2H^+(aq) \rightarrow Ba^{2+}(aq) + CO_2(g) + H_2O(l)$
Charge check: left +2, right +2.

EXAM TIP

Work ionic equations out every time; don't try to recall them. Underlined words such as "including state symbols" in the stem mean the second mark is for states.

MARK-SCHEME LANGUAGE

- Usually 2 marks: **M1** correct formulae and balancing; **M2** state symbols (only if M1 is correct or a near miss).
- "Allow multiples" of a balanced equation.
- In ionic equations, **cancel spectator ions**: "do not award uncancelled chloride ions" (WCH11 Oct 2022). One paper let uncancelled ions through, but don't rely on it.
- Acid + carbonate: the products are CO_2 + H_2O. "Do not award H_2CO_3 as a product".
- Water in a combustion equation may be (l) or (g).

Often asked as: combustion of an alkane; correct a wrong equation; build an ionic equation for a precipitation, displacement or acid reaction; MCQ "which is the ionic equation for…".

COMMON MISTAKE

- State symbols left out, even when the question underlines them. A liquid fuel such as heptane is **(l)**, not (aq). Ions in solution are **(aq)**, not (l).
- Halogens written as atoms: it is **Cl_2**, **Br_2**, not Cl or Br.
- Wrong ion charges (Ag^+, not Ag^{2+}; Cu^{2+} in copper(II) compounds). Learn the table above.
- Writing water as the product of a metal + acid. The gas is **hydrogen**.

BEYOND THE SPEC, BUT EXAMINED

- Predicting formulae of unfamiliar products from ion charges, e.g. Cr_2O_3, $Cu(NO_3)_2$, Na_2HPO_4 (WCH11 Oct 2019).
- Balancing a decomposition: $2FeSO_4{\cdot}7H_2O \rightarrow Fe_2O_3 + SO_2 + SO_3 + 14H_2O$ (WCH11 Jun 2021).
- Equations for thermal and photo-decomposition, e.g. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ (WCH11 Jun 2024).

SPEC 1.4 Relative masses, molar mass and parts per million

WHAT THIS SPEC POINT ASKS

(i) Relative atomic mass on the ^{12}C scale. (ii) Relative molecular mass and relative formula mass, calculated from Ar values ("formula mass" for giant structures). (iii) Molar mass in g mol^{-1}. (iv) Parts per million (ppm), including gases in the atmosphere.

1.4 The ^{12}C scale

Atoms are far too light to weigh in grams, so their masses are compared with **one-twelfth of the mass of a carbon-12 atom**, which is defined as exactly 12.

Term	Definition (exam wording)
Relative isotopic mass	the mass of an atom of one isotope compared with 1/12 of the mass of an atom of carbon-12
Relative atomic mass, A_r	the **weighted mean (average) mass of an atom** of an element compared with **1/12 of the mass of an atom of carbon-12**
Relative molecular mass, M_r	the sum of the A_r values of all the atoms in one **molecule** (simple molecular substances)
Relative formula mass	the same sum for one **formula unit** of a giant structure (ionic or giant covalent: NaCl, MgO, SiO_2)
Molar mass, M	the mass of **one mole** of a substance, in **g mol^{-1}**; numerically equal to A_r or M_r

- A_r is a **weighted mean** because most elements are mixtures of isotopes (2.6 shows how to calculate it). That's why chlorine's A_r is 35.5, not a whole number.
- Relative masses have **no units**. Molar mass has units of g mol^{-1}.

WORKED EXAMPLE: RELATIVE FORMULA MASS OF A HYDRATE

$CuSO_4 \cdot 5H_2O$ (A_r: Cu 63.5, S 32.1, O 16.0, H 1.0)
$CuSO_4$ = 63.5 + 32.1 + (4 × 16.0) = 159.6
$5H_2O$ = 5 × 18.0 = 90.0
Relative formula mass = 159.6 + 90.0 = **249.6** (molar mass 249.6 g mol^{-1})
The dot means "plus": add the water, don't multiply by it.

1.4 Parts per million (ppm)

- **ppm = parts per million**: the number of parts of a substance in every million (10^6) parts of the mixture.
- ppm = (amount of the substance ÷ total amount) × 10^6, using the **same units** top and bottom.
- For solutions and solids it is usually **by mass** (1 ppm = 1 mg per kg, or about 1 mg per dm^3 of water). For **gases in the atmosphere** it is **by volume** (1 ppm = 1 cm^3 of the gas in every 10^6 cm^3 = 1 m^3 of air).
- Useful conversion: 1% = 10 000 ppm. CO_2 in air at about 420 ppm is 0.042%.

WORKED EXAMPLE: PPM

(a) A toothpaste contains 1.45 mg of fluoride in every 1.00 g of paste.
ppm = (1.45 × 10^{-3} g ÷ 1.00 g) × 10^{6} = **1450 ppm**
(b) The safe limit of a gas is 1.1 ppm by volume. What volume of the gas can a 50.0 m^3 room hold?
Volume of gas = 50.0 × 1.1 ÷ 10^{6} = 5.5 × 10^{-5} m^3 = **55 cm^3**
Check that the answer makes sense: a few tens of cm^3 in a whole room is plausible.

MARK-SCHEME LANGUAGE

- A_r definition (2 marks): **M1** "weighted mean mass / average mass of an **atom**" (the word *atom* must be there); **M2** "compared to 1/12 (the mass) of a ^{12}C atom". "Element" and "isotopes" are ignored.
- Give an A_r to the number of s.f. or d.p. asked: 35.49, not 35.4938.
- ppm: write out "parts per million", then set out as part ÷ whole × 10^{6}.

Often asked as: define relative atomic mass; calculate A_r from abundances (2.6); say what ppm means; ppm of fluoride in toothpaste, of a pollutant in a room, or a gas "mixing ratio".

COMMON MISTAKE

- Leaving out "mean", "carbon-12" or "1/12" in the A_r definition. Writing "mass of an element" instead of "mass of an atom".
- Saying "relative molecular mass" for an ionic compound: use relative **formula** mass.
- Getting a ppm answer that is obviously silly (a room volume of 10^{-9} dm^3) and not checking it.

BEYOND THE SPEC, BUT EXAMINED

- A weighted mean in a new context, e.g. the average property of a fuel blend such as E10 (WCH11 Oct 2023).
- Unfamiliar concentration units for gases, such as g m^{-3} (WCH11 Oct 2022).

SPEC 1.5

Concentration in mol dm^{-3} and g dm^{-3}

WHAT THIS SPEC POINT ASKS

Calculate the concentration of a solution in mol dm^{-3} and in g dm^{-3}. (Titration calculations come later.)

1.5 The equations

- **Concentration (mol dm^{-3}) = moles ÷ volume in dm^3**, so $n = c \times V$.
- **Concentration (g dm^{-3}) = mass ÷ volume in dm^3**.
- Convert between them with the molar mass: g dm^{-3} = mol dm^{-3} × M.
- **1 dm^3 = 1000 cm^3**. Divide cm^3 by 1000 to get dm^3 before you use these equations.

WORKED EXAMPLE: MAKING A SOLUTION

(a) 2.00 g of NaOH (M = 40.0) is dissolved to make 250 cm^3 of solution.
Moles = 2.00 ÷ 40.0 = 0.0500 mol. Volume = 250 ÷ 1000 = 0.250 dm^3.
c = 0.0500 ÷ 0.250 = **0.200 mol dm^{-3}**; in g dm^{-3}: 2.00 ÷ 0.250 = **8.00 g dm^{-3}**.
(b) What mass of $CuSO_4 \cdot 5H_2O$ (M = 249.6) makes 100 cm^3 of 0.150 mol dm^{-3} solution?
Moles = 0.150 × 0.100 = 0.0150 mol. Mass = 0.0150 × 249.6 = **3.74 g**.
Use the hydrate's molar mass because the hydrate is what you weigh.

WORKED EXAMPLE: ION CONCENTRATIONS AND DILUTION

(a) 0.100 mol dm^{-3} $Al_2(SO_4)_3$: $[Al^{3+}]$ = 2 × 0.100 = **0.200 mol dm^{-3}**; $[SO_4^{2-}]$ = 3 × 0.100 = **0.300 mol dm^{-3}**.
(b) 25.0 cm^3 of 2.00 mol dm^{-3} HCl is made up to 250.0 cm^3 with water. Moles don't change:
c = 2.00 × 25.0 ÷ 250.0 = **0.200 mol dm^{-3}**.

WORKED EXAMPLE: MINIMUM CONCENTRATION OF ACID

0.200 g of $MgCO_3$ (M = 84.3) must react completely with 25.0 cm^3 of hydrochloric acid.
$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$
Moles of $MgCO_3$ = 0.200 ÷ 84.3 = 2.37 × 10^{-3} mol, so HCl needed = 2 × 2.37 × 10^{-3} = 4.74 × 10^{-3} mol.
Minimum c = 4.74 × 10^{-3} ÷ 0.0250 = **0.190 mol dm^{-3}**.
A "minimum" must not be rounded down: 0.18 would not be enough acid.

MARK-SCHEME LANGUAGE

- $n = c \times V$ with V in dm^3 (cm^3 ÷ 1000); g dm^{-3} = mass ÷ volume in dm^3.
- Use the mole ratio from the equation before converting.

Often asked as: MCQs on concentration or the mass to weigh out; the concentration of one ion; minimum concentration of acid in a practical context.

COMMON MISTAKE

- Leaving the volume in cm^3 but quoting the answer in mol dm^{-3} (a factor of 1000 out).
- Forgetting the 1 : 2 ratio between carbonate and HCl.
- Ion concentrations: forgetting that one formula unit can release 2 or 3 of an ion.

SPEC 1.6

Empirical and molecular formulae from experimental data

WHAT THIS SPEC POINT ASKS

Use experimental data (masses or percentages) to calculate empirical formulae and then molecular formulae.

1.6 The method (set it out as a table)

1. Write the **mass** (or %) of each element. If one element isn't given, find it **by difference** (often oxygen).
2. Divide each by its **A_r** (not its atomic number) to get moles of atoms.
3. Divide every answer by the **smallest** one.
4. If you get a number like 1.5, 1.33 or 1.25, **multiply all of them** by 2, 3 or 4 to get whole numbers. Don't round 1.5 to 2.
5. Write the empirical formula.
6. **Molecular formula**: work out the M_r of the empirical formula, divide the real M_r by it, and multiply the empirical formula by the answer.

WORKED EXAMPLE: PERCENTAGES, THEN MOLECULAR FORMULA

A hydrocarbon is 92.3% carbon; M_r = 78.0.

	C	H
%	92.3	100 − 92.3 = 7.7
÷ A_r	92.3 ÷ 12.0 = 7.69	7.7 ÷ 1.0 = 7.7
÷ smallest	1	1

Empirical formula **CH** (M_r 13.0). 78.0 ÷ 13.0 = 6, so molecular formula **C_6H_6**.

WORKED EXAMPLE: A RATIO THAT ISN'T WHOLE

An oxide of iron is 70.0% Fe and 30.0% O.
Fe: 70.0 ÷ 55.8 = 1.25; O: 30.0 ÷ 16.0 = 1.875.
Divide by the smallest: Fe 1, O 1.50. Multiply both by 2: Fe 2, O 3 → **Fe_2O_3**.

WORKED EXAMPLE: OXYGEN BY DIFFERENCE

A compound contains C 54.5%, H 9.1% and oxygen only. M_r = 88.0.
O = 100 − 54.5 − 9.1 = 36.4%.
C 54.5 ÷ 12.0 = 4.54; H 9.1 ÷ 1.0 = 9.1; O 36.4 ÷ 16.0 = 2.275 → ratio 2 : 4 : 1 → **C_2H_4O** (M_r 44.0).
88.0 ÷ 44.0 = 2, so **$C_4H_8O_2$**. Showing the 44.0 is what earns the last mark.

WORKED EXAMPLE: WATER OF CRYSTALLISATION

2.46 g of hydrated magnesium sulfate, $MgSO_4 \cdot xH_2O$, is heated to constant mass. 1.20 g of anhydrous $MgSO_4$ (*M* = 120.4) is left.
Mass of water = 2.46 − 1.20 = 1.26 g → 1.26 ÷ 18.0 = 0.0700 mol
Moles of $MgSO_4$ = 1.20 ÷ 120.4 = 0.00997 mol
x = 0.0700 ÷ 0.00997 = 7.02 → **x = 7**, $MgSO_4 \cdot 7H_2O$. Waters of crystallisation are always whole numbers.

WORKED EXAMPLE: FROM COMBUSTION PRODUCTS

0.840 g of a hydrocarbon burns to give 2.64 g CO_2 and 1.08 g H_2O. Its M_r is 84.0.
Moles of C = moles of CO_2 = 2.64 ÷ 44.0 = 0.0600 mol (mass of C = 0.720 g)
Moles of H = 2 × moles of H_2O = 2 × (1.08 ÷ 18.0) = 0.120 mol (mass of H = 0.120 g)
0.720 + 0.120 = 0.840 g, so there is no oxygen in the compound.
C : H = 0.0600 : 0.120 = 1 : 2 → **CH_2** (M_r 14.0). 84.0 ÷ 14.0 = 6 → **C_6H_{12}**.

MARK-SCHEME LANGUAGE

- Marks for: moles of each element (mass ÷ A_r), the simplest whole-number ratio, the formula. "Correct formula with no working scores 1 mark only."
- Molecular formula: show the M_r of the empirical formula and the multiplier.
- TE (error carried forward) is given if you use a wrong A_r but the method is right.

Often asked as: empirical formula from % composition (oxygen by difference); molecular formula from M_r (sometimes M_r comes from pV = nRT, 1.8); water of crystallisation from heating data.

COMMON MISTAKE

- Not calculating the oxygen at all; using atomic numbers instead of A_r.
- Rounding 1 : 1.5 to 1 : 2.
- Stopping at the empirical formula when the molecular formula was asked; not simplifying ($C_4H_8O_2$ is not an empirical formula).
- Calculating moles of O_2 when you need moles of O atoms.
- Swapping the masses of water and anhydrous salt in a hydrate question.

BEYOND THE SPEC, BUT EXAMINED

- The third element found by difference (WCH11 Jun 2024, Jan 2024).
- Ratios such as 1 : 1 : 1.5 that must be doubled (WCH11 Oct 2021).
- Formula of a hydrate from a % of the anhydrous salt, e.g. 78.5% anhydrous: use ×100 ÷ 78.5, not × 78.5 (WCH11 Oct 2019).

SPEC 1.7

Reacting masses

WHAT THIS SPEC POINT ASKS

Use balanced equations to calculate reacting masses (and work back from masses), using amount of substance and molar mass.

1.7 Three steps, every time

1. **Moles of what you know**: $n = m \div M$.
2. **Mole ratio** from the balanced equation: moles of what you want.
3. **Convert**: mass = $n \times M$ (or volume, or concentration).

The **mole** ratio is not the mass ratio. Never use the equation's numbers on masses directly.

WORKED EXAMPLE: MASS OF PRODUCT

What mass of calcium oxide forms when 25.0 g of calcium carbonate is heated?
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
Moles of $CaCO_3$ = 25.0 ÷ 100.1 = 0.250 mol
Ratio 1 : 1, so 0.250 mol of CaO
Mass = 0.250 × 56.1 = **14.0 g**

WORKED EXAMPLE: LIMITING REAGENT

2.43 g of Mg is added to 50.0 cm^3 of 1.00 mol dm^{-3} HCl. What volume of hydrogen forms at r.t.p.?
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
Moles of Mg = 2.43 ÷ 24.3 = 0.100 mol. Moles of HCl = 1.00 × 0.0500 = 0.0500 mol.
0.100 mol of Mg would need 0.200 mol of HCl, but only 0.0500 mol is there. **HCl is limiting**; Mg is in excess.
Moles of H_2 = 0.0500 ÷ 2 = 0.0250 mol. Volume = 0.0250 × 24.0 = **0.600 dm^3**
Always calculate the product from the **limiting** reagent.

WORKED EXAMPLE: IDENTIFY AN UNKNOWN METAL

1.56 g of a Group 1 metal M reacts with water to give 0.480 dm^3 of hydrogen at r.t.p.
$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$
Moles of H_2 = 0.480 ÷ 24.0 = 0.0200 mol, so moles of M = 2 × 0.0200 = 0.0400 mol
A_r = 1.56 ÷ 0.0400 = 39.0 → **potassium**. Check the name you give matches the number you got.

WORKED EXAMPLE: INDUSTRIAL SCALE

Aluminium is extracted by electrolysis: $2Al_2O_3 \rightarrow 4Al + 3O_2$. Mass of Al from 1.00 tonne of Al_2O_3 (M = 102.0)?
Moles of Al_2O_3 = $1.00 \times 10^6 \div 102.0 = 9.80 \times 10^3$ mol; moles of Al = $2 \times 9.80 \times 10^3 = 1.96 \times 10^4$ mol
Mass = $1.96 \times 10^4 \times 27.0 = 5.29 \times 10^5$ g = **0.529 tonnes**
If you keep the same mass unit on both sides (tonnes in, tonnes out) you can skip the conversion to grams.

MARK-SCHEME LANGUAGE

- Marked as a chain: moles of one substance; ratio from the equation; moles or mass of the other. Each step is a mark, with TE.
- "Correct answer with no working scores full marks" is common, but a wrong answer with no working scores nothing.

Often asked as: maximum (theoretical) mass of product; identify a metal or compound from its A_r or M_r; limiting reagent in a salt preparation.

COMMON MISTAKE

- Ignoring the mole ratio in the equation.
- Using the mass of a hydrate when the anhydrous mass is needed (or the reverse).
- Rounding intermediate values; working out 38.8 correctly then naming the wrong element.

BEYOND THE SPEC, BUT EXAMINED

- Identifying an unknown metal or gas by calculating A_r or M_r (WCH11 Jan 2026, Oct 2023, Jun 2023).
- % by mass of an element in a compound, e.g. % nitrogen in a fertiliser = (mass of N in the formula ÷ M_r) × 100 (WCH11 Jan 2026).

SPEC 1.8

Gas volumes, molar volume and pV = nRT

WHAT THIS SPEC POINT ASKS

Use equations to calculate gas volumes (and back), using (i) amount of substance, (ii) the molar volume of a gas, (iii) pV = nRT for gases and volatile liquids.

1.8 (i) and (ii) Molar volume and Avogadro's law

- **Avogadro's law**: equal volumes of all gases at the same temperature and pressure contain the **same number of molecules**.
- So one mole of **any** gas has the same volume, the **molar volume**. At room temperature and pressure (r.t.p., about 298 K and 1 atm) it is **24.0 dm^3 mol^{-1}** (24 000 cm^3 mol^{-1}). Use the value the question gives.
- Volume of gas = moles × 24.0 dm^3 mol^{-1}; moles = volume ÷ 24.0.

- For reactions between gases, **volume ratio = mole ratio**, so you can work with volumes directly.

WORKED EXAMPLE: VOLUME OF GAS FROM A MASS

Volume of CO_2 at r.t.p. from 5.00 g of $CaCO_3$ reacting with excess acid?
$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$
Moles of $CaCO_3$ = 5.00 ÷ 100.1 = 0.0500 mol = moles of CO_2
Volume = 0.0500 × 24.0 = **1.20 dm³**

WORKED EXAMPLE: GAS VOLUMES ONLY

20 cm³ of propane is burned in 150 cm³ of oxygen. Volumes are measured at r.t.p., so the water is a liquid.
$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
O_2 used = 5 × 20 = 100 cm³, so 50 cm³ of O_2 is left over. CO_2 made = 3 × 20 = 60 cm³.
Final gas volume = 60 + 50 = **110 cm³**. Passing it through NaOH(aq) absorbs the CO_2, leaving **50 cm³** (the excess O_2).

1.8 (iii) The ideal gas equation

$$pV = nRT$$

Symbol	Quantity	SI unit	Conversions
p	pressure	**Pa**	kPa × 10^3 → Pa
V	volume	**m³**	cm³ × 10^{-6} → m³; dm³ × 10^{-3} → m³
n	amount	mol	
R	gas constant	8.31 J mol^{-1} K^{-1}	given in the data booklet
T	temperature	**K**	°C + 273

It works for any gas, and for a **volatile liquid** once it has been vaporised (so you can find its M_r). From it, the molar volume at 298 K and 101 kPa is $V = RT \div p$ = 8.31 × 298 ÷ 101 000 ≈ 0.0245 m³, about 24 dm³.

WORKED EXAMPLE: MOLES FROM P, V AND T

How many moles of gas are in 250 cm³ at 101 kPa and 25 °C?
p = 101 × 10^3 Pa; V = 250 × 10^{-6} = 2.50 × 10^{-4} m³; T = 25 + 273 = 298 K
$n = pV \div RT$ = (1.01 × 10^5 × 2.50 × 10^{-4}) ÷ (8.31 × 298) = **0.0102 mol**

WORKED EXAMPLE: MOLAR MASS OF A VOLATILE LIQUID

0.275 g of a liquid hydrocarbon is vaporised. The vapour occupies 98.0 cm³ at 100 °C and 1.01 × 10^5 Pa.
n = (1.01 × 10^5 × 98.0 × 10^{-6}) ÷ (8.31 × 373) = 3.19 × 10^{-3} mol
M = mass ÷ moles = 0.275 ÷ 3.19 × 10^{-3} = **86.1 g mol^{-1}**
An alkane with M_r 86 is C_6H_{14} (hexane). Don't stop at *n* and call it the molar mass.

WORKED EXAMPLE: COUNTING C=C BONDS

0.0100 mol of a hydrocarbon reacts completely with 480 cm³ of hydrogen at r.t.p. (Ni catalyst).
Moles of H_2 = 0.480 ÷ 24.0 = 0.0200 mol. Each C=C takes one H_2.
0.0200 ÷ 0.0100 = 2, so there are **2 C=C bonds** per molecule.

EXAM TIP

Write the converted values before you substitute: "$V = 2.50 \times 10^{-4}$ m^3, $T = 298$ K". Each conversion is often a separate mark, and the volume to m^3 is the mark most often lost.

MARK-SCHEME LANGUAGE

- Usual structure: **M1** V in m^3; **M2** T in K; **M3** rearranged equation; **M4** answer, sometimes a mark for 2 or 3 s.f.
- 1 dm^3 = 10^{-3} m^3; 1 cm^3 = 10^{-6} m^3; kPa × 10^3 = Pa.
- For gas-only reactions the volume-ratio method is accepted. If O_2 is in excess, add the unused O_2 to the gas left at the end.
- Molar volume at r.t.p.: "24 dm^3 mol^{-1}".

Often asked as: moles from p, V, T; then use n to find M_r of a gas or volatile liquid, to count C=C bonds by hydrogenation, or to show that a vapour is a dimer (Al_2Cl_6); gas-volume ratios in combustion.

COMMON MISTAKE

- Not converting cm^3 to m^3, or multiplying by 10^6 instead of 10^{-6}. Converting cm^3 to dm^3 and leaving kPa gives the right number only by luck.
- Forgetting to add 273, or adding it twice.
- Rounding n to 1 s.f. (0.002437 → 0.002 is 18% out).
- Using 24 dm^3 mol^{-1} for a liquid or solid: molar volume is for gases only.
- Giving the answer in dm^3 when your working gives m^3.

BEYOND THE SPEC, BUT EXAMINED

- Density of a gas instead of a volume: 1.00 dm^3 of a gas has mass 0.654 g at 298 K and 101 kPa → n = 0.0408 mol → M = 16.0, so methane (WCH11 Jan 2024).
- Liquid oxygen: 1.00 kg → 1000 ÷ **32.0** = 31.3 mol → $V = nRT \div p \approx 0.766$ m^3 at 298 K and 101 kPa. Divide by 32, not 16 (WCH11 Jan 2023).
- Showing that aluminium chloride vapour is Al_2Cl_6 from its M_r (WCH11 Oct 2024).
- Predicting the volume change in a gas reaction (WCH11 Jan 2026).

SPEC 1.9

Percentage yield and atom economy

WHAT THIS SPEC POINT ASKS

Calculate percentage yields and percentage atom economies (by mass) for lab and industrial processes, from equations and experimental results.

1.9 Percentage yield

$$\%\ \text{yield} = (\text{actual mass of product} \div \text{theoretical mass}) \times 100$$

- The **theoretical** (maximum) yield comes from a reacting-mass calculation using the limiting reagent (1.7). You can use moles instead of masses.
- Yields are below 100% because: the reaction is **incomplete** or **reversible**; **side reactions** make other products; product is **lost** when transferring, filtering or purifying; reactants are impure.
- A "yield" over 100% means the product is **impure or still wet** (e.g. not dried to constant mass).

WORKED EXAMPLE: PERCENTAGE YIELD

Heating 25.0 g of $CaCO_3$ should give 14.0 g of CaO (1.7). A student gets 11.2 g.
% yield = 11.2 ÷ 14.0 × 100 = **79.9%** (use the unrounded 14.01 g)
Scaling up: to make 5.00 g of a product at 75.0% yield, plan for a theoretical yield of 5.00 × 100 ÷ 75.0 = 6.67 g. A yield below 100% means you need **more** reactant, not less.

1.9 Atom economy

$$\%\ \text{atom economy} = (\text{molar mass of the desired product} \div \text{sum of the molar masses of all products}) \times 100$$

- Use the **balanced equation**, so multiply each molar mass by its coefficient.
- The sum of the products' masses equals the sum of the reactants' masses, so you get the same answer either way, but **use products**, as the spec does.
- Atom economy is a **theoretical** measure of how much of the starting material ends up in the useful product. It doesn't depend on the yield.

WORKED EXAMPLE: ATOM ECONOMY

(a) Hydrogen from methane: $CH_4 + H_2O \rightarrow CO + 3H_2$
Desired: 3 × 2.0 = 6.0. All products: 28.0 + 6.0 = 34.0. Atom economy = 6.0 ÷ 34.0 × 100 = **17.6%**
(b) Iron: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
Desired: 2 × 55.8 = 111.6. All products: 111.6 + 3 × 44.0 = 243.6. Atom economy = **45.8%**
(c) Ethanol: hydration $C_2H_4 + H_2O \rightarrow C_2H_5OH$ is **100%** (one product). Fermentation $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ gives 92.0 ÷ 180.0 × 100 = **51.1%**.

1.9 Why high atom economy matters

- Less **waste** made and fewer raw materials used per kg of product: more **sustainable**, and cheaper.
- Addition reactions (and addition polymerisation) have **100%** atom economy; substitution and elimination reactions always make a by-product.
- If the by-product can be **sold or used**, the process is less wasteful than its atom economy suggests.
- A process needs **both** a high yield and a high atom economy to be efficient.

MARK-SCHEME LANGUAGE

- "Quoting or using the atom economy formula" is a mark on its own. Then the total of the products and the answer to 2 or 3 s.f.
- Atom economy must match the equation (use 5 × Na_2O if the equation has 5).
- Missing "× 100" is penalised only once in a paper.
- % yield: actual ÷ theoretical × 100; mass or mole routes both accepted.

Often asked as: atom economy of a given route; MCQ "which route has the highest atom economy"; % yield of a salt preparation (including hydrates and double salts); scaling up the mass needed when the yield is below 100%.

COMMON MISTAKE

- Forgetting the coefficient when a product appears more than once.
- Using the A_r of an element as the M_r of its compound (e.g. Ge, 72.6, for GeH_4).
- Dividing instead of multiplying when scaling up for a low yield.
- Quoting too many s.f.

BEYOND THE SPEC, BUT EXAMINED

- The M_r of a hydrated product ($MgSO_4 \cdot 7H_2O$) in a yield calculation (WCH11 Jan 2024).
- Double salts, e.g. ammonium iron(II) sulfate, with identifying the limiting reagent (WCH11 Oct 2025).
- Addition polymerisation has 100% atom economy (WCH11 Oct 2020).

SPEC 1.10 Determining a formula or confirming an equation by experiment

WHAT THIS SPEC POINT ASKS

Find a formula, or confirm an equation, from experimental results, and evaluate the data.

1.10 Typical experiments

Experiment	What you measure	What you work out
Heat a hydrated salt (e.g. $MgSO_4 \cdot xH_2O$)	mass before and after heating to **constant mass**	x, the water of crystallisation
Reduce copper oxide in a stream of hydrogen or methane	mass of oxide and of copper left	formula of the oxide
Burn magnesium in a crucible with a lid	mass of Mg and of MgO	formula MgO
Li or Mg with water or acid; collect H_2	mass of metal and volume of gas	mole ratio, so the equation

WORKED EXAMPLE: FORMULA OF COPPER OXIDE

1.59 g of copper oxide is reduced by hydrogen, leaving 1.27 g of copper.
Mass of O = 1.59 − 1.27 = 0.32 g
Cu: 1.27 ÷ 63.5 = 0.0200 mol; O: 0.32 ÷ 16.0 = 0.020 mol → ratio 1 : 1 → **CuO**

WORKED EXAMPLE: CONFIRMING AN EQUATION

0.0350 g of lithium reacts with water, giving 60.0 cm^3 of hydrogen at r.t.p.
Li: 0.0350 ÷ 6.9 = 5.07×10^{-3} mol; H_2: 60.0 ÷ 24 000 = 2.50×10^{-3} mol
Li : H_2 = 2.03 : 1 ≈ 2 : 1, which supports **$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$**

1.10 Evaluating the data

- **Heating to constant mass**: heat, cool, weigh, and repeat until two readings agree. This makes sure **all** the water (or other volatile product) has gone.
- **Percentage uncertainty** = (uncertainty ÷ measured value) × 100. A reading's uncertainty is half the smallest division; a mass or volume **by difference** uses two readings, so double it.
- Example: on a 2 d.p. balance each reading is ±0.005 g, so a mass by difference is ±0.01 g. For 0.32 g of oxygen that is 0.01 ÷ 0.32 × 100 = **3.1%**, much bigger than for the 1.27 g of copper (0.8%). Small masses give big percentage errors: use more sample.
- Other errors: the solid **decomposes further** (e.g. a sulfate losing SO_3) or **spits** out of the crucible; Mg **reacts with nitrogen** in air; copper **re-oxidises** if it cools in air (keep the gas flowing until cool); gas escapes or dissolves before it is measured (1.11).
- Graphs: plot one quantity against the other for several runs. A straight line through the origin confirms a fixed ratio, and its gradient gives the ratio.

MARK-SCHEME LANGUAGE

- "Heat to constant mass / until the mass does not change" and "to ensure all the water is lost". ALLOW "heat for longer". IGNORE repeats, a higher temperature or more solid. "Heat under reflux" is not awarded.
- Waters of crystallisation are given as **whole numbers**.
- Graph: axes labelled with units, points plotted to within half a small square, one ruled line of best fit.

Often asked as: hydrate calculations (masses or a graph); how to improve the method "using the same apparatus"; equation or formula from given masses and gas volumes; why a reactant is added in excess.

COMMON MISTAKE

- Answering "repeat and take a mean" when the question asks how to improve a single measurement with the same apparatus. The answer is usually **heat to constant mass**.
- Putting the masses of anhydrous salt and water the wrong way round.
- Naming evaporation or distillation to remove an excess solid metal: use **filtration**.

BEYOND THE SPEC, BUT EXAMINED

- Adding magnesium in excess so all the acid reacts, then filtering off the leftover metal (WCH11 Jan 2024).
- Deducing x in a hydrate from the ratio of decomposition products, e.g. $SO_2 : SO_3 : H_2O = 1 : 1 : 14$ (WCH11 Jun 2021).
- Interpolating on a graph of mass of anhydrous salt against mass of hydrate (WCH11 Oct 2024).

SPEC 1.11

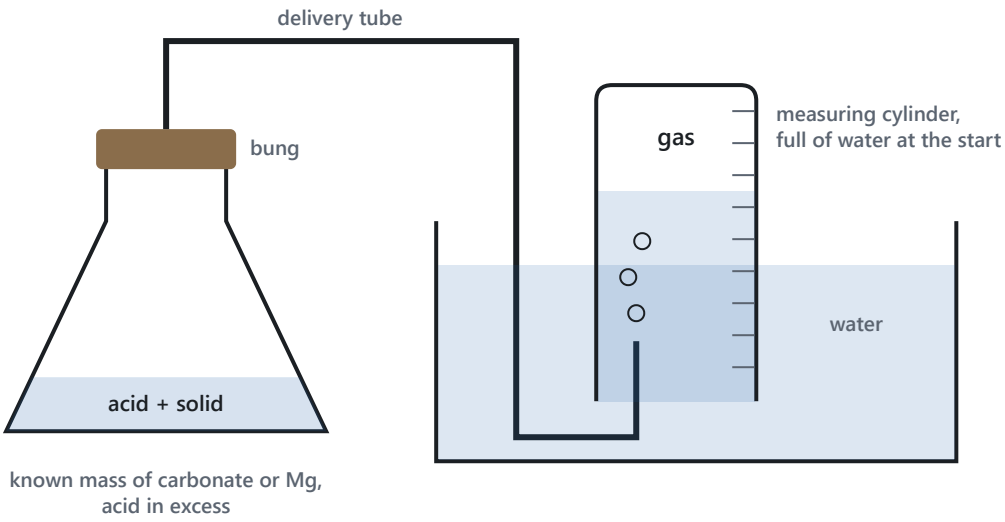
Core Practical 1: measuring the molar volume of a gas

WHAT THIS SPEC POINT ASKS

Carry out an experiment to measure the molar volume of a gas.

1.11 Principle

Measure the volume of gas made by a **known number of moles** of solid. Molar volume = volume of gas ÷ moles of gas. A carbonate ($CaCO_3$ or $MgCO_3$) with acid gives CO_2; magnesium ribbon with acid gives H_2.



Collecting gas over water. A gas syringe in place of the measuring cylinder avoids the gas dissolving in water.

CORE PRACTICAL 1: METHOD IN BRIEF

1. Weigh a small sample of the solid (e.g. about 0.05–0.25 g of $MgCO_3$) on a 2 d.p. (or 3 d.p.) balance.
2. Put a measured volume of acid in the conical flask. The acid must be **in excess** so all the solid reacts.
3. Fill a measuring cylinder (or burette) with water, turn it upside down in a trough of water, and put the end of the delivery tube under it. (Or connect a gas syringe.)
4. Add the solid, **put the bung in straight away**, and collect the gas until no more bubbles appear.
5. Read the volume of gas, with your eye level with the water level in the cylinder.
6. Repeat with different masses of solid. Plot **volume of gas** (y-axis) against **mass of solid** (x-axis).

WORKED EXAMPLE: MOLAR VOLUME

0.200 g of $MgCO_3$ (M = 84.3) gives 52.0 cm^3 of CO_2.

$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$, so moles of CO_2 = moles of $MgCO_3$ = $0.200 \div 84.3 = 2.37 \times 10^{-3}$ mol

Molar volume = $52.0 \div 2.37 \times 10^{-3} = 21\,900$ $cm^3\ mol^{-1}$ = **21.9 $dm^3\ mol^{-1}$**

From a graph: if the gradient is 260 $cm^3\ g^{-1}$, molar volume = $260 \times 84.3 = 21\,900$ $cm^3\ mol^{-1}$.

The textbook value at r.t.p. is 24.0 $dm^3\ mol^{-1}$, so this result is $(24.0 - 21.9) \div 24.0 \times 100 = 8.7\%$ low.

1.11 Why the result is often too low, and how to fix it

Problem	Effect	Improvement
CO_2 dissolves in the water in the trough	less gas collected	use a **gas syringe**, or collect H_2 (from Mg) which is much less soluble
Gas escapes before the bung is put in	less gas collected	keep the solid in a small tube inside the flask, bung the flask, then tip it over to mix
Gas leaks from joints	less gas collected	check the bung and tubing are tight
Small volume read on a large cylinder	large % uncertainty	use a smaller cylinder or a burette, or a bigger mass of solid
Temperature or pressure not at r.t.p.	molar volume differs from 24.0	record them, or correct with pV = nRT

- **Variables**: independent = mass of solid; dependent = volume of gas; control = volume and concentration of acid (in excess), temperature.
- **Safety**: eye protection; dilute acids are irritants.
- **Percentage uncertainty**: reading a 100 cm^3 cylinder to ±0.5 cm^3 for 52.0 cm^3 is 0.5 ÷ 52.0 × 100 ≈ 1%.

MARK-SCHEME LANGUAGE

- Graph: axes labelled with units, points on both sides of one ruled straight line, which should pass through the origin.
- Molar volume: moles of solid = mass ÷ *M*; molar volume = **your measured volume** ÷ moles, with units and 2 or 3 s.f. (about 21.5 to 22 $dm^3\ mol^{-1}$ in the WCH11 Jun 2019 data).
- Result lower than expected: "carbon dioxide **dissolves** in water", with a reason. "Leaks" must say how gas was lost. Vague "incomplete reaction", "impurities" or "reading errors" were ignored.

Often asked as: plot volume against mass and read a value off the line; calculate the molar volume; minimum concentration of acid needed (1.5); explain why the value is lower than 24 $dm^3\ mol^{-1}$.

COMMON MISTAKE

- Using 24 $dm^3\ mol^{-1}$ in the calculation when the question wants **your own** molar volume from the data.
- Reading an upside-down measuring cylinder the wrong way (the scale runs downwards).
- Joining the dots instead of drawing a line of best fit; ignoring the origin.

BEYOND THE SPEC, BUT EXAMINED

- Reading values from a gradient or table in place of the student's own graph (WCH11 Jun 2019).
- The same set-up used for other reactions, such as Mg with sulfuric acid in a salt preparation (WCH11 Jan 2024).

SPEC 1.12

Equations from simple test-tube reactions

WHAT THIS SPEC POINT ASKS

Link ionic and full equations, with state symbols, to what you see in simple test-tube experiments: (i) displacement reactions, (ii) typical reactions of acids, (iii) precipitation reactions.

Observations are what you **see**, **hear** or **feel**: colour changes, bubbles (effervescence), a solid appearing or disappearing, the tube getting warm. Naming the product ("hydrogen is made") is an inference, not an observation.

1.12 (i) Displacement

A more reactive metal pushes a less reactive metal out of its compound; a more reactive halogen pushes a less reactive halogen out of a halide.

Reaction	Ionic equation	Observations
Zn in $CuSO_4(aq)$	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$	blue colour fades; pink-brown solid forms; gets warm
Cu in $AgNO_3(aq)$	$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$	colourless solution turns **blue**; **grey/silver** solid forms on the copper
Mg in $FeSO_4(aq)$	$Mg(s) + Fe^{2+}(aq) \rightarrow Mg^{2+}(aq) + Fe(s)$	pale green colour fades; dark grey solid
$Cl_2(aq)$ + $KBr(aq)$	$Cl_2(aq) + 2Br^{-}(aq) \rightarrow 2Cl^{-}(aq) + Br_2(aq)$	colourless → **yellow-orange** (orange in cyclohexane)
Cl_2 or Br_2 + $KI(aq)$	$Br_2(aq) + 2I^{-}(aq) \rightarrow 2Br^{-}(aq) + I_2(aq)$	→ **brown** solution (**purple** in cyclohexane)

1.12 (ii) Typical reactions of acids

Reaction	Ionic equation	Observations
metal: $Mg + 2HCl \rightarrow MgCl_2 + H_2$	$Mg(s) + 2H^{+}(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$	**bubbles / effervescence**; metal gets smaller and disappears; warm
carbonate: $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$	$CaCO_3(s) + 2H^{+}(aq) \rightarrow Ca^{2+}(aq) + CO_2(g) + H_2O(l)$	effervescence; solid disappears
soluble carbonate in solution	$CO_3^{2-}(aq) + 2H^{+}(aq) \rightarrow CO_2(g) + H_2O(l)$	effervescence
hydrogencarbonate	$HCO_3^{-}(aq) + H^{+}(aq) \rightarrow CO_2(g) + H_2O(l)$	effervescence
metal oxide: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$	$CuO(s) + 2H^{+}(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$	**black** solid dissolves (on warming); **blue** solution forms
alkali: $NaOH + HCl \rightarrow NaCl + H_2O$	$H^{+}(aq) + OH^{-}(aq) \rightarrow H_2O(l)$	no visible change; temperature rises

- Acid + metal is a **redox** reaction (the metal displaces hydrogen from the acid). Acid + base, acid + metal oxide and acid + carbonate are **neutralisations**.
- Test the gases: H_2 gives a squeaky pop with a lighted splint; CO_2 turns limewater milky.

1.12 (iii) Precipitation

Two solutions mix and an **insoluble** product appears as a solid (a precipitate).

Precipitate	Ionic equation	Colour
silver chloride	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	white
silver bromide	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	cream
silver iodide	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	yellow
barium sulfate	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	white
copper(II) hydroxide	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	blue
iron(II) hydroxide	$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$	green
iron(III) hydroxide	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$	orange-brown
magnesium hydroxide	$Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$	white

Solubility rules (so you know which product is (s) and which is (aq)):

- **Soluble**: all Group 1 and ammonium compounds; all nitrates; most chlorides, bromides and iodides (**not** silver or lead); most sulfates (**not** barium or lead; calcium sulfate only slightly).
- **Insoluble**: most carbonates and hydroxides (except Group 1 and ammonium; $Ca(OH)_2$ is slightly soluble).

MARK-SCHEME LANGUAGE

- Observations, not explanations: "colourless solution turns blue"; "grey / silver solid forms"; "bubbles / effervescence" for a gas.
- Not credited: "fizzing" when no gas forms; "white precipitate" of magnesium sulfate (it is soluble); naming the gas instead of describing the bubbles.

Often asked as: copper in silver nitrate (observations plus ionic equation); Mg with acid; $CuCO_3$ with sulfuric acid; precipitating AgCl; MCQ "which type of reaction is this?" ($Zn + H_2SO_4$ is a displacement and a redox reaction).

COMMON MISTAKE

- Mixing up observation and inference: "hydrogen gas is given off" is an inference; "bubbles" is the observation.
- Wrong charge on a metal ion in solution (Cu^{2+}, not Cu^+; Ag^+, not Ag^{2+}).
- Writing (s) for a soluble salt such as $MgSO_4$, or (aq) for a precipitate.

BEYOND THE SPEC, BUT EXAMINED

- Using the reactivity series to predict whether displacement happens, e.g. copper is more reactive than silver (WCH11 Jun 2024).
- Deciding whether a product is (aq) or (s) from solubility (WCH11 Jan 2024).
- Ionic equation for an unfamiliar precipitate, such as a basic nickel carbonate (WCH11 Oct 2023).

Topic 2 – Atomic Structure and the Periodic Table

SPEC 2.1 The structure of an atom

WHAT THIS SPEC POINT ASKS

Know that an atom is made of protons, neutrons and electrons, and where each is found.

- An atom has a tiny, dense, positively charged **nucleus** at its centre, containing **protons** and **neutrons** (together called **nucleons**).
- **Electrons** occupy the space around the nucleus, arranged in **shells** (energy levels), which are split into **sub-shells** and **orbitals** (2.9).
- Almost all the **mass** is in the nucleus; almost all the **volume** is the space where the electrons are. The nucleus is roughly 10^{-5} times the diameter of the atom.
- An atom is **neutral** because it has equal numbers of protons and electrons.

EXAM TIP

Use the right word for where electrons are: "shell" (energy level, n = 1, 2, 3…), "sub-shell" (s, p, d), "orbital" (a region holding up to 2 electrons). Examiners don't treat them as the same.

MARK-SCHEME LANGUAGE

- "Protons and neutrons in the nucleus; electrons in shells / energy levels around it."
- Electrons are **not** in the nucleus: the nucleon count is protons + neutrons only.

Often asked as: part of a 2.4 counting question, or an MCQ on where most of the mass is.

COMMON MISTAKE

Counting electrons as nucleons. Describing electrons as moving in fixed circular "orbits" when the question asks about orbitals.

SPEC 2.2 Relative mass and charge of protons, neutrons and electrons

WHAT THIS SPEC POINT ASKS

Know the relative masses and relative charges of the three sub-atomic particles.

Particle	Where	Relative mass	Relative charge
proton	nucleus	1	+1
neutron	nucleus	1	0
electron	shells around the nucleus	1/1836 (about 0.0005, "negligible")	−1

- Protons and neutrons have almost the same mass; an electron is about 1/1840 of a proton.
- Use **relative** values in answers. (Actual values, such as 1.60×10^{-19} C for the charge, aren't needed.)

MARK-SCHEME LANGUAGE

- Electron mass "1/1836" or "1/1840" or "negligible"; charge "−1" (a minus sign is needed). Proton +1, neutron 0.

Often asked as: complete the table; MCQ on which particle is deflected most (lightest) or not at all (neutral).

COMMON MISTAKE

Giving the electron's relative mass as 0, or its charge without the minus sign.

SPEC 2.3 Atomic (proton) number and mass number

WHAT THIS SPEC POINT ASKS

Know what atomic (proton) number and mass number mean.

- **Atomic number (proton number), Z**: the number of **protons** in the nucleus. It defines the element: every atom of carbon has 6 protons.
- **Mass number (nucleon number), A**: the total number of **protons + neutrons** in the nucleus.
- Nuclide notation: the mass number goes top left and the atomic number bottom left, e.g. $^{23}_{11}\mathrm{Na}$. Often only the mass number is written: $^{23}\mathrm{Na}$, sodium-23.
- The Periodic Table lists elements in order of atomic number.

MARK-SCHEME LANGUAGE

- Atomic number = "number of protons (in the nucleus of an atom)". Mass number = "number of protons **plus** neutrons".

Often asked as: part of 2.4 and 2.5 questions; MCQs comparing species.

COMMON MISTAKE

Saying atomic number is "the number of electrons": that's only true for a neutral atom. Writing the mass number when the question asks for the number of neutrons.

SPEC 2.4 Working out the particles in an atom or ion

WHAT THIS SPEC POINT ASKS

Use atomic number and mass number to find the number of protons, neutrons and electrons in an atom or ion.

- **Protons = Z**.
- **Neutrons = A − Z**.
- **Electrons = Z − (charge)**: a positive ion has **lost** electrons; a negative ion has **gained** them.

WORKED EXAMPLE: COUNTING PARTICLES

Species	Protons	Neutrons	Electrons
^{25}Mg	12	25 − 12 = 13	12
$^{25}Mg^{2+}$	12	13	12 − 2 = 10
$^{35}Cl^{-}$	17	35 − 17 = 18	17 + 1 = 18
$^{56}Fe^{3+}$	26	30	23
$^{18}O^{2-}$	8	10	10

Species with the **same number of electrons** are **isoelectronic**: Na^{+}, Mg^{2+}, F^{-} and O^{2-} all have 10 (3.7).

MARK-SCHEME LANGUAGE

- Give the three **numbers**, labelled. A mass or atomic number in place of a particle count scores nothing.

Often asked as: complete a table for an atom and its ion; MCQ "which ion is not isoelectronic with the others?" (P^{3-}, Cl^{-} and Ca^{2+} have 18 electrons; Ti^{3+} has 22 − 3 = 19).

COMMON MISTAKE

Changing the number of protons when an ion forms: only **electrons** change. Adding electrons for a positive ion. Using the wrong isotope's mass number.

SPEC 2.5

Isotopes

WHAT THIS SPEC POINT ASKS

Understand what isotopes are.

- **Isotopes** are atoms of the **same element** with the **same number of protons** but **different numbers of neutrons** (so different mass numbers).
- Example: chlorine-35 (17 p, 18 n) and chlorine-37 (17 p, 20 n). Hydrogen has three: ^{1}H (no neutrons), ^{2}H (deuterium, 1 neutron) and ^{3}H (tritium, 2 neutrons).

2.5 Same chemistry, different physical properties

- Isotopes of an element have the **same electronic configuration**, so they react in the same way. Chemical reactions involve electrons, not neutrons.
- They have **different masses**, so physical properties that depend on mass differ slightly: density, rate of diffusion, melting and boiling points of their compounds, and how far they are **deflected** in a mass spectrometer (2.6).

MARK-SCHEME LANGUAGE

- Isotope: "atoms of the same element with the same number of protons but different numbers of neutrons". The word **number** must be there. Electrons can be ignored if they equal the protons.
- "Same element" on its own does not replace "same number of protons".
- Same chemical reactions because they have the "same **electronic configuration**". "Same number of electrons" or "same outer electrons" did not score.
- In a mass spectrometer: different mass → different deflection (both halves needed).

Often asked as: define isotope; give a similarity and a difference between two isotopes; explain why isotopes react identically.

COMMON MISTAKE

Writing "same protons, different neutrons" without "number of". Saying isotopes have "different numbers of electrons" or "different chemical properties".

BEYOND THE SPEC, BUT EXAMINED

Atoms of **different** elements with the **same mass number** (e.g. germanium-76 and selenium-76): same number of nucleons but different numbers of protons, so they are different elements, not isotopes (WCH11 Jan 2025).

SPEC 2.6 The mass spectrometer and interpreting mass spectra

WHAT THIS SPEC POINT ASKS

Know the basic principles of a mass spectrometer and use mass spectra to: (i) find the isotopic composition of an element; (ii) calculate A_r from isotope abundances and vice versa; (iii) find the M_r of a molecule and identify it; (iv) understand that some ions have a 2+ charge.

2.6 How it works

Stage	What happens	Why
Vaporisation	the sample is turned into a gas	so particles can move separately
Ionisation	atoms or molecules are **bombarded with high-energy electrons** from an electron gun, knocking an electron out: $X(g) + e^- \rightarrow X^+(g) + 2e^-$	only **charged** particles can be accelerated and deflected
Acceleration	positive ions are accelerated by an **electric field** (negatively charged plates)	to give a fast beam
Deflection	a **magnetic field** bends the ions' path; **lighter** ions (lower m/z) and more highly charged ions are deflected **more**	separates ions by mass-to-charge ratio
Detection	ions hit the detector and make a current; the bigger the current, the more ions	measures relative abundance

- The whole instrument is under a **vacuum**: otherwise ions would **collide with air molecules** and be deflected or lose their charge.
- Output: a **mass spectrum**, relative abundance (or intensity) against **m/z** (mass ÷ charge). For 1+ ions, m/z equals the mass.

2.6 (i) Isotopic composition

Each peak in the spectrum of an element is one isotope. The m/z gives its mass number; the height gives its relative abundance.

2.6 (ii) Relative atomic mass from a spectrum

$$A_r = \Sigma\ (\text{isotopic mass} \times \text{abundance}) \div \Sigma\ \text{abundances}$$

WORKED EXAMPLE: A_R FROM PERCENTAGES AND FROM PEAK HEIGHTS

(a) Magnesium: ^{24}Mg 78.99%, ^{25}Mg 10.00%, ^{26}Mg 11.01%.
$A_r = (24 \times 78.99 + 25 \times 10.00 + 26 \times 11.01) \div 100 = 2432.02 \div 100 =$ **24.32**
(b) Chlorine peaks: m/z 35, height 12.0; m/z 37, height 3.9.
$A_r = (35 \times 12.0 + 37 \times 3.9) \div (12.0 + 3.9) = 564.3 \div 15.9 =$ **35.49**
Divide by the **total abundance** (15.9), not by the number of isotopes.

WORKED EXAMPLE: WORKING BACKWARDS

(a) Boron (A_r 10.8) contains only ^{10}B and ^{11}B. Let the % of ^{10}B be x.
$10x + 11(100 - x) = 1080 \rightarrow 1100 - x = 1080 \rightarrow x =$ **20% ^{10}B**, 80% ^{11}B.
(b) Magnesium (A_r 24.3) is 79% ^{24}Mg; the other 21% is ^{25}Mg and ^{26}Mg. Let the fraction of ^{26}Mg be y.
$24.3 = 24 \times 0.79 + 25(0.21 - y) + 26y = 18.96 + 5.25 + y \rightarrow y = 0.09$
So **9% ^{26}Mg** and 12% ^{25}Mg.

2.6 (iii) M_r of a molecule

- When a molecule is ionised whole, it gives the **molecular ion, M^+**. The molecular ion peak is the peak with the **highest m/z** (ignoring the tiny M+1 peak from carbon-13), and its m/z = **M_r**.
- Molecular ions can also break up into **fragment ions**, which give peaks at lower m/z. The tallest peak (the **base peak**) is often a fragment, not the molecular ion.
- Example: ethanol, C_2H_5OH, has M^+ at m/z **46**; propanal (M_r 58) and butanal (M_r 72) are told apart by their molecular ion peaks.

2.6 (iv) 2+ ions

- If two electrons are knocked out, the ion is **2+** and appears at **half** its mass: m/z = mass ÷ 2.
- $^{24}Mg^{2+}$ appears at m/z **12**; $^{28}Si^{2+}$ at m/z **14**; the ^{35}Cl–$^{37}Cl^{2+}$ ion (mass 72) appears at m/z **36**.
- 2+ ions are deflected more than 1+ ions of the same mass.

MARK-SCHEME LANGUAGE

- Ionisation: "bombarding / firing (high-energy / high-speed) electrons" at the sample, or the equation $X + e^- \rightarrow X^+ + 2e^-$. "Bombarded with electrons" without "high energy" lost a mark.
- Acceleration: "electric field / potential difference / charged plates". A **magnetic** field for acceleration is not accepted.
- Vacuum: "air molecules would be present" and "ions would collide with them" (both points).
- Heavier ions are deflected **less**.
- Show the **charge** on every ion you name; A_r to the s.f. or d.p. asked.

Often asked as: label the stages; why a vacuum is needed; which isotope is deflected most; A_r from a spectrum or table; find an unknown abundance with algebra; identify the molecular ion; explain a peak at half the expected m/z.

COMMON MISTAKE

- Describing the ions formed instead of **how** ionisation happens. Saying a magnetic field accelerates the ions.
- Taking the base peak as the molecular ion.
- For a 2+ ion of Cl_2 at m/z 36, writing "$^{36}Cl^+$" (there is no such isotope in the sample).
- Rounding A_r too early, or dividing by the number of isotopes.

BEYOND THE SPEC, BUT EXAMINED

- Fragment ion vs molecular ion, and the base peak as the most abundant ion (WCH11 Jan 2021).
- Identifying a molecule from its molecular ion peaks, e.g. m/z 78 and 80 for a chloropropane (WCH11 Jan 2024).
- Labelling each peak with the isotopes it contains (WCH11 Jan 2026).

SPEC 2.7

Predicting the mass spectra of diatomic molecules

WHAT THIS SPEC POINT ASKS

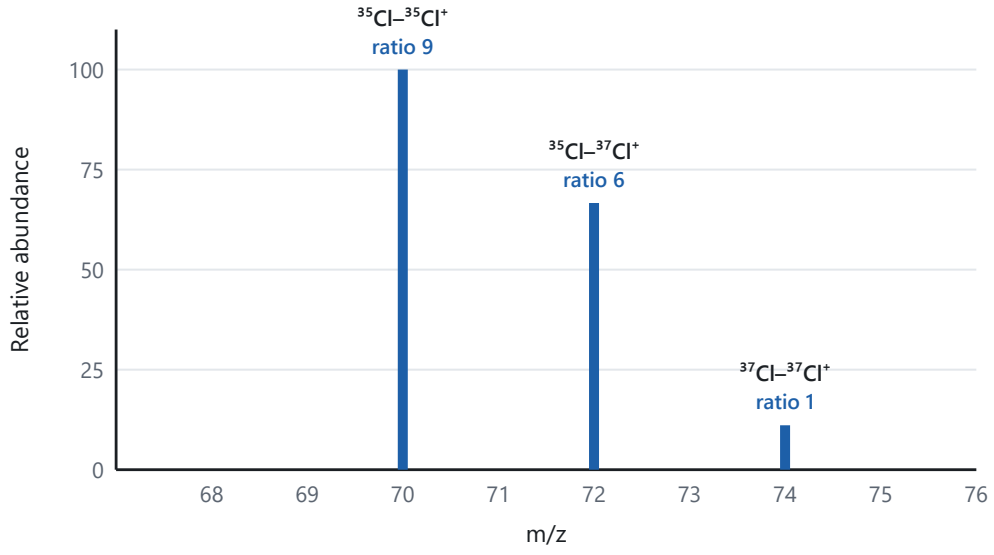
Predict mass spectra, including relative peak heights, for diatomic molecules such as Cl_2, given the isotopic abundances.

2.7 Chlorine: a probability grid

Chlorine is 75% ^{35}Cl and 25% ^{37}Cl, a ratio of **3 : 1**. Each Cl_2 molecule is two atoms picked at random:

	^{35}Cl (3)	**^{37}Cl (1)**
^{35}Cl (3)	$^{35}Cl^{35}Cl$, mass 70: 3 × 3 = **9**	$^{35}Cl^{37}Cl$, mass 72: 3 × 1 = 3
^{37}Cl (1)	$^{37}Cl^{35}Cl$, mass 72: 1 × 3 = 3	$^{37}Cl^{37}Cl$, mass 74: 1 × 1 = **1**

The mixed pair turns up twice, so its total is 3 + 3 = 6. Molecular ion peaks: **m/z 70 : 72 : 74 = 9 : 6 : 1**. (With percentages: $0.75^2 = 0.5625$, $2 \times 0.75 \times 0.25 = 0.375$, $0.25^2 = 0.0625$, the same ratio.)



Molecular-ion region of the Cl_2 spectrum. A real spectrum also has Cl^+ peaks at m/z 35 and 37 (3 : 1) from molecules that split.

2.7 Bromine

^{79}Br and ^{81}Br are about **1 : 1**, so Br_2 gives three peaks at **m/z 158 : 160 : 162 = 1 : 2 : 1**, plus Br^+ peaks at 79 and 81.

EXAM TIP

Draw vertical lines at the exact m/z values, with heights in the right ratio, and label each with its isotopes and a **+** charge. For a molecule with more halogen atoms, expand $(3 + 1)^n$ for chlorine: three Cl atoms give 27 : 27 : 9 : 1.

MARK-SCHEME LANGUAGE

- **M1** peaks at the correct m/z; **M2** (needs M1) in the correct ratio. Any heights are accepted if the ratio is right.
- Explaining the ratio: "3 × 3 : 2 × 3 × 1 : 1 × 1 = 9 : 6 : 1", or a flow diagram of the four combinations.
- Br_2: middle peak double the outer two. The isotopes are ^{79}Br and ^{81}Br (there is no ^{80}Br).

Often asked as: sketch the Cl_2 or Br_2 spectrum; explain the 9 : 6 : 1 ratio; predict peaks for ClF_3 or NCl_3.

COMMON MISTAKE

- Only two peaks (70 and 74, or 158 and 162): you've missed the mixed molecule.
- Repeating 3 : 1 without multiplying out.
- Extra peaks at the wrong m/z; charges left off.

BEYOND THE SPEC, BUT EXAMINED

- ClF_3^+: fluorine has one isotope (^{19}F), so just two peaks, m/z 92 and 94, in a 3 : 1 ratio (WCH11 Oct 2019).
- NCl_3: four molecular ion peaks, 27 : 27 : 9 : 1 (WCH11 Jan 2019).

SPEC 2.8 First, second and third ionisation energies

WHAT THIS SPEC POINT ASKS

Define the first, second and third ionisation energies, and know that all ionisation energies are endothermic.

2.8 Definitions

First ionisation energy: the energy needed to remove **one electron** from **each atom** in **one mole of gaseous atoms**, to form **one mole of gaseous 1+ ions**.

$$Na(g) \rightarrow Na^+(g) + e^- \quad \Delta H = +496 \text{ kJ mol}^{-1}$$

Second ionisation energy: the energy needed to remove one electron from each ion in one mole of **gaseous 1+ ions**, to form one mole of gaseous 2+ ions.

$$Na^+(g) \rightarrow Na^{2+}(g) + e^- \quad \Delta H = +4562 \text{ kJ mol}^{-1}$$

Third ionisation energy: the same, starting from one mole of gaseous 2+ ions.

$$Na^{2+}(g) \rightarrow Na^{3+}(g) + e^- \quad \Delta H = +6910 \text{ kJ mol}^{-1}$$

2.8 Always endothermic

- The electron is **attracted** to the positive nucleus, so energy must be **put in** to pull it away. ΔH is always positive.
- Each successive ionisation energy is **larger**, because the electron is removed from an increasingly **positive** ion: the same number of protons pulls on fewer electrons, which are held more tightly and are closer to the nucleus.

EXAM TIP

Write equations for **one** electron removed, with **(g)** on both species. Start from the right species: the second ionisation of lithium starts from $Li^+(g)$, the third of barium from $Ba^{2+}(g)$.

MARK-SCHEME LANGUAGE

- Equations: **M1** correct species, balanced; **M2** state symbols (g) (depends on M1). Ignore state symbols on the electron. ALLOW $M(g) - e^- \rightarrow M^+(g)$.
- **Do not award multiples**: the definition is per mole of atoms, one electron each.

Often asked as: equation for the first IE of Cl or S, the second IE of Li, the third IE of Ba, $Ca^{2+} \rightarrow Ca^{3+}$, $F^+ \rightarrow F^{2+}$; define first IE.

COMMON MISTAKE

- Leaving out (g); writing S(s) for sulfur.
- Starting from Cl_2 instead of Cl(g).
- Putting the electron on the left (that's electron affinity), or confusing IE with bombarding atoms in a mass spectrometer.
- Writing $Na(g) \rightarrow Na^{2+}(g) + 2e^-$ for the second IE: that is the sum of the first and second.

SPEC 2.9 Orbitals

WHAT THIS SPEC POINT ASKS

Know that an orbital is a region within an atom that can hold up to two electrons with opposite spins.

- An **orbital** is a **region within an atom** where there is a high probability of finding an electron. It can hold **up to two electrons**, and if it holds two they have **opposite spins**.
- Opposite spins (drawn ↑↓) reduce the repulsion between the two electrons enough for them to share the orbital.

2.9 Shells, sub-shells and orbitals

Shell (n)	Sub-shells	Orbitals	Max electrons
1	1s	1	2
2	2s, 2p	1 + 3	2 + 6 = 8
3	3s, 3p, 3d	1 + 3 + 5	2 + 6 + 10 = 18
4	4s, 4p, 4d, 4f	1 + 3 + 5 + 7	2 + 6 + 10 + 14 = 32

A **sub-shell** is a group of orbitals of the same type and energy in one shell (an s sub-shell has 1 orbital, p has 3, d has 5, f has 7).

MARK-SCHEME LANGUAGE

- "A region (**within** an atom) that can hold (up to) two electrons" and "with **opposite spins**". ALLOW spin +½ and −½.

Often asked as: define an orbital; why the two arrows in a box point opposite ways; MCQ on the number of orbitals or electrons in a sub-shell.

COMMON MISTAKE

- "A region **around** the atom": say **within**.
- Describing an orbital as an orbit or a path, or listing s, p, d instead of defining it.
- Using shell, sub-shell and orbital as if they meant the same thing.

SPEC 2.10

What affects the size of an ionisation energy

WHAT THIS SPEC POINT ASKS

Understand how ionisation energy depends on the number of protons, electron shielding and the sub-shell the electron is removed from.

An ionisation energy measures how strongly the **nucleus attracts** the electron being removed. These factors set it:

Factor	Effect on IE	Why
Nuclear charge (number of protons)	more protons → **higher** IE	stronger attraction for the electron
Distance from the nucleus (which shell)	further out → **lower** IE	attraction falls quickly with distance
Shielding by inner electrons	more inner shells → **lower** IE	inner electrons repel the outer electron and reduce the nucleus's pull
Sub-shell the electron is in	a p electron is easier to remove than an s electron in the same shell	the p sub-shell is at a slightly **higher energy** (and partly shielded by the s electrons)
Spin-pair repulsion	an electron in a **paired** orbital is easier to remove	the two electrons in one orbital repel each other

KEY IDEA

The template for any IE comparison: (1) say which has the **higher** value; (2) compare the **number of protons**; (3) compare the **shell** and **shielding**; (4) mention the **sub-shell** or **pairing** if it is an exception; (5) conclude with **attraction** between the nucleus and the outer electron.

MARK-SCHEME LANGUAGE

- Use "nuclear charge" or "number of protons", "shielding (by inner shells)", "distance / energy level / shell", "sub-shell".
- "Effective nuclear charge" is allowed by some mark schemes and discouraged by some examiner reports: the safer wording is **"nuclear charge"** plus **"shielding"** as two separate points.

Often asked as: inside 2.11 and 2.18 explanations (trend across a period, down a group, the exceptions).

COMMON MISTAKE

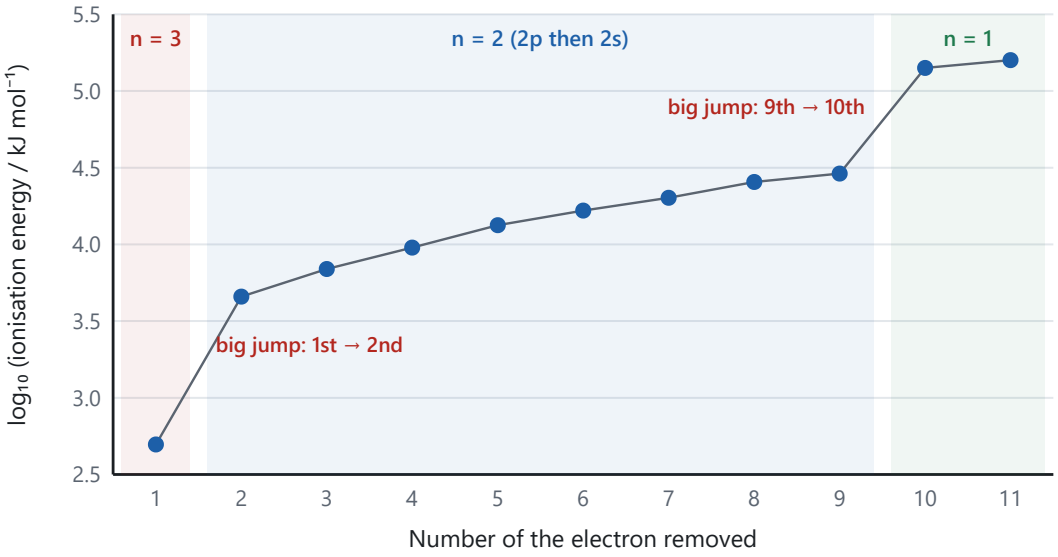
- Repeating one idea three times ("bigger radius, more shells, further away") and never mentioning nuclear charge.
- Saying shielding comes from electrons in the **same** sub-shell: shielding comes mainly from **inner** shells.

SPEC 2.11 Evidence for shells and sub-shells from ionisation energies

WHAT THIS SPEC POINT ASKS

Know how ideas about electronic structure grew from (i) successive ionisation energies, which show quantum shells and the group, and (ii) the first ionisation energies of successive elements, which show sub-shells.

2.11 (i) Successive ionisation energies: shells and the group



Sodium's 11 successive ionisation energies (log scale). Big jumps separate the shells: 1 electron in n = 3, 8 in n = 2, 2 in n = 1, so 2,8,1.

- Each IE is a bit bigger than the last (the ion gets more positive).
- A **big jump** means the next electron comes from a **new shell closer to the nucleus**: less shielded and much more strongly attracted.
- The **number of electrons removed before the first big jump = the number of outer-shell electrons = the group** (for s- and p-block elements).
- Smaller steps inside a shell show **sub-shells**: within n = 2 of sodium, the six 2p electrons (2nd–7th) come off before the two 2s electrons (8th and 9th).

WORKED EXAMPLE: WHICH GROUP?

Successive IEs of an element (kJ mol^{-1}): 578, 1817, 2745, **11 577**, 14 842…
The big jump is between the **3rd and 4th**. So 3 electrons are in the outer shell: **Group 13** (this is aluminium).
Magnesium's 738, 1451, **7733** jumps between the 2nd and 3rd: Group 2.

2.11 Why use a log scale?

- The values range from hundreds to over 100 000 kJ mol^{-1}: on an ordinary scale the small ones would all squash onto the axis. Taking **$\log_{10}$** makes them all fit, and the jumps still show.
- To get a value back from a log, use the **antilog**: IE = 10^{x}. If $\log_{10}$(IE) = 3.98, IE = $10^{3.98}$ = 9550 kJ mol^{-1}.

2.11 (ii) First ionisation energies of successive elements: sub-shells

Across Period 2 (and 3) the first IE rises overall, but it **drops** in two places (see the graph in 2.17 and 2.18):

- **Be → B** (and Mg → Al): boron's outer electron is in **2p**, a higher-energy sub-shell than 2s, and is shielded by the 2s electrons. This dip shows that shells contain **sub-shells of different energy**.
- **N → O** (and P → S): oxygen's outer electron is the first to **pair up** in a 2p orbital; spin-pair repulsion makes it easier to remove. This dip shows that sub-shells are made of **orbitals** that hold two electrons each.

MARK-SCHEME LANGUAGE

- Group: the big jump shows "the electron is removed from a new / inner shell (closer to the nucleus)", and the number of electrons before it is the group number. Quote the actual electrons ("between the 3rd and 4th").
- General increase: "each electron is removed from a more positive ion" / "less repulsion between the remaining electrons".
- Log graph marks: axes labelled with units, e.g. "$\log_{10}$ (IE / kJ mol^{-1})"; points plotted; a scale using at least half the grid; the line does **not** go through the origin (there is no "electron 0").
- Why logs: "the range of values is too large" to plot on a normal scale.

Often asked as: identify the group or element from a set of successive IEs; plot $\log_{10}$ IE against electron number and read off a value by antilog; explain the jump and the general increase; why two successive IEs are close together (same sub-shell).

COMMON MISTAKE

- Linking the jump to "the group" rather than to a **new shell**.
- Not being able to do the antilog (10^{x} on the calculator).
- Misreading 4.06 as 4.60 on a log graph.
- Writing 2nd → 4th instead of the two electrons either side of the jump.

BEYOND THE SPEC, BUT EXAMINED

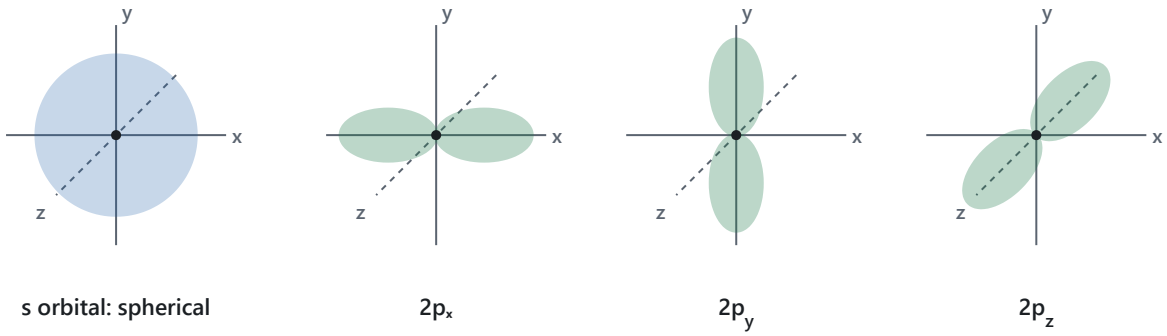
- Writing the shell structure (2,8,1) from the position of the jumps, not spd notation (WCH11 Oct 2020).
- Explaining a small rise inside a shell, e.g. between two electrons in the same sub-shell (WCH11 Jan 2022).

SPEC 2.12 Shapes of s and p orbitals

WHAT THIS SPEC POINT ASKS

Describe the shapes of s and p orbitals.

p orbitals: dumbbell shapes along the x, y and z axes, at right angles



The nucleus is at the centre of each orbital, where the axes cross.

- **s orbitals** are **spherical**, centred on the nucleus. Each shell has one; it gets bigger as the shell number rises (2s is bigger than 1s).
- **p orbitals** are **dumbbell-shaped** (two lobes either side of the nucleus). There are **three** in each p sub-shell, at **right angles** to each other along the x, y and z axes (p_x, p_y, p_z). They have the same energy. The first p orbitals are in shell 2.
- You don't need the shapes of d orbitals.

MARK-SCHEME LANGUAGE

- "Sphere / spherical". "Circle" or "round" alone is not awarded.
- p: "dumbbell / figure-of-eight", drawn along an axis with the nucleus where the lobes meet.

Often asked as: sketch an s and a p orbital; MCQ on shape and number of electrons in a p orbital.

COMMON MISTAKE

Drawing a p orbital as two separate blobs not touching at the nucleus. Saying a p orbital holds 6 electrons: the **sub-shell** holds 6; each **orbital** holds 2.

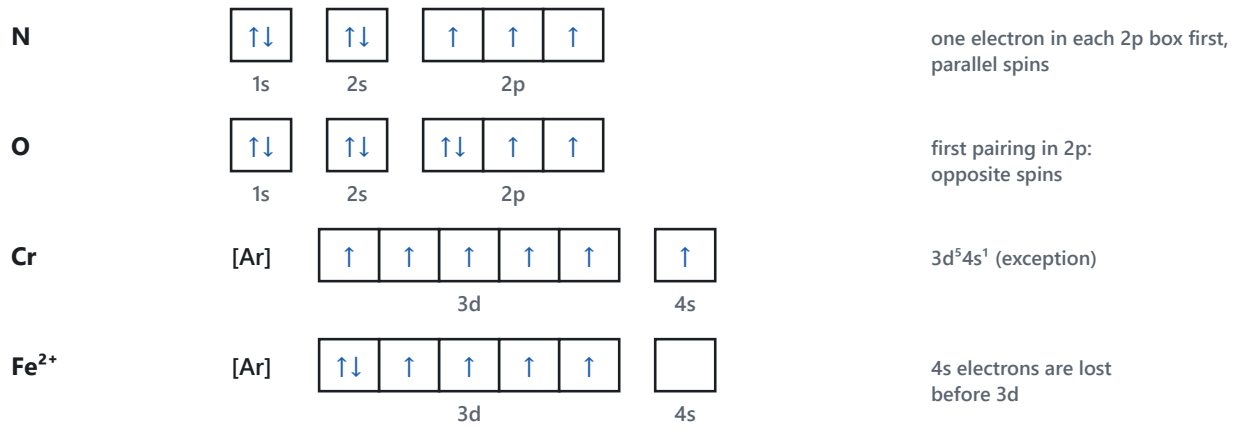
SPEC 2.13

How electrons fill orbitals

WHAT THIS SPEC POINT ASKS

Know that electrons in a sub-shell occupy orbitals singly before pairing up, and that paired electrons have opposite spins.

- Electrons fill the **lowest-energy** sub-shell available first.
- Within a sub-shell (e.g. the three 2p orbitals), electrons go **one into each orbital first**, with **parallel spins**, before any pair up. Being in separate orbitals keeps them further apart, so there is less repulsion. (This is Hund's rule.)
- When electrons pair up in an orbital, they have **opposite spins**. No orbital holds more than two.



Electrons-in-boxes: one box per orbital, one arrow per electron. Label every sub-shell.

MARK-SCHEME LANGUAGE

- Boxes: **M1** sub-shells correctly labelled; **M2** boxes correctly filled. ALLOW half-headed arrows, labels $2p_x/2p_y/2p_z$, and any p box for the pair.
- DO NOT AWARD vertical lines without arrowheads, or pairs in more than one p box when the sub-shell isn't full.

Often asked as: draw sulfur or phosphorus in boxes; MCQ on the number of unpaired electrons (Cr has six).

COMMON MISTAKE

Pairing electrons in a p orbital while another p orbital in the same sub-shell is still empty (e.g. drawing S with two paired 3p boxes). Both arrows in a pair pointing the same way.

SPEC 2.14

Electronic configurations from hydrogen to krypton

WHAT THIS SPEC POINT ASKS

Predict the electronic configurations of atoms and ions from H to Kr, using s, p, d notation and electrons-in-boxes.

2.14 Filling order

1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p

- **4s fills before 3d** because the 4s sub-shell is slightly lower in energy when it is empty.
- Shorthand: write the previous noble gas in square brackets, e.g. K = [Ar]$4s^1$.

Element	Configuration
N (7)	$1s^2\ 2s^2\ 2p^3$
Na (11)	$1s^2\ 2s^2\ 2p^6\ 3s^1$
Cl (17)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
Ca (20)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$
Ti (22)	[Ar] $3d^2\ 4s^2$
Cr (24)	**[Ar] $3d^5\ 4s^1$** (not $3d^4 4s^2$)
Fe (26)	[Ar] $3d^6\ 4s^2$
Cu (29)	**[Ar] $3d^{10}\ 4s^1$** (not $3d^9 4s^2$)
Zn (30)	[Ar] $3d^{10}\ 4s^2$
Br (35)	[Ar] $3d^{10}\ 4s^2\ 4p^5$
Kr (36)	[Ar] $3d^{10}\ 4s^2\ 4p^6$

- **Chromium and copper** are the two exceptions: one 4s electron moves into 3d, giving a **half-filled** ($3d^5$) or **full** ($3d^{10}$) d sub-shell, which is a lower-energy (more stable) arrangement.
- Either order is accepted when writing: [Ar]$3d^6 4s^2$ or [Ar]$4s^2 3d^6$.

2.14 Ions

- **Negative ions**: add electrons to the next available orbital. Cl^- = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (same as Ar).
- **Positive ions of s- and p-block elements**: remove the outer electrons. Na^+ = $1s^2\ 2s^2\ 2p^6$.
- **Transition metal ions**: the **4s electrons are lost first**, before any 3d electrons.

Ion	Configuration
Fe^{2+}	[Ar] $3d^6$
Fe^{3+}	[Ar] $3d^5$
Cu^{2+}	[Ar] $3d^9$ (one 4s and one 3d electron removed)
Ni^{2+}	[Ar] $3d^8$
Ti^{3+}	[Ar] $3d^1$
Zn^{2+}	[Ar] $3d^{10}$

MARK-SCHEME LANGUAGE

- Cl: "$1s^2 2s^2 2p^6 3s^2 3p^5$"; ALLOW $2p_x^2 2p_y^2 2p_z^2$ for $2p^6$; [Ne] accepted for $1s^2 2s^2 2p^6$.
- Cu^{2+} = [Ar]$3d^9$: one electron from 4s and one from 3d, **not** both from 3d.

Often asked as: configuration of Cl, Ca, Al, P or S (in boxes); Cu and Cu^{2+}; Ni^{2+}; MCQ "which ion has this configuration?"

COMMON MISTAKE

- Writing 3d before 4s **in the filling**, or removing 3d electrons before 4s when forming an ion.
- Giving the GCSE form 2,8,8,2 when s, p, d notation is asked (or the reverse).
- Giving the atom's configuration when the ion was asked.
- Writing 4d for 3d; untidy superscripts.

BEYOND THE SPEC, BUT EXAMINED

Ions isoelectronic with argon (P^{3-}, S^{2-}, Cl^-, K^+, Ca^{2+}) all have the configuration $1s^2 2s^2 2p^6 3s^2 3p^6$. Ti^{3+} is the odd one out: $[Ar]3d^1$, 19 electrons (WCH11 Jan 2026).

SPEC 2.15 Electronic configuration determines chemical properties

WHAT THIS SPEC POINT ASKS

Understand that an element's electronic configuration determines its chemical properties.

- Chemical reactions involve the **outer electrons** (they are lost, gained or shared). So elements with the **same outer-shell configuration** react in similar ways: that's why elements in a **group** behave alike.

Group	Outer configuration	Typical behaviour
1	ns^1	lose 1 electron → M^+ (Na^+, K^+)
2	ns^2	lose 2 electrons → M^{2+}
13	$ns^2\ np^1$	Al forms Al^{3+}; three outer electrons
16	$ns^2\ np^4$	gain 2 electrons → X^{2-} (O^{2-}, S^{2-})
17	$ns^2\ np^5$	gain 1 electron → X^-; or share one electron in a covalent bond
18	$ns^2\ np^6$ (He $1s^2$)	full outer sub-shells, very high IE: very unreactive

- The **number of shells** changes down a group, so reactivity changes (e.g. Group 1 metals become more reactive because the outer electron is further away and easier to lose).
- Isotopes have identical chemistry because their configurations are identical (2.5).

MARK-SCHEME LANGUAGE

- Same group → "same number of electrons in the outer shell / same outer electronic configuration". Different reactivity → "different number of shells".

Often asked as: MCQ predicting a formula (sulfur gains 2 electrons, so it forms K_2S with potassium); compare Group 13 elements (three outer electrons, different numbers of shells).

COMMON MISTAKE

Saying elements in a period have similar chemistry: it's elements in a **group**.

SPEC 2.16 Blocks of the Periodic Table and sub-shell capacities

WHAT THIS SPEC POINT ASKS

Know that the Periodic Table is divided into s, p and d blocks, and know how many electrons s, p and d sub-shells hold in the first four shells.

- An element's **block** is the sub-shell its **highest-energy electron** goes into.

Block	Groups	Sub-shell being filled
s	1 and 2 (and He)	s (holds 2)
p	13 to 18	p (holds 6)
d	the transition elements, Sc to Zn in Period 4	d (holds 10)
f	lanthanides and actinides	f (holds 14)

- Capacities: **s 2, p 6, d 10, f 14**. Shell totals: n = 1 holds 2, n = 2 holds 8, n = 3 holds 18, n = 4 holds 32.
- Helium is placed in Group 18 because it has a full outer shell, but its configuration ($1s^2$) makes it an s-block element.

MARK-SCHEME LANGUAGE

- "s holds 2, p holds 6, d holds 10"; order of filling 4s before 3d.

Often asked as: MCQs on block, number of sub-shells, or electrons in a shell; state the size and order of sub-shells.

COMMON MISTAKE

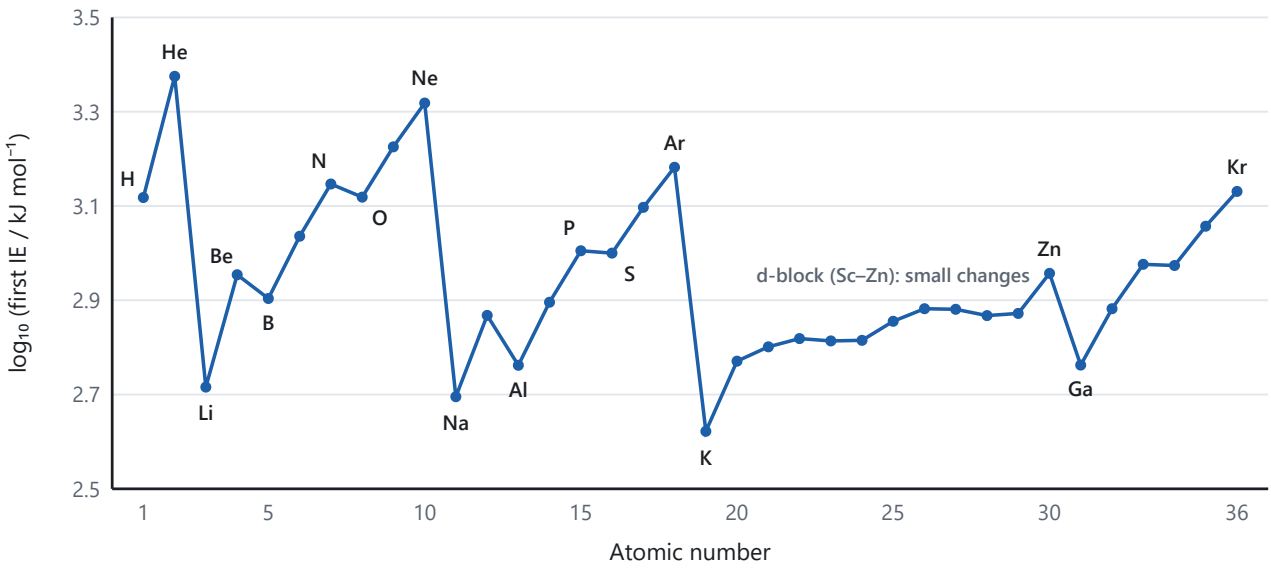
Saying the third shell holds 8 electrons: it **can** hold 18 (3s, 3p and 3d). The "2,8,8" pattern only describes the order of filling up to calcium.

SPEC 2.17

Graphs of ionisation energy and periodic properties

WHAT THIS SPEC POINT ASKS

Plot data for elements 1 to 36 as graphs (including $\log_{10}$ of first ionisation energies) and use them to explain what a "periodic property" is.



$\log_{10}$ (first ionisation energy) for H to Kr. Peaks at the noble gases, troughs at the Group 1 metals, small dips at Groups 13 and 16.

2.17 What the graph shows

- **Peaks** at the noble gases (He, Ne, Ar, Kr): a full outer shell, so the electron is very hard to remove.

- **Troughs** at the Group 1 metals (Li, Na, K): one electron in a new shell, well shielded.
- The same **pattern repeats** across each period: a general rise, with dips at Group 13 (start of the p sub-shell) and Group 16 (first paired p electron).
- Across the d block (Sc to Zn) the first IE hardly changes: the electrons removed are 4s electrons, and the extra protons are largely shielded by the new 3d electrons.

2.17 Periodic property

A **periodic property** is one that shows a **regular repeating pattern** when plotted against **atomic number**, with the same trend repeated in each **period**. First ionisation energy, atomic radius and melting temperature are all periodic properties.

EXAM TIP

When plotting: choose a scale that uses at least half the grid on both axes, label axes with quantity and unit (e.g. "$\log_{10}$ (IE / kJ mol^{-1})"), plot every point within half a square, and join the points with straight lines or a smooth line. Don't extend the line to the origin.

MARK-SCHEME LANGUAGE

- "Periodicity / periodic property": a "repeating pattern" or "trend repeated across each period".
- Graph: axes labelled, sensible linear scale, all points plotted.

Often asked as: plot first IEs or atomic radii against atomic number; describe the pattern; read a value off the graph.

COMMON MISTAKE

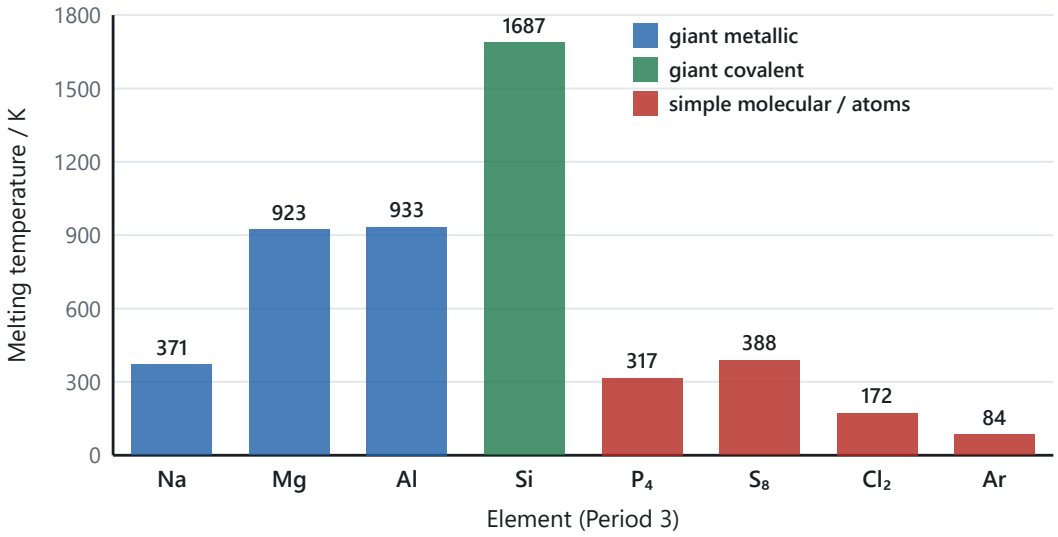
Describing a periodic property as "the property increases across the period" without saying the pattern **repeats**.

SPEC 2.18 Trends across Periods 2 and 3 and down a group

WHAT THIS SPEC POINT ASKS

Explain: (i) the trends in melting and boiling temperatures across Periods 2 and 3, in terms of structure and bonding; (ii) the general increase and the specific dips in ionisation energy across Periods 2 and 3; (iii) the decrease in first ionisation energy down a group.

2.18 (i) Melting and boiling temperatures

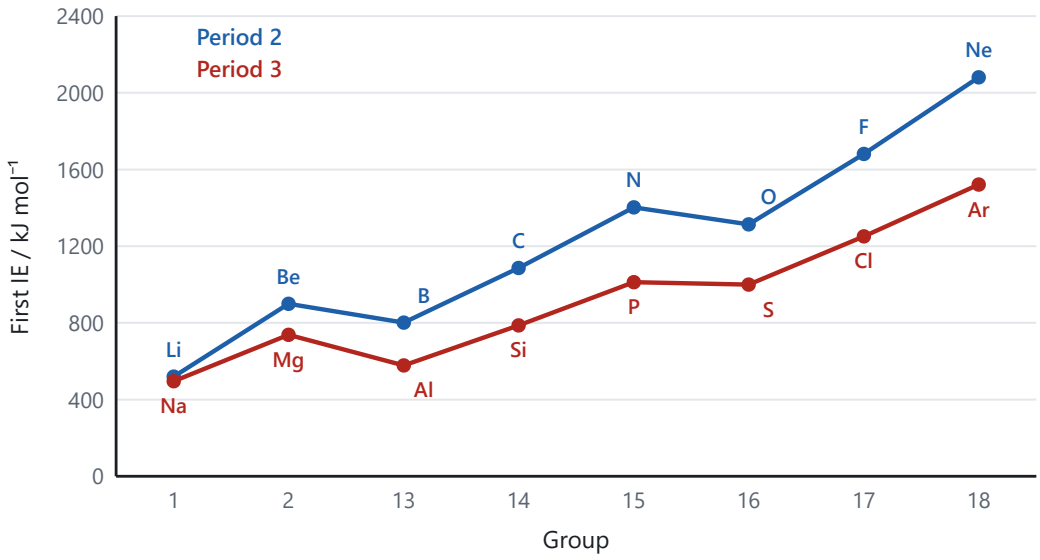


Melting temperatures across Period 3, coloured by structure.

Element (s)	Structure	What must be overcome on melting	Result
Na, Mg, Al	giant **metallic** lattice	strong attraction between metal ions and delocalised electrons	high; rises Na → Al
Si	giant **covalent** lattice	**many strong covalent bonds**	highest in the period
P_4, S_8, Cl_2	**simple molecular**	**weak London forces between molecules** (covalent bonds inside molecules don't break)	low
Ar	single **atoms**	very weak London forces between atoms	lowest

- **Na → Mg → Al**: the ions get **more highly charged** (Na^+, Mg^{2+}, Al^{3+}) and **smaller**, and each atom gives **more delocalised electrons** (1, 2, 3). So the metallic bonding gets stronger. (Boiling temperatures show this most clearly: Na 1156 K, Mg 1363 K, Al 2792 K.)
- **S_8 > P_4 > Cl_2 > Ar**: bigger molecules with **more electrons** have stronger London forces. S_8 (128 electrons) beats P_4 (60), Cl_2 (34) and Ar (18).
- **Period 2** follows the same pattern: Li (454 K) and Be (1560 K) are metallic, with Be stronger (Be^{2+}, two delocalised electrons per atom); **B** (2349 K) and **C** (about 3800 K, graphite/diamond) are giant covalent; N_2 (63 K), O_2 (54 K), F_2 (53 K) are simple molecules and Ne (25 K) is atoms: all very low.

2.18 (ii) Ionisation energy across Periods 2 and 3



First ionisation energies across Periods 2 and 3. Same shape in both; Period 3 values are lower (one more shell).

General increase across a period

- The **nuclear charge increases** (one more proton each time).
- The extra electrons go into the **same shell**, so **shielding stays about the same** and the atoms get smaller.
- So the outer electron is **attracted more strongly** and the IE rises.

Dip from Group 2 to Group 13 (Be → B, Mg → Al)

- B: $1s^2\ 2s^2\ \mathbf{2p^1}$. Its outer electron is in the **2p** sub-shell, which is **higher in energy** than 2s (and is partly shielded by the 2s electrons). It is easier to remove, despite the extra proton.

Dip from Group 15 to Group 16 (N → O, P → S)

- N: 2p has three **unpaired** electrons, one in each orbital. O: $2p^4$, so one 2p orbital now holds a **pair**.
- The **repulsion between the paired electrons** makes one of them easier to remove, so O's first IE is lower than N's.

2.18 (iii) Down a group

The first IE **decreases** down a group (e.g. Li 520 > Na 496 > K 419 kJ mol^{-1}; Cl 1251 > Br 1140 kJ mol^{-1}):

- The outer electron is in a **shell further from the nucleus** (larger atomic radius).
- There are **more inner shells**, so **more shielding**.
- The nuclear charge does increase, but this is outweighed by the extra distance and shielding.
- So there is **less attraction** between the nucleus and the outer electron.

MARK-SCHEME LANGUAGE

- Melting: name the **structure and bonding** of each element, say what is broken or overcome, then compare the energy needed. Si: "giant covalent (lattice), strong covalent bonds throughout, a lot of energy to break". P_4: "simple molecular; weak intermolecular (London) forces; little energy to overcome". Giant covalent answers must not mention ions, metallic bonds or intermolecular forces.
- Across a period (2–3 marks): "more protons / nuclear charge increases"; "electron added to the same shell / similar shielding"; sometimes also "removed from the same sub-shell". IGNORE atomic radius in that mark scheme.
- N → O: O "loses a paired electron (from a 2p orbital)"; N "loses an electron from a singly occupied orbital"; "repulsion between paired electrons". The reverse argument is allowed.
- Mg → Al: Al's outer electron is in **3p**, Mg's in **3s**; 3p is "further from the nucleus / higher energy / shielded by the 3s electrons".
- Down a group: state **which is higher** first, then three of: further from the nucleus / more shells; more shielding; less attraction (despite more protons).

Often asked as: explain the trend across Period 3; explain one dip (often N/O or Mg/Al); compare two elements in a group; explain the melting temperatures of Si and P, or spot a wrongly plotted point on a melting-temperature graph.

COMMON MISTAKE

- Saying **covalent bonds break** when phosphorus or sulfur melts, or calling silicon "simple molecular".
- Writing "weak covalent bonds" or "strong structures" instead of naming the forces.
- Saying "a half-full p **orbital** is stable": it's a half-full **sub-shell** (one electron in each orbital), and you must say **which** electrons repel.
- Explaining a dip with the general trend, or stating a trend without explaining it.
- Answers with no comparator ("it is higher") or with three versions of the same distance point.

BEYOND THE SPEC, BUT EXAMINED

- Estimating a missing melting temperature from the pattern: N_2 is a gas at room temperature, so its melting temperature is far below 298 K (WCH11 Oct 2024).
- Boron has a giant covalent structure and a very high melting temperature (WCH11 Oct 2024).
- Spotting impossible values on a plotted graph, e.g. argon at 500 K (WCH11 Jan 2020).

Topic 3 – Bonding and Structure

SPEC 3.1 Evidence for the existence of ions

WHAT THIS SPEC POINT ASKS

Know and interpret the evidence that ions exist: the physical properties of ionic compounds, electron density maps, and the migration of ions.

3.1 Physical properties

Property of ionic compounds	What it tells you
High melting and boiling temperatures	strong forces act in every direction through a giant structure
Conduct electricity when molten or dissolved, but not when solid	charged particles exist; they are **fixed in place** in the solid but **free to move** in the liquid or solution
Often **soluble in water**, insoluble in non-polar solvents	charged particles are attracted to polar water molecules
Brittle: they shatter when hit	a small shift lines up like charges, which repel (3.4)
Electrolysis decomposes them, with products at both electrodes	positive particles go to the cathode, negative ones to the anode

3.1 Electron density maps

- X-ray diffraction measures how electron density is spread through a crystal. The result is drawn as a contour map, like a map of hills.
- In **sodium chloride** each ion has its **own separate set of roughly circular contours**, and the electron density falls to **almost zero between the ions**. There is no build-up of electrons between them, so electrons are not shared: they belong to one ion or the other.
- The Cl^- contours enclose a bigger region than the Na^+ contours: Cl^- is the larger ion. Compare this with a covalent bond (3.10), where contours **surround both nuclei** and density stays high between them.

3.1 Migration of ions

If coloured ions move towards an electrode when a direct current flows, the colour must be carried by **charged** particles.

- **On a microscope slide**: wet a strip of filter paper with an electrolyte (e.g. sodium chloride or copper(II) sulfate solution), clip it to a slide, connect each end to a low-voltage d.c. supply, and put a crystal of **potassium manganate(VII)** in the middle. A **purple** colour moves towards the **positive electrode (anode)**: the MnO_4^- ion is negative. With a crystal of copper(II) sulfate, a **blue** colour moves towards the **negative electrode (cathode)**: Cu^{2+} is positive.
- **In a U-tube**: put green **copper(II) chromate(VI)** solution at the bottom and carefully layer a colourless electrolyte (e.g. dilute acid) on top in each arm. With a d.c. supply, a **blue** colour (Cu^{2+}) rises towards the **cathode** and a **yellow** colour (CrO_4^{2-}) towards the **anode**. The green colour was the two ions together.

MARK-SCHEME LANGUAGE

- Migration (3 marks): electrolysis using a **d.c.** supply; a crystal or solution of copper(II) sulfate on **damp filter paper** (on a microscope slide), or a U-tube; "**blue** copper(II) ions move to the **negative** electrode / cathode". A labelled diagram is accepted.
- Solid ionic compound doesn't conduct because "the **ions** are fixed / not free to move". Not "there are no delocalised electrons".

Often asked as: describe an experiment to show ions migrate; MCQ on reading an electron density map; explain why an ionic compound conducts when molten but not when solid.

COMMON MISTAKE

Saying a solid ionic compound doesn't conduct because it has "no free electrons". It has **ions**; they just can't move. Saying the electrons move in a molten ionic compound: the **ions** move.

BEYOND THE SPEC, BUT EXAMINED

Colours as evidence: blue Cu^{2+} moves to the cathode, purple MnO_4^- to the anode (WCH11 Oct 2025).

SPEC 3.2

Forming ions by losing or gaining electrons

WHAT THIS SPEC POINT ASKS

Describe how ions form when atoms lose or gain electrons.

- **Metals** (few outer electrons) **lose** electrons to form **positive ions (cations)**. Na: 2,8,1 → Na^+: 2,8.
- **Non-metals** (nearly full outer shell) **gain** electrons to form **negative ions (anions)**. Cl: 2,8,7 → Cl^-: 2,8,8.
- In forming an ionic compound, electrons are **transferred** from the metal atoms to the non-metal atoms. The ions usually have the configuration of a **noble gas**.
- In s, p, d notation: Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$ → Mg^{2+} $1s^2\ 2s^2\ 2p^6$ (two 3s electrons lost); O $1s^2\ 2s^2\ 2p^4$ → O^{2-} $1s^2\ 2s^2\ 2p^6$ (two electrons gained into 2p).
- The ion's charge follows from the group: Group 1 → 1+, Group 2 → 2+, Group 13 (Al) → 3+, Group 15 (N) → 3−, Group 16 → 2−, Group 17 → 1−.

MARK-SCHEME LANGUAGE

- "Electrons are transferred from the metal to the non-metal"; give the number of electrons and the charges ("each Mg atom loses two electrons to form Mg^{2+}").

Often asked as: describe what happens to the electrons when sodium reacts with fluorine; part of a dot-and-cross question.

COMMON MISTAKE

Saying atoms "share" electrons in an ionic compound. Describing the ion as the atom ("magnesium has a 2+ charge"): say **magnesium ion**.

SPEC 3.3 Dot-and-cross diagrams for ionic compounds

WHAT THIS SPEC POINT ASKS

Draw dot-and-cross diagrams to show the electrons in cations and anions.

• electrons from the non-metal × electrons from the metal



sodium chloride, NaCl

magnesium oxide, MgO

Outer shells only. The cation's outer shell is shown empty here; drawing its full inner shell of 8 is also accepted.

How to draw one:

1. Draw **each ion separately**, in **square brackets**, with its **charge** at the top right.
2. Show the anion with a **full outer shell of 8**: its own electrons as dots, the electron(s) it gained as crosses.
3. Show the cation with **no outer electrons** (or with the full shell beneath).
4. Show the right **number** of each ion: $CaCl_2$ needs one Ca^{2+} and **two** Cl^- (or write "2" in front of $[Cl]^-$).

Compound	Ions drawn
NaF	$[Na]^+$ and $[F]^-$ (8 electrons: 7 •, 1 ×)
MgO	$[Mg]^{2+}$ and $[O]^{2-}$ (8 electrons: 6 •, 2 ×)
$CaCl_2$	$[Ca]^{2+}$ and 2 $[Cl]^-$
Al_2O_3	2 $[Al]^{3+}$ and 3 $[O]^{2-}$
Li_3N	3 $[Li]^+$ and $[N]^{3-}$

MARK-SCHEME LANGUAGE

- One mark per ion, each with its charge. Na^+ with 0 or 8 outer electrons; F^- with 8. Circles and brackets are not required.
- ALLOW all dots or all crosses, or dots and crosses swapped. A missing charge may lose one mark.

Often asked as: dot-and-cross of NaF, NaCl, MgO, $CaCl_2$, $BaCl_2$, aluminium oxide.

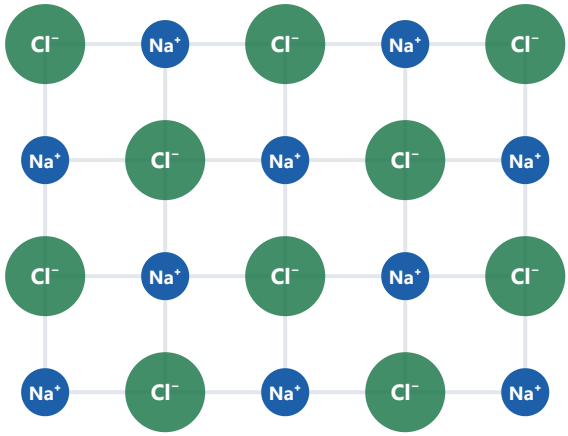
COMMON MISTAKE

- Drawing a **covalent** (shared-pair) diagram for an ionic compound, or the reverse.
- Wrong charges (Ca^+), no charges at all, or a cation with a half-filled outer shell.

SPEC 3.4 Ionic crystals are giant lattices

WHAT THIS SPEC POINT ASKS

Describe ionic crystals as giant lattices of ions.



One layer of the NaCl lattice. In 3D, each ion is surrounded by 6 ions of the opposite charge.

- An ionic solid is a **giant ionic lattice**: a **regular, repeating** 3D arrangement of a huge number of positive and negative ions, alternating so that each ion is surrounded by ions of the opposite charge.
- In NaCl each Na^+ has **6** Cl^- neighbours and each Cl^- has 6 Na^+ (a cubic arrangement). MgO has the same structure.
- The formula (NaCl) is only the **ratio** of ions; there are no molecules.
- **Brittleness**: if a force shifts one layer by one ion, ions of the **same charge** line up next to each other, repel, and the crystal splits.

MARK-SCHEME LANGUAGE

- Drawing: "at least 9 correctly labelled ions", alternating + and −, in a regular (cubic) array.
- Say "giant ionic lattice", not "molecule" or "macromolecule".

Often asked as: draw a section of the NaCl lattice; state the structure of a solid such as barium chloride.

COMMON MISTAKE

Calling NaCl a molecule. Drawing ions of the same charge next to each other, or leaving ions unlabelled.

SPEC 3.5

Ionic bonding is a net electrostatic attraction

WHAT THIS SPEC POINT ASKS

Know that ionic bonding is the strong net electrostatic attraction between ions.

- **Ionic bonding** is the **strong electrostatic attraction between oppositely charged ions**.
- In the lattice every ion is attracted to all the oppositely charged ions around it and repelled by the like-charged ions slightly further away. The attractions are bigger than the repulsions, so there is a **net attraction**.
- The attraction acts **in all directions** (it is non-directional), and there are huge numbers of ions, so a lot of energy is needed to separate them: **high melting temperatures**.

MARK-SCHEME LANGUAGE

- "(Strong) electrostatic attraction between **oppositely charged ions**" (or "between positive and negative ions").

Often asked as: define ionic bonding; explain why an ionic compound has a high melting temperature.

COMMON MISTAKE

Saying "attraction between atoms", or "between the nucleus and electrons" (that's covalent or metallic). Saying ionic bonds are "weak".

SPEC 3.6

How ionic charge and radius affect the strength of ionic bonding

WHAT THIS SPEC POINT ASKS

Understand how ionic radius and ionic charge affect the strength of ionic bonding.

- **Higher charge** on the ions → **stronger** attraction.
- **Smaller ions** → the centres of the ions are **closer** → **stronger** attraction.
- Stronger ionic bonding → more energy needed to separate the ions → **higher melting temperature** (and a more exothermic lattice energy, which you meet in Unit 4).

Compare	Melting temperature / °C	Reason
NaF vs NaCl	993 vs 801	F^- is **smaller** than Cl^-
NaCl vs KCl	801 vs 771	Na^+ is **smaller** than K^+
MgO vs NaCl	2852 vs 801	Mg^{2+} and O^{2-} are **doubly charged** (and Mg^{2+} is smaller)
$BaCl_2$ vs CsCl	962 vs 645	Ba^{2+} has a **bigger charge** and is **smaller** than Cs^+

WORKED EXAMPLE: A 4-MARK COMPARISON

Explain why barium chloride has a higher melting temperature than caesium chloride.

1. The barium **ion** has a greater charge (2+) than the caesium **ion** (1+).
2. The barium ion is **smaller** than the caesium ion.
3. So the **electrostatic attraction** between the ions is **stronger** in $BaCl_2$.
4. **More energy** is needed to overcome it, so the melting temperature is higher.

MARK-SCHEME LANGUAGE

- Name the **ions** and compare **charge** and **size**, then "stronger electrostatic attraction between the ions", then "more energy needed".
- Comparing MgO with $MgSO_4$: "oxide ion is smaller than the sulfate ion" so stronger attraction between Mg^{2+} and O^{2-}. Comparisons with sulfur or sulfide are not awarded.
- "Higher charge density" is accepted.

Often asked as: compare the melting temperatures of two ionic compounds; MCQ ranking lattice strength.

COMMON MISTAKE

- Writing about **atoms** ("magnesium has a higher charge"): it's the magnesium **ion**.
- Using atomic radius, electronegativity or intermolecular forces when the question says "consider the ions".
- Confusing ionic charge with nuclear charge, or ionisation energy with bond strength.

SPEC 3.7 Trends in ionic radius

WHAT THIS SPEC POINT ASKS

Understand the trends in ionic radius down a group, and across an isoelectronic series such as N^{3-} to Al^{3+}.

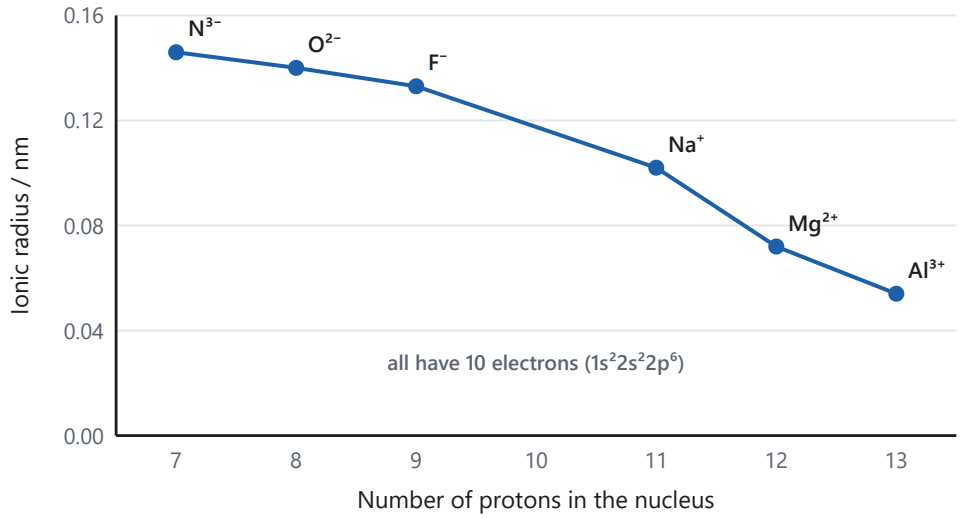
3.7 Ions compared with their atoms

- A **cation is smaller** than its atom: it has lost its whole outer shell, and the same number of protons now pulls on fewer electrons. Na atom 0.186 nm → Na^+ 0.102 nm.
- An **anion is bigger** than its atom: extra electrons increase the repulsion between electrons, and the same nuclear charge holds more electrons. Cl atom 0.099 nm → Cl^- 0.181 nm.

3.7 Down a group

Ionic radius **increases** down a group (Li^+ 0.076 < Na^+ 0.102 < K^+ 0.138 nm; F^- 0.133 < Cl^- 0.181 < Br^- 0.196 < I^- 0.220 nm), because each ion has **one more shell** of electrons.

3.7 An isoelectronic series



Isoelectronic ions N^{3-} to Al^{3+} (approximate radii; values differ slightly between data sources).

- N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} are **isoelectronic**: all have **10 electrons**, $1s^2\ 2s^2\ 2p^6$, with the same shielding.
- From N^{3-} to Al^{3+} the **number of protons increases** from 7 to 13. The bigger nuclear charge pulls the same 10 electrons **closer**, so the **radius decreases**.

MARK-SCHEME LANGUAGE

- Isoelectronic trend: "ionic radius **decreases** as atomic number increases" (a **trend**, not a comparison of two ions), "same number of electrons", "more protons" so electrons are pulled closer. DO NOT AWARD "atomic radius".
- Na^+ smaller than F^-: same number of electrons, but Na^+ has **more protons**, so the electrons are attracted more strongly. ALLOW "greater nuclear charge". IGNORE shielding.

Often asked as: plot the radii of N^{3-} to Al^{3+} and read off a value; describe and explain the trend; explain why Na^+ is smaller than F^-.

COMMON MISTAKE

- Comparing just N^{3-} and Al^{3+} when a **trend** for the whole series is asked.
- Writing "atomic radius" or "smaller atom" for an ion.
- Saying shielding changes across an isoelectronic series: it doesn't; only the number of protons changes.

BEYOND THE SPEC, BUT EXAMINED

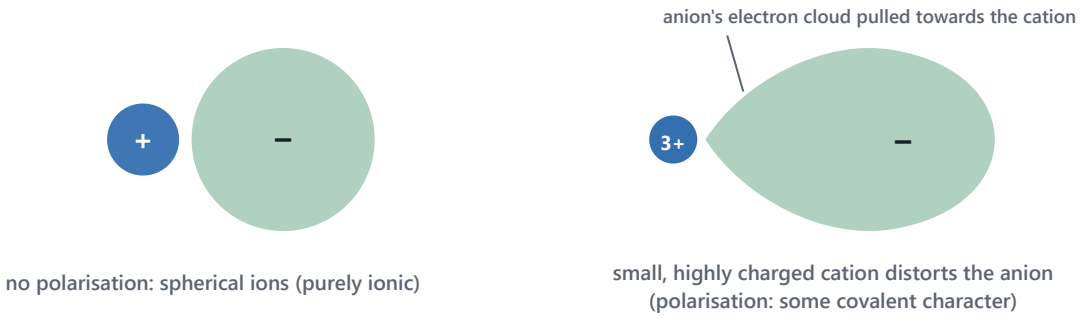
Reading an unknown ionic radius from a line of best fit through the series (a range is accepted) (WCH11 Jan 2023).

SPEC 3.8

Polarisation of ions

WHAT THIS SPEC POINT ASKS

Understand what "polarisation" means for ions.



Left: the purely ionic model. Right: a polarised anion.

- The simple ionic model treats ions as **perfect spheres** of charge.
- In reality a **cation attracts the electron cloud** of a nearby anion and **distorts** it towards itself. This is **polarisation**.
- Some electron density now sits **between** the two ions, a bit like a shared pair. So the bond gets some **covalent character**.
- The more the anion is polarised, the more covalent the compound: lower melting temperature than expected, or even a molecular structure ($AlCl_3$ sublimes at about 180 °C; AlF_3 stays solid until about 1290 °C).
- In Unit 4 you'll see the evidence: lattice energies calculated from the pure ionic model don't match experimental values for polarised compounds.

MARK-SCHEME LANGUAGE

- "The cation **distorts / polarises** the electron cloud of the anion", giving "covalent character".

Often asked as: explain why aluminium chloride has covalent character; why fluoride is hard to polarise (3.9).

COMMON MISTAKE

Saying the cation is polarised. In this unit it is the **anion's** electron cloud that is distorted, by the cation.

SPEC 3.9

Polarising power and polarisability

WHAT THIS SPEC POINT ASKS

Understand that the polarising power of a cation depends on its radius and charge, and that the polarisability of an anion also depends on its radius and charge.

	Bigger effect when the ion is...	Examples (increasing)
Polarising power of a cation	**small** and **highly charged** (high charge density)	$K^+ < Na^+ < Li^+$; $Na^+ < Mg^{2+} < Al^{3+}$; $Ba^{2+} < Ca^{2+} < Be^{2+}$
Polarisability of an anion	**large** and **highly charged**	$F^- < Cl^- < Br^- < I^-$; $Cl^- < S^{2-}$

- A small, highly charged cation concentrates its charge in a small volume, so it pulls hard on nearby electrons.
- In a large anion the outer electrons are far from its nucleus and well shielded, so they are easily pulled away. Extra negative charge adds to this.
- **Most covalent character**: small, highly charged cation + large, highly charged anion (e.g. AlI_3, $BeCl_2$, LiI). **Most ionic**: large, singly charged cation + small, singly charged anion (e.g. CsF, KF).

WORKED EXAMPLE: RANKING COVALENT CHARACTER

Put LiF, LiI and KI in order of increasing covalent character.
Li^+ is smaller than K^+, so it is more polarising. I^- is larger than F^-, so it is more polarisable.
KI (large cation) < LiF (small anion) < **LiI** (small cation **and** large anion) has the most covalent character.
(KI vs LiF can be argued either way; questions usually avoid that pair or give data.)

MARK-SCHEME LANGUAGE

- Covalent character in $AlCl_3$: "the aluminium ion is (very) **small** and **highly charged**", so it "**polarises / distorts** the chloride ion / anion".
- Why F^- isn't polarised: "the fluoride ion has a **single charge** and a **small** ionic radius", so "the electron cloud is difficult to distort". The reverse argument for large, highly charged anions is allowed.
- $BeCl_2$ vs $CaCl_2$: Be^{2+} is smaller / has a higher charge density, so it is more polarising (or: Be is more electronegative).

Often asked as: explain covalent character in $AlCl_3$ or $BeCl_2$; why the fluoride ion isn't easily polarised; MCQ on which cation is most polarising.

COMMON MISTAKE

- Using the **cation's** size to explain the **anion's** polarisability.
- Saying fluoride is "large with a small charge". Not using the word **distort**.

SPEC 3.10 Covalent bonding and the evidence for it

WHAT THIS SPEC POINT ASKS

Understand that a covalent bond is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them, based on evidence from (i) the physical properties of giant atomic structures and (ii) electron density maps of simple molecules.

3.10 What a covalent bond is

- A **covalent bond** is the **strong electrostatic attraction between the two nuclei and the shared pair of electrons between them**.
- Each atom usually contributes one electron to the pair. The shared pair sits between the nuclei and is attracted to both, which holds them together.
- A double bond is two shared pairs; a triple bond is three.

3.10 (i) Evidence from giant atomic (giant covalent) structures

- **Diamond**, **graphite**, **silicon** and **silicon dioxide** (SiO_2) are giant lattices of atoms joined by covalent bonds.
- They have **very high melting temperatures**, are **hard** (except graphite), **insoluble**, and (except graphite) **don't conduct**.
- To melt them you must break **many strong covalent bonds**, which needs a lot of energy. So covalent bonds must be **strong**. Their fixed shapes (e.g. tetrahedral around each atom in diamond and SiO_2) show the bonds are **directional**.

3.10 (ii) Evidence from electron density maps

- In an electron density map of H_2 or Cl_2, the contours **surround both nuclei** together, and the electron density stays **high in the region between the nuclei**.
- That build-up of electrons between the nuclei is the shared pair. Compare NaCl (3.1), where density falls to almost zero between the ions.
- In HCl the contours are pulled towards the Cl atom, showing a **polar** bond (3.14).

MARK-SCHEME LANGUAGE

- Definition: "(electrostatic) **attraction**" between "the shared pair of electrons" and "the **nuclei** (of the two atoms)". Both the word *attraction* and the nuclei are needed.
- Electron density map of a covalent molecule: contours drawn around **both** nuclei, enclosed by a shared contour; not two separate circles.
- Giant covalent: "giant lattice", "many strong covalent bonds", "a lot of energy to break them".

Often asked as: define a covalent bond (often poorly answered); sketch or interpret an electron density map; explain the high melting temperature of SiO_2 or carbon.

COMMON MISTAKE

- Defining a covalent bond as "a shared pair of electrons" with no attraction and no nuclei.
- Mentioning ions, metallic bonding or intermolecular forces for a giant covalent structure.
- Drawing a contour map that looks like two separate atoms.

BEYOND THE SPEC, BUT EXAMINED

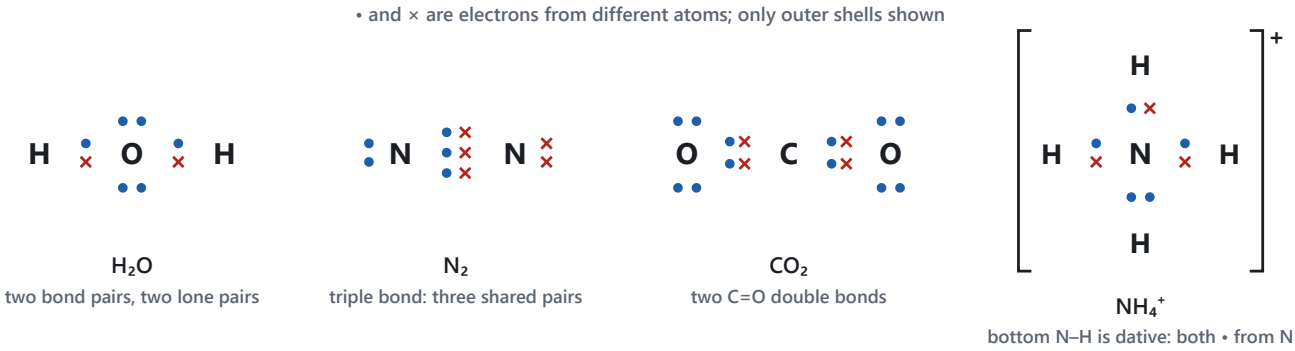
- **σ and π bonds** (spec 5.1): a σ bond is the head-on overlap of orbitals between the nuclei; a π bond is the sideways overlap of p orbitals, above and below the σ bond. A double bond is one σ + one π (WCH11 Jan 2022).
- SiO_2 is giant covalent: each Si bonded to 4 O atoms tetrahedrally, each O to 2 Si; no mobile electrons, so it doesn't conduct (WCH11 Oct 2021).

SPEC 3.11 Dot-and-cross diagrams for covalent substances, including dative bonds

WHAT THIS SPEC POINT ASKS

Draw dot-and-cross diagrams for covalent substances: (i) molecules with single, double and triple bonds; (ii) species with dative covalent (coordinate) bonds, including Al_2Cl_6 and the ammonium ion.

3.11 (i) Single, double and triple bonds



Draw every outer electron. Each shared pair has one • and one × (except a dative pair). Lone pairs belong to one atom.

Rules:

1. Use **dots** for one atom's electrons and **crosses** for the other's.
2. Put each **shared pair** between the two atoms (in the overlap if you draw circles). One pair = single bond, two = double, three = triple.
3. Show **all the non-bonding (lone) pairs**, so you can count 8 outer electrons round each atom (2 for H).
4. Then count: every atom's own electrons must add up to its group number.

Molecule	Bonds	Lone pairs
Cl_2	one single bond	3 on each Cl
H_2O	two O–H	2 on O
NH_3	three N–H	1 on N
CH_4	four C–H	none
O_2	**O=O** (two shared pairs)	2 on each O
CO_2	**two C=O**	2 on each O, none on C
C_2H_4	C=C and four C–H	none
N_2	**N≡N** (three shared pairs)	1 on each N

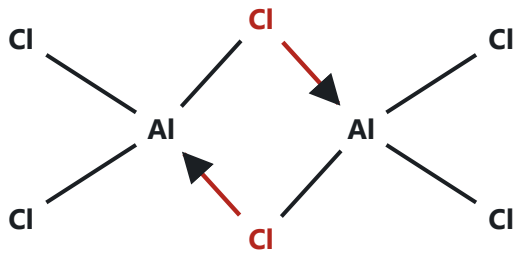
Exceptions to the octet rule

- **Electron-deficient** central atoms have fewer than 8: $BeCl_2$ (4 electrons round Be), BCl_3 and $AlCl_3$ (6 round B or Al).
- **Expanded octets**: elements in Period 3 and below can have more than 8, because they have d orbitals available: PCl_5 (10 round P), SF_6 (12 round S). Nitrogen can't do this, so NCl_5 doesn't exist.

3.11 (ii) Dative covalent (coordinate) bonds

- A **dative covalent bond** is a shared pair of electrons in which **both electrons come from the same atom**.
- You need a **donor** with a **lone pair** and an **acceptor** with an **empty orbital** (often an electron-deficient atom, or H^+).
- Show it with an **arrow from the donor atom to the acceptor atom**. In a dot-and-cross diagram, both electrons in that pair are the same symbol (both from the donor).
- Once formed, a dative bond is **identical** to any other covalent bond: all four N–H bonds in NH_4^+ are the same.

Species	Donor → acceptor
NH_4^+	lone pair on N of NH_3 → H^+
H_3O^+	lone pair on O of H_2O → H^+
Al_2Cl_6	lone pair on a Cl of one $AlCl_3$ → the Al of the other $AlCl_3$ (two dative bonds)
NH_3BF_3	lone pair on N → B (which had only 6 electrons)



arrows: dative bonds, a lone pair on Cl donated to Al

Al_2Cl_6: two $AlCl_3$ units joined by two bridging Cl atoms. Each Al ends up with 8 outer electrons. There is no Al–Al bond.

- $AlCl_3$ has only 6 electrons round Al. At lower temperatures in the vapour, two $AlCl_3$ molecules join: a lone pair on a Cl of each molecule is donated to the Al of the other, giving the **dimer Al_2Cl_6**.

MARK-SCHEME LANGUAGE

- Dot-and-cross: any combination of dots, crosses or circles; inner shells ignored. $BeCl_2$: "2 single covalent bonds" and "6 non-bonding electrons on each chlorine"; no lone pair on Be. An ionic diagram for a covalent molecule scores 0.
- Dative bond definition: "a (shared) pair of electrons" where "both electrons come from the same atom".
- Dative arrow goes **from the donor (Cl) to the acceptor (Al)**. In a dot-and-cross, the two dative electrons "must appear the same".
- NH_4^+: square brackets and a + charge, with one N–H shown as coordinate.
- Expanded octet (ClF_3): "10 electrons / 5 pairs in the outer shell of chlorine", "has expanded its octet".

Often asked as: dot-and-cross of N_2, Br_2, CCl_4, SCl_2, $BeCl_2$, NCl_3, $B(OH)_3$, ClF_3, $SbCl_5$, SF_4; Al_2Cl_6 with dative bonds; NH_4^+; define a dative bond; why N can't form NCl_5.

COMMON MISTAKE

- Lone pairs left off Cl or Br; an extra lone pair added to Be or B.
- A single bond in N_2, or drawing a halogen molecule when N_2 was asked.
- Dative arrows from Al to Cl; an Al–Al bond; extra Cl^- ions in Al_2Cl_6.
- Drawing NH_3 when NH_4^+ was asked, or leaving off the charge. Calling the dative pair a "lone pair" after it has bonded.
- Merging O and H into one symbol in $B(OH)_3$; diagrams too small to count.

BEYOND THE SPEC, BUT EXAMINED

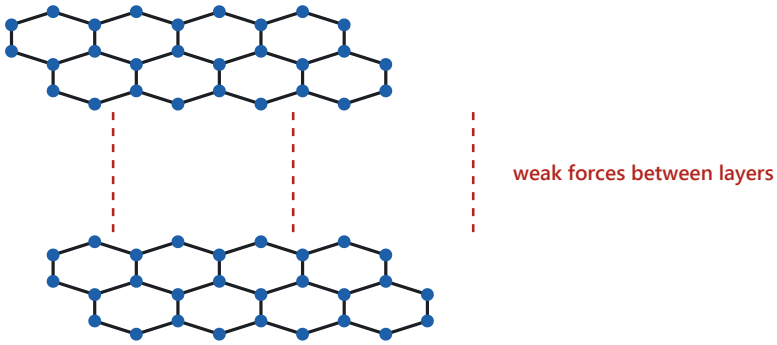
- Expanded octets in unfamiliar molecules: ClF_3, $SbCl_5$, SF_4, S_2F_{10} (WCH11 Oct 2019, Jan 2021, Jan 2025).
- Why nitrogen can't form NCl_5: it has **no 2d orbitals** (no available orbitals in the second shell), so it can't expand its octet (WCH11 Jan 2021).
- Negative ions: add the extra electron(s) to the central atom's count, e.g. GeH_3^- has a lone pair on Ge (WCH11 Oct 2022). Oxoanions such as SO_4^{2-} (WCH11 Oct 2025).

SPEC 3.12

Giant lattices of carbon: graphite, diamond and graphene

WHAT THIS SPEC POINT ASKS

Describe the giant structures of carbon (graphite, diamond and graphene) and discuss their uses.



each C forms 3 strong covalent bonds; 4th electron delocalised across the layer

Graphite: layers of hexagons. When drawing it, show at least two layers with at least three hexagons each, and label the forces between the layers.

	Diamond	Graphite	Graphene
Structure	giant covalent; each C bonded to **4** others, **tetrahedral** (109.5°)	giant covalent **layers** of hexagons; each C bonded to **3** others (120°)	a **single layer** of graphite, one atom thick
Delocalised electrons	**none**: all 4 outer electrons in bonds	**one per C**, free to move along the layers	one per C, across the sheet
Between layers	n/a	**weak London forces**; layers 0.335 nm apart	n/a
Electrical conductivity	**doesn't** conduct	**conducts** (along the layers)	conducts very well
Hardness	**very hard**	**soft**, slippery: layers slide	very strong for its mass; flexible
Density	3.5 g cm^{-3} (higher)	2.3 g cm^{-3} (lower: layers far apart)	very light; almost transparent
Uses	cutting tools, drill tips, jewellery	electrodes, lubricant, pencils	electronics, touchscreens, light strong composites (e.g. sports equipment), conductive inks, sensors

- **Both** diamond and graphite have **very high melting temperatures**: graphite's layers are held by weak forces, but to **melt** it you must break the **strong covalent bonds within the layers**.
- Each use should be linked to a property: graphite **electrodes** because it conducts, has a high melting temperature and is fairly unreactive; **lubricant** because the layers slide; diamond **cutting tools** because it is the hardest substance.

MARK-SCHEME LANGUAGE

- Diamond: "each carbon covalently bonded to **four** others"; "no delocalised electrons", so no conduction; hard. Graphite: "layers", "each carbon bonded to **three** others", "one delocalised electron per atom", so it conducts; "weak forces between layers", so they slide.
- Differences must be **properties** (conductivity, hardness, density). **Melting temperature is not accepted** as a difference between diamond and graphite.
- Graphene: "a single layer of graphite"; a use **with** the property that suits it (strong, light, conducts, transparent). "Lubricant" alone is not credited for graphene.
- Drawing graphite: "at least two layers of hexagons (at least 3 hexagons per layer)" and "labelled forces of attraction between the layers".

Often asked as: compare diamond and graphite (properties and structure); relate graphene to graphite and give a use; why graphite is used as an electrode; draw graphite layers; explain the high melting temperature of carbon.

COMMON MISTAKE

- Calling graphite or diamond "molecular", or saying diamond has "intermolecular forces".
- Mixing up the **weak forces between layers** with **weak bonds within** the layers.
- Writing about diamond, then graphite, in separate paragraphs in a "compare" question: compare in the same sentence.
- Confusing graphene with graphite.

BEYOND THE SPEC, BUT EXAMINED

- Showing the electrons in diamond as a dot-and-cross section: each C with four shared pairs (WCH11 Jun 2024).
- Graphite as the electrode in aluminium extraction at about 950 °C: giant covalent, high melting temperature, delocalised electrons carry the current (WCH11 Jun 2023).

SPEC 3.13 Electronegativity

WHAT THIS SPEC POINT ASKS

Understand what electronegativity means for atoms in a covalent bond.

- **Electronegativity** is the **ability of an atom to attract the bonding pair of electrons** in a covalent bond.
- It is measured on the **Pauling scale**. Fluorine is the most electronegative element (4.0). Noble gases have no value (they don't usually bond).

	Group 1	2	13	14	15	16	17
Period 1	H 2.1						
Period 2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Period 3	Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
Period 4	K 0.8	Ca 1.0					Br 2.8

3.13 Trends

- **Increases across a period**: more protons (bigger nuclear charge), similar shielding, smaller atom, so the bonding pair is attracted more strongly.
- **Decreases down a group**: more shells, more shielding and a larger distance from the nucleus to the bonding pair, so the attraction is weaker.

MARK-SCHEME LANGUAGE

- "The ability of an atom to attract the **bonding pair** of electrons (in a covalent bond)." "Attract electrons" alone is too vague.

Often asked as: define electronegativity; use values to decide which end of a bond is δ−.

COMMON MISTAKE

Defining electronegativity as "how easily an atom gains an electron" (that's electron affinity) or leaving out "in a covalent bond / bonding pair".

SPEC 3.14 The bonding continuum: from covalent to ionic

WHAT THIS SPEC POINT ASKS

Know that ionic and covalent bonding are the two extremes of a continuum, and explain this through electronegativity differences: polar bonds, polar molecules, and ionic bonding when the difference is large enough.

Electronegativity difference	Bond	Example
zero	**pure (non-polar) covalent**: pair shared equally	H–H, Cl–Cl
small	**polar covalent**: pair pulled towards the more electronegative atom, which gets **δ−**; the other gets **δ+**	$H^{\delta+}$–$Cl^{\delta-}$ (difference 0.9)
large	**ionic**: the electron is effectively transferred	NaCl (2.1), NaF (3.1)

- Write polar bonds as **$H^{\delta+}$–$Cl^{\delta-}$**. The δ− goes on the **more electronegative** atom.
- There is **no sharp cut-off** between the types. A polar covalent bond has some ionic character; an ionic bond with a polarised anion has some covalent character (3.8). Bonds sit somewhere on a **continuum**.
- The bigger the difference, the more **ionic character**: $BaCl_2$ (3.0 − 0.9 = 2.1) is more ionic than $BeCl_2$ (3.0 − 1.5 = 1.5).

MARK-SCHEME LANGUAGE

- "Ionic and covalent are extremes of a **continuum**"; the electronegativity difference decides where a bond lies. Both compounds have "some ionic and some covalent character"; one is "more ionic / less covalent".
- State which element is **more electronegative**, then put δ− on it.

Often asked as: use electronegativity values to compare two compounds; mark δ+ and δ− on a bond.

COMMON MISTAKE

- Treating a difference as a fixed cut-off ("1.7 or more is ionic"). Examiners mark that as a misconception.
- Putting δ+ and δ− the wrong way round; drawing a dipole arrow when δ+/δ− was asked.

SPEC 3.15

Polar bonds and polar molecules

WHAT THIS SPEC POINT ASKS

Distinguish between polar bonds and polar molecules, and predict whether a molecule is likely to be polar.

- A molecule can contain **polar bonds** and still be **non-polar overall**. It is a **polar molecule** only if the bond dipoles **don't cancel**: one end of the molecule is slightly positive and the other slightly negative (a permanent dipole).
- **Symmetrical** shapes with identical outer atoms and no lone pair on the central atom cancel out.

Molecule	Polar bonds?	Shape	Polar molecule?
CO_2	yes (C=O)	linear	**no**: dipoles cancel
CCl_4, CH_4	C–Cl yes; C–H nearly non-polar	tetrahedral	**no**
BF_3	yes	trigonal planar	**no**
SF_6	yes	octahedral	**no**
H_2O	yes	bent	**yes**
NH_3	yes	trigonal pyramidal	**yes**
HCl	yes	linear (two atoms)	**yes**
$CHCl_3$, CH_3Cl	yes	tetrahedral but not symmetrical	**yes**

3.15 Method

1. Work out the **shape** (3.18 and 3.19).
2. Find the **polar bonds** (electronegativity difference).
3. Ask whether the bond dipoles **cancel** because of symmetry.

3.15 The charged rod test

Run a thin jet of liquid from a burette past a **charged rod** (rubbed polythene or acetate). Jets of **water** and **ethanol** bend towards the rod: their molecules are polar and turn so their oppositely charged end faces the rod. A jet of **cyclohexane** isn't deflected: its molecules are non-polar. The polar jet bends towards the rod whether the rod is positive or negative.

MARK-SCHEME LANGUAGE

- CCl_4: "chlorine is **more electronegative** than carbon (so the bonds are polar)" **but** "the molecule is not polar because the **dipoles cancel** / the molecule is **symmetrical**". IGNORE "the polar bonds cancel".
- Polar molecule: "one end slightly positive and the other slightly negative".
- CO_2: oxygen is more electronegative, so δ+ on C and δ− on each O; but linear, so non-polar.

Often asked as: is CCl_4 / CO_2 polar, and why; define a polar molecule; MCQ from jet deflection.

COMMON MISTAKE

- Arguing that a difference is "not big enough" to be polar.
- Getting the shape wrong in one part and carrying it into the polarity answer.
- Calling C–H bonds strongly polar.

BEYOND THE SPEC, BUT EXAMINED

- Predicting polarity of an unfamiliar molecule from its shape: SF_4 (see-saw) is polar, SF_6 (octahedral) is not (WCH11 Jan 2025).
- **Induced dipoles**: the π electrons of an alkene repel the electrons in a Cl_2 or Br_2 molecule, distorting its electron cloud and inducing a dipole (WCH11 Jan 2025; see Topic 5).

SPEC 3.16

Electron-pair repulsion theory

WHAT THIS SPEC POINT ASKS

Understand the principles of electron-pair repulsion theory, used to explain and predict the shapes of simple molecules and ions.

3.16 The principles

1. The **electron pairs** in the outer shell of the central atom (both **bonding pairs** and **lone pairs**) **repel** each other.
2. They arrange themselves **as far apart as possible**, to **minimise repulsion**. That arrangement fixes the shape.
3. **Lone pairs repel more** than bonding pairs, because a lone pair is held closer to the central nucleus and spreads out more. The order is: **lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair**.
4. Each lone pair squeezes the bonds closer together, cutting the bond angle by **roughly 2.5°** (109.5° → 107° → 104.5°).
5. A **double (or triple) bond** counts as **one** bonding region for working out the shape.
6. The **shape is named from the atoms** only, but the lone pairs still affect it.

3.16 Counting the pairs

For a central atom bonded only by single bonds:

- electrons = central atom's outer electrons + **1 for each bonded atom** + **1 for each negative charge** − **1 for each positive charge**
- pairs = electrons ÷ 2; **lone pairs** = pairs − bonding pairs.
- If double bonds are present, draw the dot-and-cross diagram first and count each double bond as one region.

Total pairs (regions)	Arrangement	Angle with no lone pairs
2	linear	180°
3	trigonal planar	120°
4	tetrahedral	109.5°
5	trigonal bipyramidal	120° and 90°
6	octahedral	90°

MARK-SCHEME LANGUAGE

- "Electron pairs repel (each other)" and "get as far apart as possible / to minimise repulsion". ALLOW "maximum separation".
- "Lone pairs repel more than bonding pairs."
- Teach and use **electron pairs**. One recent mark scheme ignored "bonds" or "electron regions"; another accepted regions of electron density for a double bond. So say "pairs of electrons", and say a double bond "counts as one region".

Often asked as: inside every shapes question (3.18, 3.19).

COMMON MISTAKE

- "The **atoms** (or bonds) repel", or "the molecule moves to minimise repulsion". It is the **electron pairs** that repel.
- Saying lone pairs repel "the atoms".
- Quoting "2.5° per lone pair" without the reason (lone pairs repel more).

SPEC 3.17

Bond length and bond angle

WHAT THIS SPEC POINT ASKS

Understand the terms bond length and bond angle.

- **Bond length**: the distance between the **nuclei** of two atoms joined by a covalent bond. It is where attraction (nuclei to the shared pair) and repulsion (nucleus to nucleus) balance.
- **Bond angle**: the angle between **two bonds** that meet at the **same atom** (between the lines joining the nuclei).
- More shared pairs pull the nuclei closer: multiple bonds are **shorter and stronger**.

Bond	Length / nm	Mean bond enthalpy / kJ mol^{-1}
C–C	0.154	347
C=C	0.134	612
C≡C	0.120	838

MARK-SCHEME LANGUAGE

- Bond length: "distance between the **nuclei** (of two bonded atoms)" (not "between the atoms").
- Bond angle: label it at the **central** atom, between two bonds.

Often asked as: definitions; MCQ comparing bond lengths; label an angle on a diagram.

COMMON MISTAKE

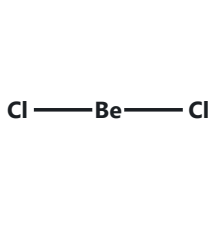
Labelling a bond angle in the wrong place, or giving two different angles for one label (the second one cancels the mark).

SPEC 3.18

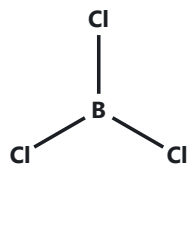
Shapes of named molecules and ions

WHAT THIS SPEC POINT ASKS

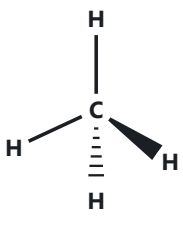
Know and explain the shapes and bond angles of $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4.



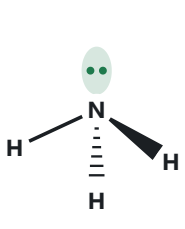
$BeCl_2$
linear
180°



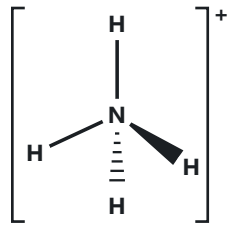
BCl_3
trigonal planar
120°



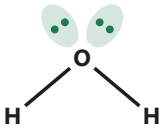
CH_4
tetrahedral
109.5°



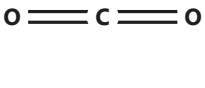
NH_3
trigonal pyramidal
107°



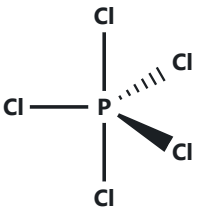
NH_4^+
tetrahedral
109.5°



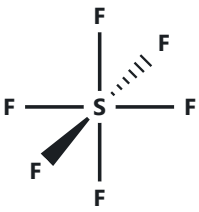
H_2O
bent (V-shaped)
104.5°



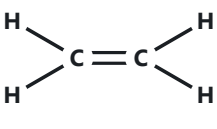
CO_2
linear
180°



PCl_5
trigonal bipyramidal
120° and 90°



SF_6
octahedral
90°



C_2H_4
planar; trigonal planar at each C
120°

A solid wedge comes towards you; a hashed wedge goes away; plain lines are in the plane of the page. Green lobes are lone pairs.

Species	Bonding pairs	Lone pairs	Shape	Bond angle
$BeCl_2$	2	0	**linear**	180°
BCl_3	3	0	**trigonal planar**	120°
CH_4	4	0	**tetrahedral**	109.5°
NH_4^+	4 (one dative)	0	**tetrahedral**	109.5°
NH_3	3	1	**trigonal pyramidal**	107°
H_2O	2	2	**bent / V-shaped**	104.5°
CO_2	2 double bonds (2 regions)	0	**linear**	180°
PCl_5 (gas)	5	0	**trigonal bipyramidal**	120° (equatorial) and 90° (axial to equatorial)
SF_6	6	0	**octahedral**	90°
C_2H_4	3 regions round each C (2 C–H, 1 C=C)	0	**trigonal planar** at each C; the whole molecule is flat	120°

WORKED EXAMPLE: THE FULL EXPLANATION FOR NH_3

1. N has 5 outer electrons; 3 H atoms add 3 → 8 electrons = **4 pairs**: **3 bonding pairs and 1 lone pair**.
2. The electron pairs **repel** and get **as far apart as possible**: a tetrahedral arrangement of pairs.
3. The **lone pair repels more** than the bonding pairs, so it squeezes the N–H bonds together.
4. Shape: **trigonal pyramidal**; bond angle **107°** (less than 109.5°).
   When NH_3 gains H^+ to become NH_4^+, the lone pair becomes a bonding pair: **4 bonding pairs**, all repel equally, **tetrahedral, 109.5°**.

WORKED EXAMPLE: WHY CH_4, NH_3 AND H_2O DIFFER

All three have **4 electron pairs** round the central atom.
CH_4: 4 bonding pairs, 0 lone pairs → 109.5°.
NH_3: 3 bonding pairs, **1** lone pair → 107°.
H_2O: 2 bonding pairs, **2** lone pairs → 104.5°.
Lone pairs repel more than bonding pairs, so **each extra lone pair reduces the angle**.

EXAM TIP

Sketch 3D shapes with wedges and dashes, label the angle **between two bonds at the central atom**, and give **one** value per angle. Write "trigonal **bi**pyramidal" in full: "trigonal pyramidal" is a different shape.

MARK-SCHEME LANGUAGE

- Standard 2-mark answer: the angle (e.g. 109.5°) **and** "(four) electron pairs / bonding pairs repel to get maximum separation / minimum repulsion".
- Always state the numbers of **bonding pairs and lone pairs**.
- Water-like angles: 103° to 104.5° accepted in recent mark schemes. Learn **104.5°**.
- CO_2: two C=O regions, no lone pairs on C → linear, 180°.
- PCl_5: "trigonal bipyramidal", "**90°** and **120°**". SF_6: "octahedral", "90°".

Often asked as: a table of bonding pairs, lone pairs, shape and angle (CH_4, NH_3, H_2O); explain why the angles differ; explain the shape of a named molecule; NH_3 vs NH_4^+.

COMMON MISTAKE

- Forgetting a lone pair, which turns NH_3 or NCl_3 into "trigonal planar".
- Calling the dative pair in NH_4^+ a lone pair.
- Mentioning lone pair–lone pair repulsion in a molecule with only **one** lone pair (NH_3, PCl_3).
- "Trigonal pyramidal" for BCl_3 or $AlCl_3$ (no lone pair: trigonal planar).
- Not knowing both angles in PCl_5.

SPEC 3.19

Predicting the shapes of other molecules and ions

WHAT THIS SPEC POINT ASKS

Apply electron-pair repulsion theory to predict the shapes and bond angles of molecules and ions similar to those in 3.18.

3.19 Method (show every step)

1. Find the central atom's outer electrons (its group number).
2. Add one for each atom bonded by a single bond; add the negative charge, or subtract the positive charge.
3. Halve it to get the **number of pairs**; split them into **bonding pairs** and **lone pairs**.
4. Pick the arrangement from the number of pairs (3.16), then name the shape from the atoms.
5. Adjust the angle for lone pairs (about 2.5° less per lone pair, from 109.5°).

WORKED EXAMPLE: FOUR ANALOGUES

PCl_4^+: 5 + 4 − 1 = 8 → 4 pairs, 4 bonding, 0 lone → **tetrahedral, 109.5°** (like NH_4^+).
PCl_6^-: 5 + 6 + 1 = 12 → 6 pairs, 6 bonding → **octahedral, 90°** (like SF_6).
NCl_3: 5 + 3 = 8 → 4 pairs, 3 bonding, 1 lone → **trigonal pyramidal, 107°** (like NH_3).
ClF_3: 7 + 3 = 10 → 5 pairs, 3 bonding, 2 lone → **T-shaped**, about 87.5° (just under 90°). The two lone pairs sit in the equatorial positions, where they have most room.

3.19 Analogues to recognise

Species	Like	Bonding pairs	Lone pairs	Shape	Angle
H_3O^+, NCl_3, PCl_3, GeH_3^-	NH_3	3	1	trigonal pyramidal	107°
NH_2^-, SCl_2, HOCl (at O)	H_2O	2	2	bent	104.5°
BF_4^-, $SiCl_4$, CCl_4, PCl_4^+	CH_4	4	0	tetrahedral	109.5°
BF_3, $AlCl_3$	BCl_3	3	0	trigonal planar	120°
CS_2	CO_2	2 regions	0	linear	180°
CO_3^{2-}, NO_3^-	BCl_3	3 regions	0	trigonal planar	120°
SO_4^{2-}	CH_4	4 regions	0	tetrahedral	109.5°
PF_5, $SbCl_5$	PCl_5	5	0	trigonal bipyramidal	120° and 90°
PCl_6^-, S_2F_{10} (at each S)	SF_6	6	0	octahedral	90°
SF_4, $TlCl_4^{3-}$	(new)	4	1	see-saw	about 102° and 173° (below 120° and 180°)
ClF_3	(new)	3	2	T-shaped	about 87.5°
XeF_4	(new)	4	2	square planar	90°

MARK-SCHEME LANGUAGE

- Quote "bond pairs **and** lone pairs" for each centre: e.g. in a ring with an O atom, "four bond pairs around C (109.5°); two bond pairs and two lone pairs around O (104.5°)".
- Shape "and justification": "electron pairs arranged to minimise repulsion", with the count of pairs.
- SF_4: see-saw, angles about 90° and 180° (about 173° accepted). $TlCl_4^{3-}$: has a **lone pair** on Tl, so see-saw.
- Deduce the **charge** of an ion such as PCl_4^+ or PCl_6^- from its shape.

Often asked as: shapes of NCl_3, PCl_3, SCl_2, GeH_3^-, HOCl, $AlCl_3$, $TlCl_3$, $TlCl_4^{3-}$, SF_4, S_2F_{10}, PCl_4^+/PCl_6^-; angles at C and O in an organic ring.

COMMON MISTAKE

- Not counting the lone pairs from the dot-and-cross diagram you've just drawn.
- Forgetting to add or subtract electrons for the ion's charge.
- Giving 180° for an O–B–O angle in $B(OH)_3$ (it's trigonal planar at B, 120°).

BEYOND THE SPEC, BUT EXAMINED

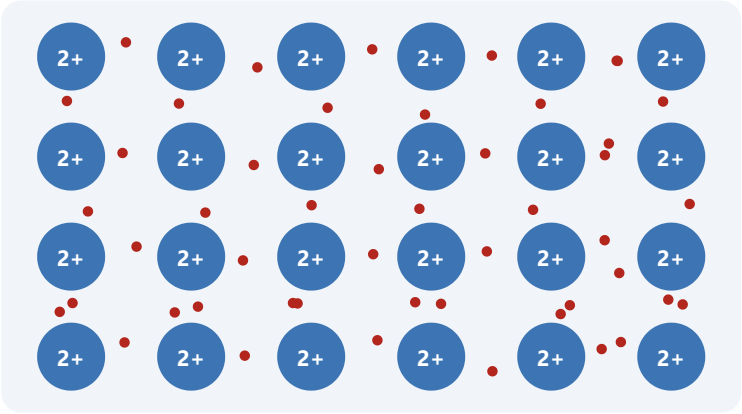
- The beryllium chloride **polymer** in the solid: each Be has four bonding pairs (two of them dative), so it is tetrahedral round Be (WCH11 Oct 2021).
- Shapes with 5 or 6 pairs including lone pairs: ClF_3, SF_4, $TlCl_4^{3-}$ (WCH11 Oct 2019, Jan 2025, Oct 2024).
- Angles in organic molecules: 109.5° at a saturated C, 104.5° at an O with two lone pairs (WCH11 Jan 2020).

SPEC 3.20

The structure of metals

WHAT THIS SPEC POINT ASKS

Understand that metals are giant lattices of metal ions in a sea of delocalised electrons.



Magnesium: a regular lattice of Mg^{2+} ions, with two delocalised electrons per ion moving freely between them.

- In a metal, each atom releases its **outer electrons**, which become **delocalised**: free to move through the whole structure, not tied to any one atom.
- What's left is a **giant lattice of positive metal ions** (cations), arranged in regular layers, held in a "**sea**" of delocalised electrons.

- The number of delocalised electrons per ion equals the ion's charge: Na gives 1 (Na^+), Mg gives 2 (Mg^{2+}), Al gives 3 (Al^{3+}).

MARK-SCHEME LANGUAGE

- Diagram: a **regular** array of cations **with charges**, plus electrons in an **irregular** arrangement between them, about **twice as many** electrons as ions for a 2+ ion. Label the electrons or show them as e^- or dots.

Often asked as: draw a labelled diagram of calcium, silver or magnesium.

COMMON MISTAKE

Drawing too many or too few electrons, or arranging them in a regular grid so the diagram looks ionic. Calling the particles "atoms" instead of **ions**.

SPEC 3.21

Metallic bonding

WHAT THIS SPEC POINT ASKS

Know that metallic bonding is the strong electrostatic attraction between metal ions and delocalised electrons.

- **Metallic bonding** is the **strong electrostatic attraction between the positive metal ions and the delocalised electrons**.
- It acts throughout the lattice and isn't directional.
- It gets **stronger** when the ions have a **higher charge** and there are **more delocalised electrons** per ion (and when the ions are smaller). So: Na < Mg < Al.

WORKED EXAMPLE: MAGNESIUM VS SODIUM

Explain why magnesium has a higher melting temperature than sodium.
1. Magnesium forms **Mg^{2+}** ions; sodium forms **Na^+** ions (higher charge).
2. Magnesium has **two delocalised electrons per atom**; sodium has one.
3. So the **attraction between the ions and the delocalised electrons** is **stronger** in magnesium, and more energy is needed to overcome it.

MARK-SCHEME LANGUAGE

- "(**Strong**) electrostatic attraction between (metal) **cations / positive ions** and **delocalised electrons**". The word **strong** was required (WCH11 Jan 2023). IGNORE protons or nuclei. Any mention of another bond type scores 0.
- Mg vs Na (2 marks): **greater charge** on the ion **and more delocalised electrons**. In this mark scheme ionic size was ignored and "just a higher charge" wasn't enough.

Often asked as: describe metallic bonding in sodium or aluminium; compare the melting temperatures of two metals.

COMMON MISTAKE

Saying "free electrons" or "charge carriers" instead of **delocalised electrons**. Describing attraction between positive and **negative ions** in a metal. Leaving out "strong".

SPEC 3.22 Explaining the properties of metals

WHAT THIS SPEC POINT ASKS

Use the metallic bonding model to explain simple properties of metals, including electrical conductivity and high melting temperature.

Property	Explanation
Conduct electricity (solid **and** liquid)	the **delocalised electrons can move** through the structure; when a potential difference is applied they flow and carry the charge
High melting and boiling temperatures	the **strong attraction** between the ions and the delocalised electrons needs a lot of energy to overcome
Good **thermal** conductors	delocalised electrons carry kinetic energy quickly through the lattice
Malleable and ductile	layers of ions can **slide** over each other; the delocalised electrons move with them, so the **bonding isn't broken**

3.22 Alloys

An **alloy** is a mixture of a metal with other elements. The **ions of different sizes** disrupt the regular layers, so the layers can't slide over each other as easily: alloys are **harder and stronger** than the pure metal.

MARK-SCHEME LANGUAGE

- Conductivity: "**delocalised electrons** that **move / flow**" in both the solid and the liquid. Any reference to **ions** moving in a metal is not awarded.
- Malleability: "layers of cations slide over each other" and "the bonding is not broken".
- Alloys: "ions of **different size**" so "layers don't slide as easily". The word **ion** is needed.
- A stated use needs a matching property (e.g. aluminium cables: good conductor, low density, ductile).

Often asked as: why a metal conducts when solid and when molten; why it is malleable; why an alloy is harder; why aluminium is used for a given job.

COMMON MISTAKE

- "Electrons are delocalised" without saying they **move**.
- Explaining a metal's conductivity by the charge on its ions alone.
- Mixing models: "delocalised electrons in the molten ionic compound" or "mobile ions in the metal".
- A general list of metal properties when the question is about one metal and one use.

BEYOND THE SPEC, BUT EXAMINED

- Alloys: silver alloyed with copper is harder because the copper ions are a different size and stop the layers sliding (WCH11 Jun 2024).
- Choosing aluminium for a use from its conductivity, ductility and low density (WCH11 Oct 2021).

Topic 4 – Introductory Organic Chemistry and Alkanes

SPEC 4.1 Hazard and risk

WHAT THIS SPEC POINT ASKS

Know the difference between a hazard and a risk.

4.1 Two different ideas

- A **hazard** is something that **could** cause harm. It is a property of the substance or the equipment itself: bromine is toxic, hexane is flammable, concentrated acid is corrosive. You can't change a hazard.
- A **risk** is the **chance** (likelihood) that the hazard actually causes harm, and how bad that harm would be, **in the way you are using it**. It depends on the amount, the concentration, the conditions (heating, open bench or fume cupboard) and who is doing the work.
- Safety precautions **reduce the risk**. They never remove the hazard. Goggles don't make an acid less corrosive; they make it less likely the acid reaches your eyes.

	Hazard	Risk
What it is	the potential to cause harm	the chance of harm happening in a given use
Can you change it?	no, it belongs to the substance	yes, by changing how you work
Example	ethanol is flammable	low with a few cm^3 in a water bath; high if heated over a Bunsen flame

4.1 Hazard symbols

Bottles carry the internationally agreed hazard pictograms (red-bordered diamonds). The ones you meet most in organic chemistry:

Pictogram	Meaning	Organic examples
Flame	flammable	alkanes, alkenes, alcohols, cyclohexane
Flame over a circle	oxidising (makes other things burn more fiercely)	potassium manganate(VII), potassium dichromate(VI)
Skull and crossbones	acutely toxic	bromine, carbon monoxide, methanol
Corrosive (liquid eating a hand and a surface)	corrosive	bromine, concentrated acids
Exclamation mark	harmful / irritant	dilute bromine water, many halogenoalkanes
Health hazard (person with a star on the chest)	carcinogenic, mutagenic, long-term harm	benzene, some halogenoalkanes, dichromates
Dead tree and fish	harmful to the environment	halogenated solvents
Gas cylinder, exploding bomb	gas under pressure; explosive	gas cylinders; organic peroxides

MARK-SCHEME LANGUAGE

- "Flammable / inflammable" for a fuel or solvent; "toxic / corrosive / poisonous / irritant" for HCl gas. The word must match the hazard: flammable for HCl **negated** the mark.
- Pick only the symbols that apply to the substance named. Ticking every symbol scores nothing.

Often asked as: MCQ on which pictogram fits a substance; "identify a hazard of X and a precaution".

COMMON MISTAKE

Confusing the **flammable** and **oxidising** symbols. Flammable = a flame on its own (the substance burns). Oxidising = a flame on top of a circle (the substance makes something else burn).
Saying a precaution "removes the hazard". It lowers the risk only.

SPEC 4.2

Hazards of organic compounds and risk assessments

WHAT THIS SPEC POINT ASKS

Know the typical hazards of organic compounds, and why a risk assessment is needed before you work with hazardous materials.

4.2 Why organic compounds are often hazardous

- **Flammable.** Most organic compounds burn. Small molecules are **volatile**: they evaporate easily at room temperature, so their vapour can reach a flame some distance away and ignite.
- **Toxic or harmful.** Many vapours are harmful to breathe (halogenoalkanes, bromine, benzene). Some are **carcinogenic** (benzene) or cause long-term damage.
- **Irritant or corrosive.** Bromine, acids used as catalysts (phosphoric(V), sulfuric).
- **Products can be hazardous too.** Incomplete combustion makes **carbon monoxide**; burning PVC releases **hydrogen chloride** gas.

4.2 What a risk assessment does

A risk assessment is done **before** the practical. You:

1. List every substance used and made, and look up its **hazards** (for example on a safety data sheet).
2. Decide how big the **risk** is in *this* procedure (amounts, heating, gases given off, who is doing it).
3. Choose **control measures** that lower each risk (see 4.3).
4. Plan what to do if something goes wrong (spills, skin contact, fire).

Why it is needed: so you know the hazards in advance, can reduce the risk to an acceptable level, and are ready for an accident. In a school or industry, it is also a legal requirement.

MARK-SCHEME LANGUAGE

- A two-mark "hazard and precaution" question: M1 names the hazard; M2 gives a precaution **linked to that hazard**. Jan 2020: "flammable" + "avoid naked flames / use a heating mantle / electric water bath / small amounts".
- IGNORE generic answers: fume cupboard (for a flammable liquid), gloves, tie hair back, fire extinguisher, "do not heat".
- Oct 2020: an oxidising agent must not be mixed with alkanes because it **promotes burning** and alkanes are **flammable**.

Often asked as: name a hazard and a precaution; explain why two chemicals must be stored apart; everyday safety in an unfamiliar context.

COMMON MISTAKE

"Wear goggles and a lab coat" scores nothing on its own: these are worn in every practical. Give a precaution aimed at **this** hazard (WCH11 Jan 2020 ER).
Writing that mixing an oxidiser and a fuel "starts a fire". It **makes a fire burn more fiercely** (WCH11 Oct 2020 ER).

SPEC 4.3

Reducing risk

WHAT THIS SPEC POINT ASKS

Suggest ways to reduce risk and carry out reactions safely: (i) work on a smaller scale, (ii) take precautions that suit the particular hazard, (iii) use a different method with less hazardous substances.

4.3 Three ways to cut the risk

Method	How it helps	Example
(i) **Smaller scale**	less substance → less vapour, smaller fire, smaller spill	microscale tests in a spotting tile; a few drops of bromine water
(ii) **Precaution matched to the hazard**	targets the way harm would happen	flammable → no naked flames, heat with a **water bath** or **electric heating mantle**; toxic gas or vapour → **fume cupboard**; corrosive → gloves and goggles
(iii) **Less hazardous alternative**	swaps a dangerous chemical for a safer one	**bromine water** instead of liquid bromine; cyclohexane instead of benzene; a plant-based fuel instead of a toxic solvent

4.3 Hazards in the Unit 1 organic practicals

Practical	Hazard	Precaution
Testing with bromine	liquid bromine: very toxic, corrosive vapour	use dilute **bromine water**, small volumes, fume cupboard, gloves
Burning or heating alkanes, alkenes, cyclohexane	flammable, volatile	no naked flames near the bottle; small amounts; heat in a water bath
Halogenation in UV light	chlorine is toxic; UV damages eyes	fume cupboard; UV-protective screen or glasses
Cracking liquid paraffin	hot glass; **suck-back** of cold water; flammable gases	remove the delivery tube from the water **before** you stop heating; no flames near collected gas
Oxidation with acidified $KMnO_4$	oxidising; acid is an irritant	gloves and goggles; keep away from flammables
Incomplete combustion	carbon monoxide is toxic	good ventilation; fume cupboard

MARK-SCHEME LANGUAGE

Oct 2025 (carbon monoxide exposure), two of: "use a fume cupboard"; "open windows / doors (ventilation)"; "wear a respirator / oxygen mask"; "take breaks, to avoid continuous exposure".

Often asked as: suggest a safer method; suggest how to reduce the risk in a given preparation.

EXAM TIP

For "suggest" questions, think: **what is the hazard → how would it hurt someone → what stops that route.** Vapour that ignites → keep flames away. Gas that poisons → fume cupboard. Liquid that burns skin → gloves.

SPEC 4.4 Homologous series and functional groups

WHAT THIS SPEC POINT ASKS

Understand what a homologous series is and what a functional group is.

4.4 Definitions

- A **functional group** is the atom or group of atoms that gives a compound its characteristic chemical reactions. Examples: C=C in alkenes, –OH in alcohols, –Br in bromoalkanes.
- A **homologous series** is a family of organic compounds that:
  - have the **same functional group**;
  - have the **same general formula**;
  - differ from the next member by **CH_2**;
  - have **similar chemical properties** (they react in the same way);
  - show a **gradual trend in physical properties** (for example, boiling temperature rises as the chain gets longer).

4.4 Series you meet in Unit 1

Series	Functional group	General formula	Example	Name ending
Alkanes	none (C–C and C–H only)	C_nH_{2n+2}	$CH_3CH_2CH_3$ propane	-ane
Cycloalkanes	ring of C–C bonds	C_nH_{2n}	cyclohexane	cyclo…ane
Alkenes	C=C	C_nH_{2n}	$CH_2{=}CHCH_3$ propene	-ene
Halogenoalkanes	C–X (X = F, Cl, Br, I)	$C_nH_{2n+1}X$	CH_3CH_2Br bromoethane	prefix chloro-, bromo- …
Alcohols	–OH	$C_nH_{2n+1}OH$	CH_3CH_2OH ethanol	-ol

4.4 Trend in physical properties

Alkane	Formula	Boiling temperature / °C	State at 20 °C
methane	CH_4	−162	gas
ethane	C_2H_6	−89	gas
propane	C_3H_8	−42	gas
butane	C_4H_{10}	−1	gas
pentane	C_5H_{12}	36	liquid
hexane	C_6H_{14}	69	liquid

Each extra CH_2 adds more electrons, so the London forces between molecules get stronger and more energy is needed to separate them. (Intermolecular forces are covered fully in Unit 2.)

MARK-SCHEME LANGUAGE

Jan 2025 (same homologous series), two from: "same functional group"; "similar chemical properties"; "same general formula"; "differ by CH_2".

Often asked as: state two features of a homologous series; MCQ on which pair is in the same series; deduce a general formula for an unfamiliar series.

COMMON MISTAKE

Saying members of a series have the "same physical properties". Their physical properties **change gradually**; it is the **chemical** properties that are similar.

SPEC 4.5 IUPAC naming and drawing organic compounds

WHAT THIS SPEC POINT ASKS

(i) Name the compounds in this specification using IUPAC rules. (ii) Draw them as structural, displayed and skeletal formulae. Know the stems for C1 to C10.

4.5 Stems

Carbons	1	2	3	4	5	6	7	8	9	10
Stem	meth	eth	prop	but	pent	hex	hept	oct	non	dec

The same stems give the side-chain (alkyl) names: methyl $–CH_3$, ethyl $–C_2H_5$, propyl $–C_3H_7$.

4.5 Naming a branched alkane: the routine

1. Find the **longest continuous carbon chain**. It may bend round corners on the page. This gives the stem.
2. Number the chain from the end that gives the side chains the **lowest numbers** (lowest locants).
3. Name each side chain and give its number. Two or more of the same group: **di-, tri-, tetra-**, and **every** group needs its own number (2,2-dimethyl, not 2-dimethyl).
4. Different groups go in **alphabetical order** (ethyl before methyl; bromo before chloro before methyl). Ignore di- and tri- when you alphabetise.
5. Punctuation: **commas between numbers**, **hyphens between numbers and letters**, and the last prefix runs straight into the stem (2-methylpentane).

WORKED EXAMPLE: 2,2,4-TRIMETHYLPENTANE

Structure: $(CH_3)_3CCH_2CH(CH_3)CH_3$

1. Longest chain: 5 carbons → **pentane**.
2. Three methyl groups. Numbered from the left they sit on C2, C2 and C4. From the right they sit on C2, C4, C4.
3. Compare the first point of difference: 2,**2**,4 beats 2,**4**,4. Number from the left.
4. Three methyls → **tri**methyl, and each one gets a number.

Name: **2,2,4-trimethylpentane** (this is "iso-octane", the petrol standard).

WORKED EXAMPLE: ALPHABETICAL ORDER

$CH_3CH_2CH(C_2H_5)CH_2CH(CH_3)CH_3$

Longest chain: 6 carbons (hexane). Numbering from the right gives methyl at 2 and ethyl at 4 (from the left it would be 3 and 5).

Alphabetical: **e**thyl before **m**ethyl → **4-ethyl-2-methylhexane**.

4.5 Naming other families

Family	Rule	Examples
Cycloalkanes	add **cyclo-** to the stem; number round the ring so substituents get the lowest numbers	cyclohexane; methylcyclopentane; 1,2-dimethylcyclobutane
Alkenes	stem + **-ene**; number the C=C from the end that gives it the **lowest number**, then place side chains	but-1-ene $CH_2{=}CHCH_2CH_3$; but-2-ene $CH_3CH{=}CHCH_3$; 2-methylbut-2-ene $(CH_3)_2C{=}CHCH_3$; 3-methylbut-1-ene $CH_2{=}CHCH(CH_3)_2$
Halogenoalkanes	prefix fluoro-, chloro-, bromo-, iodo-, alphabetical with alkyl groups	1,2-dibromopropane $CH_2BrCHBrCH_3$; 2-bromo-2-methylpropane $(CH_3)_3CBr$; 2-chloro-3-methylbutane
Alcohols	stem + **-ol** with the position of the –OH	propan-1-ol $CH_3CH_2CH_2OH$; propan-2-ol $CH_3CH(OH)CH_3$
Diols	keep the "e" of the stem, add **-diol**	ethane-1,2-diol $HOCH_2CH_2OH$; propane-1,2-diol $CH_3CH(OH)CH_2OH$
Polymers	poly(monomer name)	poly(ethene); poly(propene); poly(chloroethene)

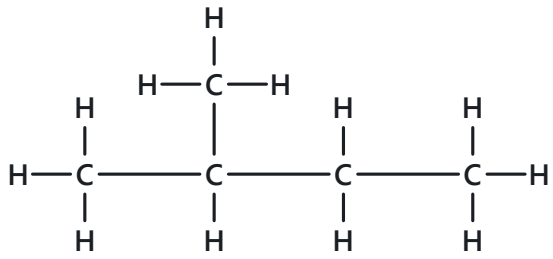
When two numbering directions tie, the group that comes first in the alphabet gets the lower number: $CH_3CHClCH(CH_3)CH_3$ is 2-chloro-3-methylbutane.

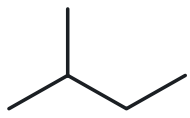
4.5 The formula types

Type	What it shows	2-methylbutane
Molecular	the actual number of each atom	C_5H_{12}
Empirical	simplest whole-number ratio	C_5H_{12} (no simpler ratio here; for ethane, C_2H_6, it is CH_3)
General	formula for a whole series	C_nH_{2n+2}
Structural	enough to identify the structure without ambiguity, group by group	$CH_3CH(CH_3)CH_2CH_3$
Displayed	**every** atom and **every** bond	see figure
Skeletal	carbon chain as zig-zag lines; each end and each corner is a carbon; H on carbon not shown; functional groups shown in full	see figure

Molecular **Structural** **Displayed** **Skeletal**

C_5H_{12} $CH_3CH(CH_3)CH_2CH_3$





One molecule, four formulae. In the skeletal formula, each line end and each corner is a carbon with enough hydrogens to make four bonds.

4.5 Drawing tips

- **Displayed**: count four bonds on every carbon. Draw the O–H bond in an alcohol (C–O–H), not just "OH".
- **Skeletal**: never write C or CH_3 at the line ends. Do write functional groups: –OH, –Br, and the C=C as a double line. Count the corners and ends afterwards to check the carbon total.
- **Structural**: brackets show a side chain on the carbon just before it: $CH_3CH(CH_3)CH_3$.

MARK-SCHEME LANGUAGE

- Skeletal formula: "IGNORE bond angles and lengths"; a skeletal formula with **CH_3 groups written in** is not credited (Jan 2019). The O–H in a skeletal alcohol must be bonded through the O: "do not allow O–H–C".
- If you give more than one type of formula for the same molecule, **all must be correct**.
- Names need every part: E/Z where relevant, the position of the double bond, the stem and the ending.

Often asked as: name a branched alkane (2,2,4-trimethylpentane in Jan 2023 and Oct 2024); draw a skeletal formula from a name; name a product such as 1,2-dibromopropane.

COMMON MISTAKE

"2,2-dimethyl-4-methylpentane": combine identical groups and number them all → 2,2,4-trimethylpentane (WCH11 Jan 2023 ER).
Counting the longest chain wrongly when it bends. Trace every possible chain.
Numbering from the wrong end (2,4,4- instead of 2,2,4-).
Missing "cyclo" or "di"; using "oct" for the wrong number of carbons (WCH11 Jun 2024 ER).
Giving a molecular formula when the skeletal formula is asked for, or a structural formula when displayed is asked for (WCH11 Oct 2020 ER).

BEYOND THE SPEC, BUT EXAMINED

Names of unfamiliar molecules built from the same rules, e.g. 3-ethyl-2,3-dimethylpentane in an MCQ (WCH11 Jan 2026).
Simple inorganic names using the same prefixes: S_2F_{10} is disulfur decafluoride (WCH11 Jan 2025).

SPEC 4.6 Classifying reactions

WHAT THIS SPEC POINT ASKS

Classify a reaction as addition, substitution, oxidation, reduction or polymerisation.

4.6 The five types

Type	What happens	Unit 1 example
Addition	two molecules join to give **one** product; a C=C becomes C–C	$CH_2{=}CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$
Substitution	one atom or group is **replaced** by another; two products	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$
Oxidation	gain of oxygen or loss of hydrogen (loss of electrons)	combustion; alkene + acidified $KMnO_4$ → diol
Reduction	gain of hydrogen or loss of oxygen (gain of electrons)	$CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$
Polymerisation	many small monomer molecules join into a long chain	$n\ CH_2{=}CH_2 \rightarrow -[CH_2{-}CH_2]-_n$

4.6 How to decide

- Count the products. **One** product from two reactants → addition. **Two** products, with an atom swapped → substitution.
- Look at what the organic molecule gains or loses. Gains O or loses H → oxidation. Gains H → reduction.
- One reaction can fit two labels. Alkene + H_2 is **addition** *and* **reduction** (also called hydrogenation).
- The mechanism name adds a word in front: **free radical** substitution (4.18); **electrophilic** addition (5.6).

MARK-SCHEME LANGUAGE

- Exact words are needed. Jan 2020, alkene + H_2 with nickel: "addition / reduction / hydrogenation". **DO NOT AWARD "electrophilic addition"** for this reaction.
- Jun 2022: an alkene + HBr needed "**electrophilic** addition"; "addition" alone was not enough there.

Often asked as: "state the type of reaction"; classify each step in a reaction scheme.

COMMON MISTAKE

Calling a substitution an "addition" (WCH11 Oct 2020 ER). Check: did the C=C disappear, or did one atom swap for another?
Calling hydrogenation "electrophilic addition". H_2 is not an electrophile; the reaction happens on the nickel surface.

SPEC 4.7 Homolytic and heterolytic bond breaking

WHAT THIS SPEC POINT ASKS

Understand that a covalent bond can break (i) homolytically, giving free radicals, or (ii) heterolytically, giving ions.

4.7 Two ways to split a bonding pair

- **Homolytic fission**: the bond breaks **evenly**. Each atom takes **one** electron from the shared pair. You get two **free radicals**.
  - Cl–Cl → Cl• + •Cl (written 2Cl•)
  - Needs energy from **UV light** or high temperature. Typical for **non-polar** bonds such as Cl–Cl, Br–Br, C–C.

- **Heterolytic fission**: the bond breaks **unevenly**. **One** atom takes **both** electrons. You get a positive ion and a negative ion.
  - H–Br → H^+ + Br^-
  - The **more electronegative** atom takes the pair. Typical for **polar** bonds.

Homolytic fission



each atom keeps one electron of the pair
→ two free radicals (Cl_2 needs UV light)

Heterolytic fission



the more electronegative atom (Br) takes
both electrons → two ions

Left: two half-headed ("fish-hook") arrows, one electron to each Cl. Right: one double-headed curly arrow, both electrons to Br.

4.7 Curly-arrow rules

Arrow	Shows the movement of	Used in
Half-headed (fish-hook), one barb	**one** electron	homolytic fission, free-radical steps
Full (double-barbed) head	a **pair** of electrons	heterolytic fission, electrophilic addition

- An arrow always **starts at a bond or a lone pair** (where the electrons are) and **ends at an atom** (or between two atoms where a new bond forms).
- Never start an arrow on an atom symbol or on a + charge.

MARK-SCHEME LANGUAGE

- Homolytic fission: "the bond breaks and **one electron goes to each atom**". Mention electrons moving; "splits equally" alone is too vague (Jan 2024 ER).
- Initiation drawn with **single-headed (half) arrows** on the bond and **dots** on both radicals; omission of the dot is penalised once only.

Often asked as: define homolytic fission; draw the homolytic fission of a given molecule (not just Cl_2); MCQ on the products of heterolytic fission.

COMMON MISTAKE

Double-headed arrows for homolytic fission (WCH11 Jun 2019, Oct 2020, Jun 2024 ER).
Showing radicals with charges (Cl^+, Cl^-) instead of dots.
Showing homolytic fission of Cl_2 when the question asked about a different molecule (WCH11 Oct 2020 ER).

SPEC 4.8 Free radicals and electrophiles

WHAT THIS SPEC POINT ASKS

Know the definitions of "free radical" and "electrophile".

4.8 Definitions to learn word for word

- **Free radical**: a species with an **unpaired electron**. Shown with a dot on the atom that carries the unpaired electron: Cl•, CH_3•, CH_3CH_2•.
- **Electrophile**: an **electron-pair acceptor**. It is attracted to a region of **high electron density** (such as a C=C double bond) and accepts a pair of electrons to form a new covalent bond. Electrophiles are positive ions or have a δ+ atom: H^+, the Hδ+ end of HBr, the Brδ+ end of a polarised Br_2.

4.8 Why radicals and electrophiles react

- A radical's unpaired electron makes it very **reactive**: it pairs up by taking an electron from another molecule.
- An electrophile is short of electrons, so it heads for electron-rich sites. "Electrophile" = *electron-loving*.

MARK-SCHEME LANGUAGE

- Free radical: "(a species with) a **single / unpaired electron**". ALLOW "free electron". IGNORE "free radical" repeated in the answer (Jan 2026).
- Electrophile: "electron pair acceptor". (Section B mark schemes rarely ask this outright; it appears in MCQs and mechanisms.)

Often asked as: define free radical (1 mark); identify the electrophile in a mechanism; MCQ.

COMMON MISTAKE

"A radical has a free electron" or "its own electron" can lose the mark; say **unpaired**. A Cl atom with a dot and no charge is a radical; Cl^- is an ion, not a radical.

BEYOND THE SPEC, BUT EXAMINED

The opposite of an electrophile, a **nucleophile**, is an **electron-pair donor** (Br^- in step 2 of electrophilic addition, OH^- in Unit 2). Nucleophile and electrophile appear side by side as MCQ options (WCH11 Jun 2022).

SPEC 4.9 Alkanes and cycloalkanes

WHAT THIS SPEC POINT ASKS

Know the general formulae of alkanes and cycloalkanes, and that they are saturated hydrocarbons (C and H only, single bonds only).

4.9 Key words

- **Hydrocarbon**: a compound of **carbon and hydrogen only**.
- **Saturated**: contains **single bonds only** (no C=C).
- **Alkane**: C_nH_{2n+2}, open chain (straight or branched). Methane CH_4, ethane C_2H_6, propane C_3H_8, butane C_4H_{10}.
- **Cycloalkane**: C_nH_{2n}, the carbons form a ring. Closing the ring removes two H atoms. Cyclohexane C_6H_{12}.

Cycloalkanes have the **same** general formula as alkenes but are still **saturated**. That's why C_6H_{12} could be hexene or cyclohexane, and only a test (5.5) tells them apart.

4.9 Structure and properties

- Each carbon makes four single bonds arranged **tetrahedrally**, bond angle **109.5°**. A chain is therefore zig-zag, not straight.
- **Unreactive** in general: C–C and C–H bonds are strong and almost **non-polar**, so electrophiles and nucleophiles are not attracted. Alkanes do burn (4.17) and react with halogens in UV light (4.18).
- **Boiling temperature** rises with chain length (more London forces). **Branching lowers** it: pentane 36 °C, 2-methylbutane 28 °C, 2,2-dimethylpropane 10 °C. Branched molecules can't pack as closely, so there is less contact between them.
- Insoluble in water; good solvents for other non-polar substances.

MARK-SCHEME LANGUAGE

- "Hydrocarbon: contains hydrogen and carbon **only**." "Saturated: **single (C–C) bonds only**."
- General formula questions often come as MCQs: C_nH_{2n+2} for alkanes, C_nH_{2n} for cycloalkanes, C_nH_{2n-2} for cycloalkenes (Oct 2024, Jan 2026).

Often asked as: MCQ on a general formula; "explain why cyclohexane is saturated"; "why does cyclohexane have the same formula as hexene?"

COMMON MISTAKE

"A hydrocarbon contains hydrogen and carbon" without **only**. Ethanol contains hydrogen and carbon too. Thinking every C_nH_{2n} compound is an alkene. Cycloalkanes share the formula.

SPEC 4.10 Structural isomerism

WHAT THIS SPEC POINT ASKS

Understand structural isomerism and draw the structural isomers of a compound from its molecular formula.

4.10 Definition

Structural isomers have the **same molecular formula** but **different structural formulae** (the atoms are joined in a different order).

4.10 Three kinds

Kind	What changes	Example (same molecular formula)
Chain	the carbon skeleton (branching)	butane and 2-methylpropane (C_4H_{10})
Position	where the functional group sits	1-bromopropane and 2-bromopropane (C_3H_7Br); but-1-ene and but-2-ene (C_4H_8)
Functional group	a different functional group	propene (alkene) and cyclopropane (cycloalkane), both C_3H_6

Structural isomers are **different compounds**: their physical properties differ (branched isomers boil lower), and with a different functional group their chemistry differs too.

4.10 How to draw all the isomers without repeats

1. Draw the **longest** chain first (all carbons in a line).

2. Shorten the chain by **one** carbon. Put that carbon on as a branch, at each **different** position. Never on an end carbon: a methyl on C1 just makes the chain longer again.
3. Shorten by **two**: add two methyls (same carbon or different carbons) or one ethyl. An ethyl branch needs to sit at least on C3 or it lengthens the chain.
4. For a functional group, move it to each distinct position on each skeleton.
5. **Name every structure.** Two drawings with the same name are the same molecule, however differently they are drawn. Also check every carbon has four bonds and the formula is right.

WORKED EXAMPLE: ISOMERS OF $C_3H_6BR_2$

Skeleton: only one (three carbons in a line). Place two Br atoms:

- both on C1: **1,1-dibromopropane**
- on C1 and C2: **1,2-dibromopropane**
- on C1 and C3: **1,3-dibromopropane**
- both on C2: **2,2-dibromopropane**

Four isomers. "3,3-" is just 1,1- named from the wrong end.

MARK-SCHEME LANGUAGE

- "**Same molecular formula**, different structural formula(e)" (Oct 2025). ALLOW, for alkenes, "same molecular formula but the double bond in a different position".
- Any formula type is accepted (displayed, structural, skeletal). If you give two types for one isomer, both must be correct.

Often asked as: define structural isomers (1–2 marks); draw the isomers of a given formula.

COMMON MISTAKE

Mixing up structural isomers with stereoisomers (E/Z, 5.2) or with isotopes (WCH11 Jan 2021 ER). Drawing the same molecule twice, flipped or bent. 3-methylpentane is the one most often repeated (WCH11 Jan 2020 ER).

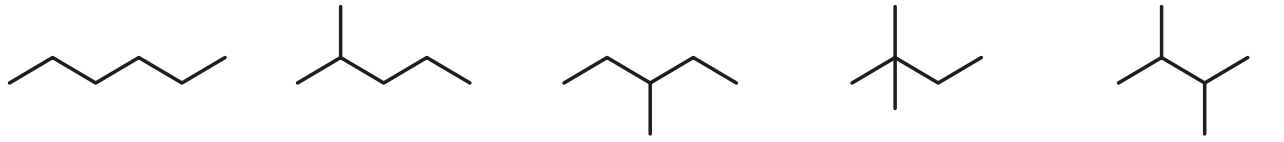
SPEC 4.11 Isomers of alkanes and cycloalkanes up to C6

WHAT THIS SPEC POINT ASKS

Draw and name the structural isomers of alkanes and cycloalkanes with up to six carbon atoms.

4.11 Alkanes

Formula	Number of isomers	Names
C_4H_{10}	2	butane; 2-methylpropane
C_5H_{12}	3	pentane; 2-methylbutane; 2,2-dimethylpropane
C_6H_{14}	5	hexane; 2-methylpentane; 3-methylpentane; 2,2-dimethylbutane; 2,3-dimethylbutane



hexane | 2-methylpentane | 3-methylpentane | 2,2-dimethylbutane | 2,3-dimethylbutane

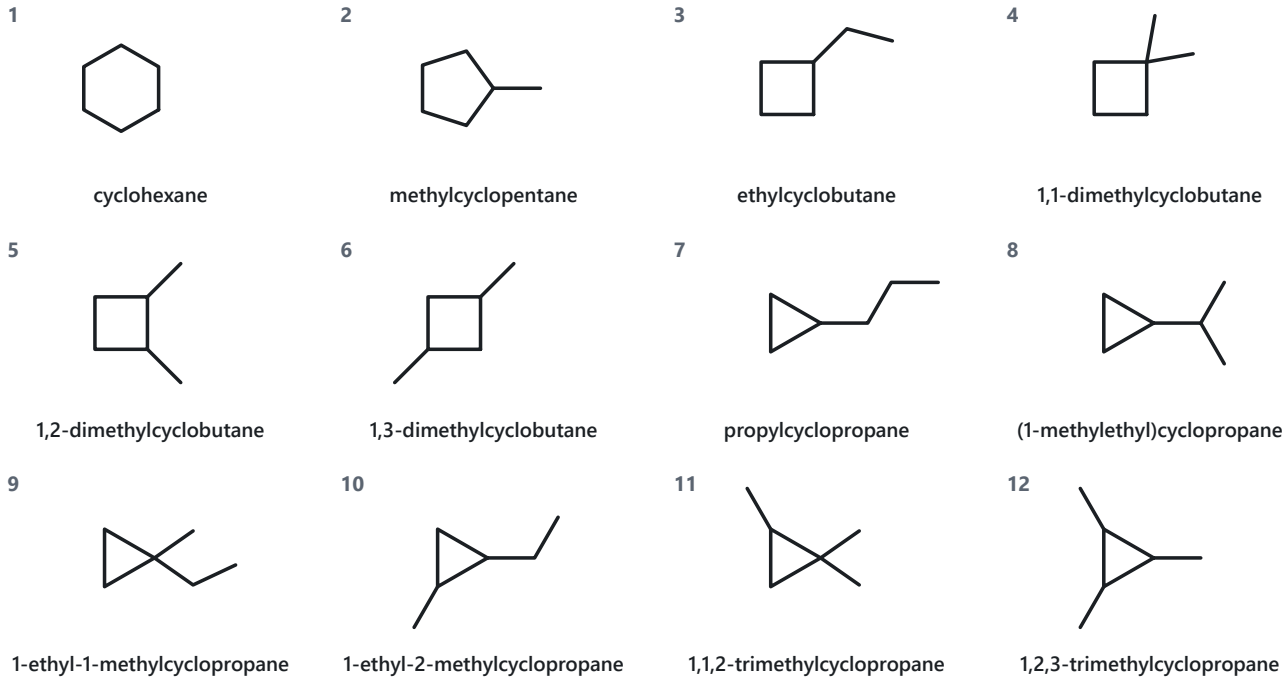
The five structural isomers of C_6H_{14} as skeletal formulae.

Check with the routine: 6 in a line (1); 5-chain + methyl on C2 or C3 (2); 4-chain + two methyls on C2,C2 or C2,C3 (2). An ethyl on a 4-chain would make a 5-chain again, so there are no more.

4.11 Cycloalkanes

Work by **ring size**, biggest first, then hang the leftover carbons on the ring.

Formula	Isomers
C_3H_6	cyclopropane
C_4H_8	cyclobutane; methylcyclopropane
C_5H_{10}	cyclopentane; methylcyclobutane; ethylcyclopropane; 1,1-dimethylcyclopropane; 1,2-dimethylcyclopropane
C_6H_{12}	12 isomers (figure below)



The twelve cycloalkane isomers of C_6H_{12}: one 6-ring, one 5-ring, four 4-rings and six 3-rings. On a ring, number so the substituents get the lowest numbers.

EXAM TIP

With a ring, "1,1-" (both groups on one carbon) is easy to forget. On a 4-ring, 1,2- (next door) and 1,3- (across) are different, but there is no 1,4-: that is 1,2- counted the other way round.

MARK-SCHEME LANGUAGE

- Jan 2020, isomers of C_6H_{14} (other than hexane): 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane, 2,3-dimethylbutane.
- Oct 2019, C_6 cycloalkanes: **DO NOT AWARD** a structure with **two or more rings** (C_6H_{12} only has room for one ring). Ignore bond angles.

Often asked as: draw the isomers of C_6H_{14} or of C_5H_{10}/C_6H_{12} with a ring; name a given isomer; MCQ: how many isomers of C_4H_9Br (Jan 2026).

COMMON MISTAKE

Including hexane when asked for isomers "of hexane". Drawing C_7H_{16} or C_6H_{12} by mistake; leaving a carbon with five bonds (WCH11 Jan 2020 ER).
Missing the cyclic isomer when the formula is C_nH_{2n}: it is the one candidates find hardest (WCH11 Jan 2023 ER). 1,1-dimethylcyclobutane was the most-missed C_6 ring isomer (WCH11 Oct 2019 ER).

BEYOND THE SPEC, BUT EXAMINED

For C_nH_{2n}, mix alkenes and cycloalkanes. Ring isomers of C_5H_{10} such as methylcyclobutane, ethylcyclopropane and 1,1-dimethylcyclopropane were asked alongside the alkenes (WCH11 Oct 2025). C_4H_9Br has four structural isomers: 1-bromobutane, 2-bromobutane, 1-bromo-2-methylpropane, 2-bromo-2-methylpropane (WCH11 Jan 2026 MCQ).

SPEC 4.12

Alkanes as fuels: fractional distillation, cracking and reforming

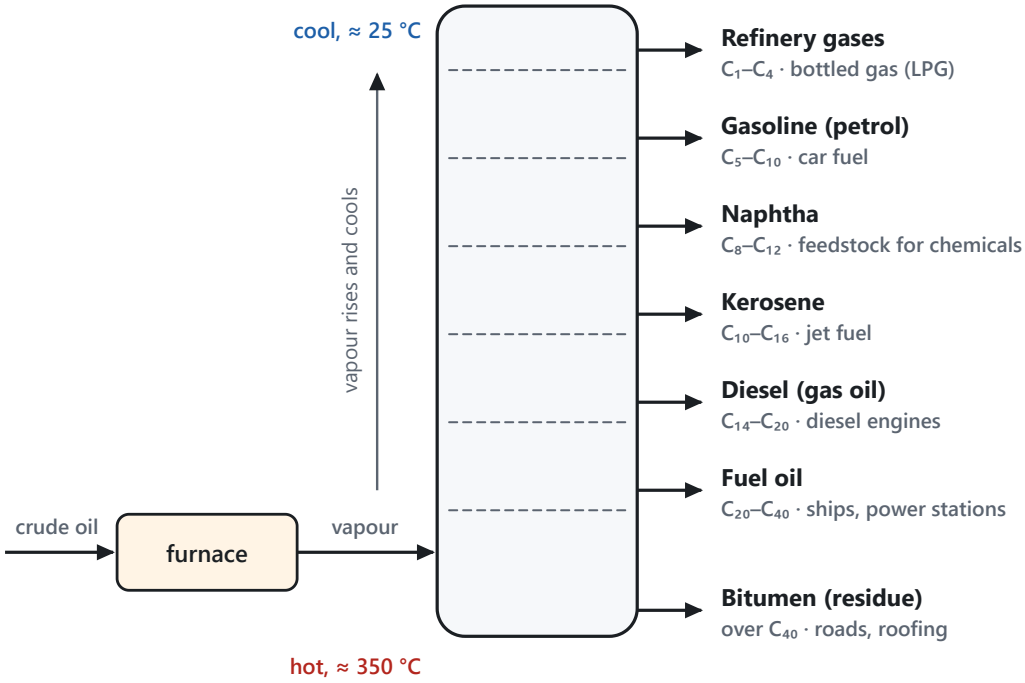
WHAT THIS SPEC POINT ASKS

Know that alkanes are used as fuels and come from crude oil by fractional distillation, cracking and reforming. Write equations for these processes.

4.12 Fractional distillation of crude oil

Crude oil is a **mixture** of hydrocarbons, mostly alkanes. Fractional distillation separates it into **fractions**: groups of compounds with similar boiling temperatures (similar chain lengths).

1. Crude oil is heated in a furnace until most of it **vaporises**.
2. The vapour enters near the bottom of a tall **fractionating column**, which is **hot at the bottom and cooler at the top** (a temperature gradient).
3. Vapour rises. Each compound **condenses** when it reaches a tray cooler than its boiling temperature, and is piped off there.
4. Longer chains have **higher boiling temperatures** (stronger London forces), so they condense **lower down**. Short chains stay as gas and leave at the top. The heaviest residue (bitumen) never vaporises.



Fractions from top to bottom: chain length and boiling temperature increase down the column. Chain-length ranges overlap and vary between refineries.

4.12 Cracking

- **What**: breaking **long-chain** alkanes into **shorter** alkanes **and alkenes**. It is thermal decomposition.
- **Why**: distillation gives too much of the long-chain fractions (fuel oil) and not enough of the short ones in demand (petrol). The alkenes made are valuable feedstock for polymers and other chemicals.
- **How**: high temperature, with or without a catalyst.
  - **Thermal cracking**: high temperature (around 450–900 °C) and high pressure; gives lots of alkenes.
  - **Catalytic cracking**: about 450 °C, low pressure, **zeolite** catalyst; gives branched alkanes and aromatic compounds for petrol.
  - In the lab: hot **aluminium oxide** or broken porcelain as the catalyst.

Equations: one alkane in, at least one shorter alkane and one alkene out. Balance carbon and hydrogen.

- $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$
- $C_{16}H_{34} \rightarrow C_8H_{18} + C_8H_{16}$
- $C_{12}H_{26} \rightarrow C_6H_{14} + 2C_3H_6$

4.12 Reforming

- **What**: turning **straight-chain** alkanes into **branched** alkanes, **cycloalkanes** or arenes, **keeping the same number of carbons**. Catalyst: platinum.
- **Why**: branched and cyclic hydrocarbons **burn more smoothly** in a petrol engine. They are less likely to **pre-ignite (knock)**, so the fuel has a higher **octane number**.
- Equations:
  - Isomerisation: $CH_3(CH_2)_6CH_3$ (octane) → $(CH_3)_3CCH_2CH(CH_3)CH_3$ (2,2,4-trimethylpentane), both C_8H_{18}
  - Ring formation: C_6H_{14} (hexane) → C_6H_{12} (cyclohexane) + H_2
  - C_7H_{16} (heptane) → C_7H_{14} (methylcyclohexane or cycloheptane) + H_2
  - To an arene: $C_7H_{16} \rightarrow C_6H_5CH_3$ (methylbenzene, C_7H_8) + $4H_2$

	Cracking	Reforming
Chain length	gets **shorter**	**stays the same**
Products	smaller alkane + alkene	branched or cyclic isomer (+ H_2 if a ring forms)
Purpose	meet demand for petrol; make alkenes	better petrol (less knocking)

SUGGESTED PRACTICAL: CRACKING LIQUID PARAFFIN

1. Soak mineral wool in **liquid paraffin** (a mixture of long-chain alkanes) at the bottom of a boiling tube clamped horizontally.
2. Put **aluminium oxide** (or broken porcelain) in the middle of the tube as the catalyst.
3. Heat the catalyst strongly, then flick the flame back to the paraffin so its vapour passes over the hot catalyst.
4. Collect the gas over water in test tubes. Discard the first tube (it is mostly air).
5. Shake a tube of gas with **bromine water**: orange → colourless shows an **alkene** was made.

Safety: lift the delivery tube **out of the water before you stop heating**, or cold water is sucked back into the hot tube and cracks it (suck-back). The gases are flammable.

MARK-SCHEME LANGUAGE

- Fractional distillation (Jun 2023): "the vapour **condenses** at different levels / temperatures"; "the **higher the boiling temperature**, the **lower** the level at which it condenses". ALLOW the temperature gradient. DO NOT AWARD melting temperature; DO NOT AWARD "simple distillation".
- Why branched / cyclic: "burns more **efficiently / smoothly**", "prevents **pre-ignition / knocking / pinking**"; ALLOW "increases the octane number". IGNORE "burns more cleanly", "ignites more easily", "more volatile".
- Reforming: ALLOW "isomerisation". IGNORE "rearrangement". DO NOT AWARD "cracking".
- A reforming equation to a cycloalkane has **one** hydrocarbon product plus **H_2** (Jun 2021: $C_7H_{16} \rightarrow C_7H_{14} + H_2$).

Often asked as: name the process from an equation; write a cracking or reforming equation; explain how fractional distillation separates crude oil; explain why branched alkanes are added to petrol.

COMMON MISTAKE

Calling reforming "cracking" or "fractional distillation" (WCH11 Oct 2022 and Jan 2023 ER). If the number of carbons stays the same, it's reforming.
If the question shows a five-membered ring product, don't write cyclohexane (WCH11 Oct 2023 ER). Use the structure in the stem.
Describing the whole refinery when asked a specific point about boiling temperatures (WCH11 Oct 2024 ER).

BEYOND THE SPEC, BUT EXAMINED

Octane number and knocking are the reason for reforming (WCH11 Oct 2022).
Fractional distillation of **liquid air**: the air is first liquefied by cooling and compressing, then separated by boiling temperature. "Temperature" alone scored nothing (WCH11 Jan 2023).

SPEC 4.13 Pollutants from burning alkane fuels

WHAT THIS SPEC POINT ASKS

Know that burning alkane fuels releases carbon monoxide, oxides of nitrogen and sulfur, carbon particulates and unburned hydrocarbons.

4.13 Where each pollutant comes from

Pollutant	How it forms	Equation
Carbon monoxide, CO	incomplete combustion: **limited oxygen**	$C_8H_{18} + 8½O_2 \rightarrow 8CO + 9H_2O$
Carbon particulates (soot)	very incomplete combustion	$C_8H_{18} + 4½O_2 \rightarrow 8C + 9H_2O$
Unburned hydrocarbons	fuel that passes through the engine without burning	(none)
Oxides of nitrogen, NO and NO_2 (NO_x)	**nitrogen and oxygen from the air** react at the **high temperature** inside the engine	$N_2 + O_2 \rightarrow 2NO$; $2NO + O_2 \rightarrow 2NO_2$
Oxides of sulfur, SO_2 (and SO_3)	**sulfur impurities** in the fuel burn	$S + O_2 \rightarrow SO_2$

Complete combustion gives only CO_2 and H_2O. CO_2 is not toxic at normal levels, but it is a greenhouse gas (4.15).

MARK-SCHEME LANGUAGE

- Pollutants (Oct 2024): nitrogen dioxide, sulfur dioxide / trioxide, carbon (particulates, soot, PM10 / PM2.5), unburnt hydrocarbons. DO NOT AWARD CO_3, SO, NO_3. IGNORE water.
- Incomplete combustion needs "**limited / insufficient** oxygen". DO NOT AWARD "no oxygen" (Jun 2021).
- NO forms "from **nitrogen and oxygen in the air** at the high temperature (of the engine)": $N_2 + O_2 \rightarrow 2NO$ (Oct 2024).

Often asked as: name a pollutant; write an equation for incomplete combustion; explain how NO forms in an engine.

COMMON MISTAKE

Saying the nitrogen comes from the fuel. Petrol contains no nitrogen; it comes from **air** (WCH11 Oct 2024 ER).
Writing a complete-combustion equation when incomplete was asked (WCH11 Jan 2023 ER).
Giving methane as the unburned hydrocarbon from a heptane fuel: the unburned hydrocarbon is the fuel itself (WCH11 Jan 2023).

SPEC 4.14 Why CO, NO_x and SO_x are a problem

WHAT THIS SPEC POINT ASKS

Understand the problems these pollutants cause, limited to why carbon monoxide is toxic and why oxides of nitrogen and sulfur are acidic.

4.14 Carbon monoxide is toxic

- CO **binds to the haemoglobin** in red blood cells (at the iron of the haem group, the site where oxygen binds).
- It binds **much more strongly** than oxygen and is hardly released (it forms carboxyhaemoglobin).
- So **less oxygen is carried** in the blood to the cells. Symptoms: headache, dizziness, drowsiness, then unconsciousness and death.
- CO is **colourless and odourless**, so you can't tell it is there.

4.14 Oxides of nitrogen and sulfur are acidic

They are **non-metal oxides**: they **dissolve in / react with water** in the atmosphere (rain, clouds) to form acids, giving **acid rain**.

- $SO_2 + H_2O \rightarrow H_2SO_3$ (sulfurous acid)
- $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$ (sulfuric acid)
- $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$ (nitrous and nitric acids)
- $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$

Effects of acid rain: damages limestone and marble buildings ($CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$), corrodes metals, lowers the pH of lakes (harming fish), and damages trees and soil.

MARK-SCHEME LANGUAGE

- CO: "toxic"; "**combines (irreversibly) with haemoglobin**" so less oxygen is carried (Oct 2021). DO NOT AWARD particulates. (In that question, if nothing else was given, "CO_2 causes suffocation" got 1 mark.)
- Acid rain (Jan 2019): oxides of nitrogen "**dissolve in / react with water**" to form "**nitric / nitrous acid(s)** / acid rain". IGNORE oxides of sulfur when nitrogen was asked about.

Often asked as: why CO is toxic (2 marks); how NO_x causes acid rain; write an equation for SO_2 or NO_2 forming an acid.

COMMON MISTAKE

Forgetting to say that the oxides **dissolve in water**. "NO_2 is acidic" is not enough (WCH11 Jan 2019 ER).
Writing about sulfur dioxide when the question named oxides of nitrogen.
Mixing up acid rain with global warming or the ozone layer. These are separate problems.

BEYOND THE SPEC, BUT EXAMINED

Catalytic converters ($2CO + 2NO \rightarrow 2CO_2 + N_2$) are **not** in this unit's spec, and they are wrong as an answer to a question about removing HCl or SO_2 from waste gases (WCH11 Jun 2019 ER). Use a base instead (5.8).

SPEC 4.15 Alternative fuels: sustainability and climate change

WHAT THIS SPEC POINT ASKS

Discuss why alternative fuels are being developed: sustainability and lower emissions, including CO_2 and its link to climate change.

4.15 The case against fossil fuels

- **Finite (non-renewable).** Crude oil took millions of years to form and will run out. Using it is not **sustainable**: it doesn't meet today's needs without harming the ability of future generations to meet theirs.
- **CO_2 and climate change.** Burning fossil fuels releases carbon that was locked underground, so the CO_2 concentration in the air rises.
  - The Earth's surface absorbs sunlight and gives out **infrared** radiation.
  - **Greenhouse gases** (CO_2, CH_4, H_2O) **absorb infrared** and re-emit some of it back towards the Earth.
  - More CO_2 → **enhanced greenhouse effect** → **global warming** → climate change.
- **Other pollutants**: CO, NO_x, SO_2, particulates (4.13).

4.15 The alternatives

Fuel	Made from	For	Against
Bioethanol	fermentation of sugars from crops (sugar cane, maize)	renewable; plants absorb CO_2 as they grow	needs farmland (food vs fuel); energy for farming and distillation; lower energy per litre than petrol
Biodiesel	vegetable oils, waste cooking oil	renewable; can use waste	land use; engine changes
Biogas (methane)	anaerobic decay of waste (landfill, digesters)	uses waste; captures methane that would escape	low energy density; needs collecting
Hydrogen	electrolysis of water, or from methane	only water when burned	storage (gas, explosive); making it uses energy
Electricity (battery cars)	power stations, renewables	no exhaust emissions; quiet	only as clean as the electricity source

MARK-SCHEME LANGUAGE

- Climate change (Jun 2019): "**carbon dioxide** (released when fossil fuels burn)" + "global warming / greenhouse effect / temperature rise". IGNORE consequences (sea level). **Acid rain and ozone** in a climate answer **negate** the mark.
- Biofuels against fossil fuels (Jun 2023): fossil fuels are **non-renewable / finite**, biofuels are **renewable**; biofuels release CO_2 that was **absorbed while the plants grew**. ALLOW "carbon neutral". DO NOT AWARD "biodegradable".
- Electric cars (Jan 2023): **state both halves**: "no CO_2 emitted (from the car)" **and** "CO_2 causes global warming"; or "can use renewable electricity". Also "no local pollution", "quieter".

Often asked as: why fossil fuels cause climate change; compare a biofuel with petrol; evaluate electric or hydrogen cars; landfill gas.

COMMON MISTAKE

Mixing acid rain, ozone and sulfur emissions into a climate-change answer (WCH11 Jun 2019 ER).
Calling biofuels "biodegradable", or claiming they release no CO_2 when burned (WCH11 Jun 2023 ER). They do release CO_2; the point is that the plants took it in first.
"Environmentally friendly", "no waste", "less pollution" with no reason. Name the gas and the effect.

BEYOND THE SPEC, BUT EXAMINED

Landfill gas: waste decomposes by **anaerobic** respiration (biodegradation) to methane and CO_2, both greenhouse gases. DO NOT AWARD "thermal decomposition" or aerobic respiration (WCH11 Jun 2022). Hexane replaced by supercritical **CO_2** as an extraction solvent: non-flammable and readily available, but needs costly high pressure (WCH11 Oct 2022).

SPEC 4.16 Carbon neutrality

WHAT THIS SPEC POINT ASKS

Apply the idea of carbon neutrality to different fuels, such as petrol, bioethanol and hydrogen.

4.16 What carbon neutral means

A fuel is **carbon neutral** if using it causes **no net change** in the amount of CO_2 in the atmosphere: the CO_2 released when it burns equals the CO_2 removed from the air to make it.

4.16 Bioethanol: neutral on paper

Stage	Equation	CO_2
Photosynthesis (crop grows)	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$	**6** taken in
Fermentation	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$	2 released
Combustion	$2C_2H_5OH + 6O_2 \rightarrow 4CO_2 + 6H_2O$	4 released

6 taken in, 2 + 4 = 6 released: in theory, no net CO_2.

In practice it is **not fully** carbon neutral, because fossil fuels supply energy for: farm machinery and fertiliser, transporting the crop and fuel, and **distilling** the ethanol out of the fermentation mixture.

4.16 Comparing three fuels

Fuel	Carbon neutral?	Why
Petrol	**no**	its carbon was locked underground for millions of years; burning it adds new CO_2 to the air and nothing takes it back out
Bioethanol	**close, not quite**	CO_2 released was absorbed by the crop recently; but energy for farming, transport and distillation usually comes from fossil fuels
Hydrogen	**depends how it is made**	$2H_2 + O_2 \rightarrow 2H_2O$, so no CO_2 when it burns. Made by electrolysis with **renewable** electricity → close to neutral; made from methane or with fossil-fuel electricity → not neutral

WORKED EXAMPLE: ENERGY IN E10 (ILLUSTRATIVE VALUES)

E10 is 10% bioethanol and 90% petrol by volume. Take petrol as 34.2 MJ dm^{-3} and ethanol as 21.2 MJ dm^{-3}.
Energy from 1 dm^3 of E10 = (0.90 × 34.2) + (0.10 × 21.2) = 30.78 + 2.12 = **32.9 MJ** (3 s.f.)
So you need slightly more E10 than petrol to drive the same distance. Exam questions give the values; this is a weighted mean.

MARK-SCHEME LANGUAGE

- Carbon neutral (Jun 2019, 2 marks): the plant **takes in carbon dioxide (by photosynthesis)** as it grows **and** the **same amount** is released when the fuel burns → **no net** CO_2. "Takes in carbon" alone, or the plant taking in CO_2 by *respiration*, does not score.
- E10 (Oct 2023), three of: the ethanol is **bioethanol from plants**; made by **fermentation** of sugars; CO_2 released on combustion is **offset by CO_2 absorbed in photosynthesis**; less impact on global warming; uses less of a **finite** resource.
- Hydrogen (Oct 2021): reduces greenhouse gas emissions; fossil fuels non-renewable. DO NOT AWARD "carbon neutral" for hydrogen without discussing how it is made.

Often asked as: define carbon neutral; is fuel X carbon neutral?; advantages of E10; compare hydrogen with fossil fuels.

COMMON MISTAKE

Half the candidates scored nothing on the carbon-neutral definition (WCH11 Jun 2019 ER). Learn both halves: CO_2 **absorbed** while growing = CO_2 **released** on burning.
Answering a sustainability question about E10 with octane number or combustion efficiency from a different question (WCH11 Oct 2023 ER).

BEYOND THE SPEC, BUT EXAMINED

Energy-density calculations for blended fuels (a weighted mean) and scaling up the volume needed (WCH11 Oct 2023).
Ethene from fruit (bananas) instead of cracking: renewable, but a low yield and large amounts of crop, competing with food. "Sustainable" or "biodegradable" alone were ignored (WCH11 Jun 2019).

SPEC 4.17

Reactions of alkanes: combustion and halogenation

WHAT THIS SPEC POINT ASKS

Understand how alkanes react with (i) oxygen in air (combustion) and (ii) halogens.

4.17 Combustion

- **Complete** combustion (plenty of oxygen): CO_2 + H_2O. Very **exothermic**, which is why alkanes are fuels.
- **Incomplete** combustion (**limited** oxygen): CO and/or C (soot) + H_2O.

Fuel	Complete	Incomplete (to CO)
methane	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	$CH_4 + 1\frac{1}{2}O_2 \rightarrow CO + 2H_2O$
hexane	$C_6H_{14} + 9\frac{1}{2}O_2 \rightarrow 6CO_2 + 7H_2O$	$C_6H_{14} + 6\frac{1}{2}O_2 \rightarrow 6CO + 7H_2O$
heptane	$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$	$C_7H_{16} + 7\frac{1}{2}O_2 \rightarrow 7CO + 8H_2O$
octane	$C_8H_{18} + 12\frac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$	$C_8H_{18} + 8\frac{1}{2}O_2 \rightarrow 8CO + 9H_2O$

WORKED EXAMPLE: BALANCING A COMBUSTION EQUATION

Complete combustion of C_7H_{16}.

1. Carbon: 7 C → **$7CO_2$**.
2. Hydrogen: 16 H → **$8H_2O$**.
3. Oxygen on the right: (7 × 2) + 8 = 22 O atoms → **$11O_2$**.

$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$
If the O_2 number comes out as a half (e.g. 12½ for octane), that is fine, or double everything: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$.

4.17 Halogenation

- Alkane + chlorine or bromine **in UV light** (or sunlight) → halogenoalkane + hydrogen halide. In the dark at room temperature there is **no reaction**.
- It is a **substitution**: a hydrogen atom is replaced by a halogen atom.
  - $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (chloromethane)
  - $C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$ (bromoethane)
- With bromine you see the **orange-brown colour fade** as the bromine is used up, and steamy fumes of HBr.

MARK-SCHEME LANGUAGE

- Combustion equations need correct formulae and balancing; a question may ask for complete **or** incomplete specifically (Jun 2021).
- Condition for halogenation: "**ultraviolet light**" (ALLOW sunlight / UV radiation; IGNORE heat).

Often asked as: write the equation for complete or incomplete combustion of a named alkane; state the condition for halogenation; overall equation for a substitution.

COMMON MISTAKE

Writing "no oxygen" for incomplete combustion. It is **limited** oxygen.
Saying alkanes react with bromine water in the dark. They don't (that is why bromine water tests for alkenes, 5.5).

SPEC 4.18 Free-radical substitution: the mechanism

WHAT THIS SPEC POINT ASKS

Understand the free-radical substitution mechanism for an alkane and a halogen: (i) radicals as species with an unpaired electron, shown by a dot; (ii) the initiation step with curly half-arrows; (iii) propagation and termination; (iv) why the reaction is of limited use in synthesis.

4.18 The mechanism for methane + chlorine (UV light)

1. Initiation: radicals are made.

- UV light gives the energy to break the Cl–Cl bond **homolytically**. Draw **two half-headed arrows** starting from the middle of the Cl–Cl bond, one ending on each Cl.
- $Cl_2 \rightarrow 2Cl\bullet$
- The Cl–Cl bond breaks (not C–H) because it is the **weakest** bond present.

2. Propagation: a chain reaction. Each step uses up one radical and makes another.

- Step 1: $Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$
  (the chlorine radical takes an H atom; a methyl radical is left)
- Step 2: $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$
  (the methyl radical takes a Cl atom; a new chlorine radical is released)
- The $Cl\bullet$ made in step 2 goes back to step 1. One initiation can cause thousands of propagation cycles.
- Add the two steps: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (the overall equation).

3. Termination: radicals are removed. Two radicals meet and their unpaired electrons form a bond.

- $Cl\bullet + Cl\bullet \rightarrow Cl_2$
- $\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$
- $\bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$ (ethane)

4.18 Any other alkane: same pattern

Put the radical dot on the carbon that **lost the H atom**. For ethane + chlorine:

- Initiation: $Cl_2 \rightarrow 2Cl\bullet$
- Propagation: $Cl\bullet + CH_3CH_3 \rightarrow CH_3CH_2\bullet + HCl$
- Propagation: $CH_3CH_2\bullet + Cl_2 \rightarrow CH_3CH_2Cl + Cl\bullet$
- Termination: $CH_3CH_2\bullet + Cl\bullet \rightarrow CH_3CH_2Cl$; $2CH_3CH_2\bullet \rightarrow C_4H_{10}$ (butane); $2Cl\bullet \rightarrow Cl_2$

Two alkyl radicals joining always give a hydrocarbon with **twice** the carbons: 2 butyl radicals → octane, C_8H_{18}.

4.18 Why it has limited use in synthesis

- **Further substitution.** The product still has C–H bonds, so $Cl\bullet$ attacks it too: $CH_3Cl \rightarrow CH_2Cl_2 \rightarrow CHCl_3 \rightarrow CCl_4$. You get a **mixture**.
  - $Cl\bullet + CH_3Cl \rightarrow \bullet CH_2Cl + HCl$, then $\bullet CH_2Cl + Cl_2 \rightarrow CH_2Cl_2 + Cl\bullet$
- **Substitution at different positions.** Propane gives both 1-chloropropane (H removed from an end carbon) and 2-chloropropane (H removed from the middle carbon).
- **Termination** adds other products (e.g. butane from ethane).
- The mixture must be separated, and the yield of any one product is low.
- **Controlling it**: an **excess of the alkane** makes the monosubstituted product more likely ($Cl\bullet$ is more likely to hit an alkane than a chloroalkane). An **excess of halogen** pushes it all the way to full substitution.

WORKED EXAMPLE: FULL SUBSTITUTION WITH EXCESS BROMINE

Hexane reacts with a large excess of bromine in UV light. Every H is replaced, so the product is C_6Br_{14} (not $C_6H_{13}Br$).
Mr = (6 × 12.0) + (14 × 79.9) = 72.0 + 1118.6 = 1190.6
% Br by mass = 1118.6 ÷ 1190.6 × 100 = **94.0%**

MARK-SCHEME LANGUAGE

- Name: "**(free) radical substitution**", both words. DO NOT AWARD heterolytic / nucleophilic / electrophilic. IGNORE homolytic, photochemical.
- Condition: "UV light" (ALLOW sunlight; IGNORE heat).
- Initiation: $Cl_2 \rightarrow 2Cl\bullet$ with **half-arrows**. Initiation = **one** equation, propagation = **two**, termination = **one** (e.g. a six-mark question naming each step, Jun 2021).
- Propagation, ethane (Jan 2026, 1 mark each): $Cl\bullet + CH_3CH_3 \rightarrow CH_3CH_2\bullet + HCl$; $CH_3CH_2\bullet + Cl_2 \rightarrow CH_3CH_2Cl + Cl\bullet$. ALLOW $C_2H_5\bullet$. Missing dots penalised **once only**.
- Limitation (Jan 2026): "**further substitution** (reactions) / more than one substitution product". IGNORE "further reactions", "low yield", "side reactions", "atom economy".

Often asked as: the whole mechanism for a named alkane (hexane, propane, butane, cyclohexane, 2,2,4-trimethylpentane); a termination step that makes a hydrocarbon; why the method is not a good way to make one product; a % by mass or % yield calculation.

COMMON MISTAKE

A propagation step that makes **two** radicals, or makes **H•** (WCH11 Jan 2020, Oct 2023 ER). Each propagation step has exactly one radical on each side.
Radicals drawn with + or − charges, or the dot missing (WCH11 Oct 2020, Jan 2020 ER).
Giving Cl_2 or a chloroalkane when the question asks for a termination step that makes **a hydrocarbon** (WCH11 Jan 2019 ER).
Using methane from memory when the question named hexane (WCH11 Oct 2023 ER). Use the formulae in the stem and balance each step.
"UV light breaks bonds" without saying **which** bond: it breaks the **Cl–Cl** (or Br–Br) bond homolytically (WCH11 Oct 2023 ER).
Writing more than one equation when the question asks for "a propagation step": only the first is marked (WCH11 Oct 2020 ER).

BEYOND THE SPEC, BUT EXAMINED

Mechanisms for alkanes other than methane: hexane with bromine (WCH11 Jan 2024), propane (WCH11 Jan 2026), butane (WCH11 Jun 2025), cyclohexane (WCH11 Oct 2019).
Two termination products that are hydrocarbons, drawn as displayed formulae (WCH11 Oct 2020).
With excess halogen, the fully substituted product (e.g. C_6Br_{14}) in a calculation (WCH11 Jan 2020).

Topic 5 – Alkenes

SPEC 5.1 Alkenes: unsaturated hydrocarbons with σ and π bonds

WHAT THIS SPEC POINT ASKS

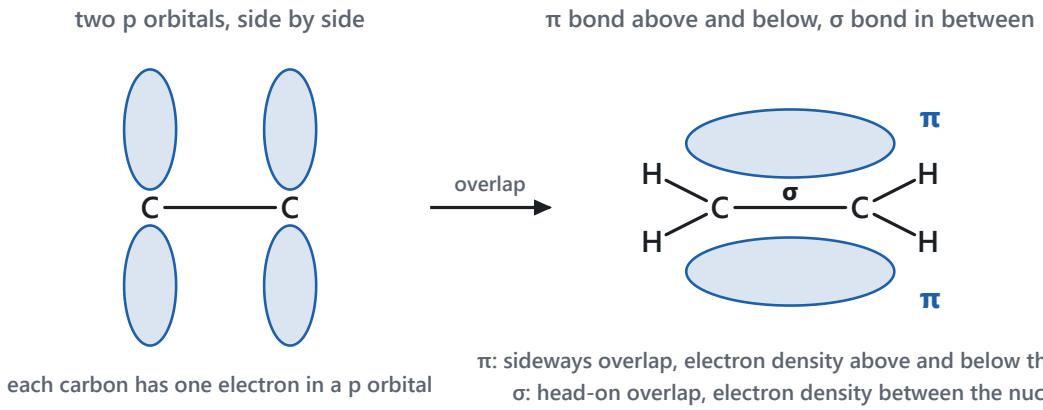
Know the general formula of alkenes. Understand that alkenes and cycloalkenes are unsaturated hydrocarbons, with a C=C double bond made of one σ bond and one π bond.

5.1 Key facts

- **Alkenes**: hydrocarbons with **one C=C double bond**. General formula **C_nH_{2n}**. Ethene C_2H_4, propene C_3H_6, butene C_4H_8.
- **Cycloalkenes** (a ring plus one C=C): **C_nH_{2n-2}**. Cyclohexene C_6H_{10}.
- **Unsaturated**: the molecule contains a C=C double bond, so more atoms can be **added** across it.
- Around each carbon of the C=C there are three bonding regions, so the shape is **trigonal planar**, bond angle **about 120°**. The two carbons and the four atoms attached to them all lie in **one plane**.

5.1 What the double bond is made of

- **σ (sigma) bond**: formed by **head-on (end-on) overlap** of two orbitals, directly between the two nuclei. The electron density sits **on the line between the nuclei**. Every single bond is a σ bond.
- **π (pi) bond**: formed by **sideways overlap of two p orbitals**, one on each carbon. The electron density sits **above and below** the plane of the molecule, not between the nuclei.
- A C=C double bond = **one σ bond + one π bond**.



Each carbon's p orbital (left) overlaps sideways with its neighbour's to make the π bond (right): two regions of electron density, above and below the σ bond.

5.1 Why the π bond makes alkenes reactive

- The π electrons are further from the nuclei and **exposed** above and below the molecule, so they are easier to reach.
- The π bond is **weaker** than the σ bond. C=C (612 kJ mol^{-1}) is **less** than twice C–C (2 × 347 = 694 kJ mol^{-1}).
- The C=C is a region of **high electron density**, so it attracts **electrophiles** (5.6). In addition reactions the π bond breaks and the σ bond stays.

WORKED EXAMPLE: COUNTING Σ AND Π BONDS

Propene, $CH_2=CHCH_3$.

- 6 C–H bonds: 6 σ
- 1 C–C bond: 1 σ
- 1 C=C bond: 1 σ + 1 π

Total: **8 σ and 1 π**.
Rule: every single bond is 1 σ; every double bond is 1 σ + 1 π.

MARK-SCHEME LANGUAGE

- Drawing the π bond (Jan 2024): **two** regions of electron density (lobes), one **above** and one **below** the σ bond, and the σ bond shown between the nuclei.
- σ = "end-on / head-on overlap"; π = "**sideways** overlap of **p** orbitals".

Often asked as: describe or draw how a π bond forms; count σ and π bonds in an unfamiliar molecule (MCQ, Jan 2026); give a general formula (cycloalkene, Oct 2024).

COMMON MISTAKE

Drawing the π bond in line with the σ bond, or as one lobe only (WCH11 Jan 2022 ER).
Drawing a single line for the σ bond with no indication of overlapping orbitals when the question asks for orbitals (WCH11 Jan 2024 ER).
Saying the π bond is "stronger" because a double bond is stronger. The **double bond** is stronger than a single bond, but the **π part** of it is weaker than the σ part.

SPEC 5.2

Geometric (E/Z) isomerism

WHAT THIS SPEC POINT ASKS

Explain geometric isomerism using restricted rotation about C=C and the nature of the groups on each carbon of the double bond.

5.2 Stereoisomers

Stereoisomers have the **same structural formula** (atoms joined in the same order) but a **different arrangement of atoms in space**. Geometric (E/Z or cis/trans) isomerism is one kind.

5.2 Two conditions, both needed

1. **Restricted rotation about the C=C.** Rotating one end of a double bond would twist the p orbitals apart and **break the π bond**. That needs a lot of energy, so it doesn't happen at room temperature. (Around a C–C single bond, rotation is free, so alkanes don't form geometric isomers.)
2. **Two different groups on each carbon of the C=C.** If either carbon carries two identical groups, swapping them changes nothing.

5.2 Check each C=C carbon

Alkene	Groups on carbon 1 of C=C	Groups on carbon 2 of C=C	Geometric isomers?
propene CH_2=$CHCH_3$	H, H	H, CH_3	**no** (two H on one carbon)
but-1-ene CH_2=$CHCH_2CH_3$	H, H	H, C_2H_5	**no**
but-2-ene CH_3CH=$CHCH_3$	H, CH_3	H, CH_3	**yes**
2-methylpropene $(CH_3)_2C$=CH_2	CH_3, CH_3	H, H	**no**
2-methylbut-2-ene $(CH_3)_2C$=$CHCH_3$	CH_3, CH_3	H, CH_3	**no**
pent-2-ene CH_3CH=$CHCH_2CH_3$	H, CH_3	H, C_2H_5	**yes**
1,2-dichloroethene CHCl=CHCl	H, Cl	H, Cl	**yes**

MARK-SCHEME LANGUAGE

- Why isomers exist (Jan 2025, Oct 2025): "**restricted rotation** around the C=C" **and** "**two different groups / atoms** on each carbon of the double bond".
- Why no isomers (Oct 2020, isoprene): "each double bond has **two hydrogen atoms on one carbon**"; ALLOW "two identical groups / atoms attached to one carbon".
- The π bond prevents rotation; rotation would need the π bond to break.

Often asked as: explain why a named alkene has geometric isomers (2 marks); explain why another does not; draw both isomers.

COMMON MISTAKE

"Similar groups" instead of **different** groups; "molecules" instead of **groups** or **atoms** (WCH11 Oct 2020, Jan 2024 ER).
"The double bond restricts rotation **of the molecule**" or "**allows** restricted rotation". Say: there is no rotation **about the C=C bond** (WCH11 Oct 2020, Jan 2024 ER).
Checking only one carbon of the C=C. Check both.

BEYOND THE SPEC, BUT EXAMINED

Molecules with **several C=C bonds**: test each double bond separately. Only C=C bonds with two different groups on both carbons give E/Z isomers. A molecule whose other C=C bonds each carry two identical groups has just two geometric isomers (WCH11 Jan 2021, α-farnesene; WCH11 Jan 2022, ocimene).

SPEC 5.3 The E–Z naming system

WHAT THIS SPEC POINT ASKS

Understand E–Z naming of geometric isomers, and why it is needed when cis/trans naming breaks down.

5.3 Cis/trans and its limit

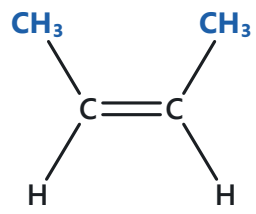
- **cis**: two identical groups on the **same side** of the C=C. **trans**: on **opposite sides**.

- This only works when the two carbons share a group to compare (usually H or CH_3). If all four groups are different (e.g. $CBrCl{=}CHCH_3$), there is nothing to compare, so cis/trans can't be used. E/Z works for every case.

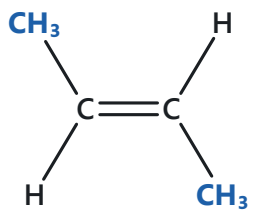
5.3 E/Z: the priority rules (Cahn–Ingold–Prelog)

1. Look at **each carbon of the C=C separately**.
2. Compare the two **atoms directly attached** to that carbon. The one with the **higher atomic number** has **higher priority**. (Br 35 > Cl 17 > O 8 > C 6 > H 1.)
3. If the two attached atoms are the same (e.g. both C), list the atoms attached **to each of them** in order of atomic number and compare at the **first point of difference**.
4. **Z** (*zusammen*, together): the two higher-priority groups are on the **same side**.
   E (*entgegen*, opposite): the two higher-priority groups are on **opposite sides**.

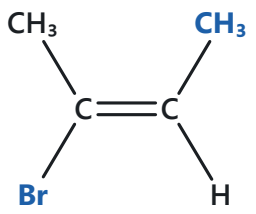
Memory aid: **Z** = "**Z**ame **Z**ide".



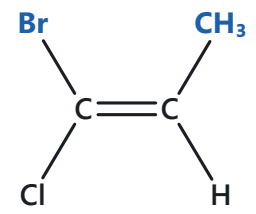
(Z)-but-2-ene
cis: CH_3 groups on the same side



(E)-but-2-ene
trans: CH_3 groups on opposite sides



(E)-2-bromobut-2-ene
CH_3 groups 'cis', but it is E



(Z)-1-bromo-1-chloroprop-1-ene
no shared group: cis/trans fails

Higher-priority group on each carbon in blue. For but-2-ene, Z = cis and E = trans. In 2-bromobut-2-ene the methyls are "cis" but Br outranks CH_3, so the isomer is E: the two systems don't always agree.

5.3 Drawing E and Z isomers: the method

1. Draw the C=C horizontally.
2. Put the two groups on each carbon at about 120°: one up and one down on the left carbon, one up and one down on the right carbon.
3. Mark the higher-priority group on each carbon.
4. Both higher-priority groups up (or both down) → **Z**. One up, one down → **E**.
5. To draw the other isomer, **swap the two groups on one carbon only**. Don't redraw the chain differently, or you may change the molecule.

WORKED EXAMPLE: (Z)-3-METHYLPENT-2-ENE

Structure: $CH_3CH{=}C(CH_3)CH_2CH_3$. The C=C is between C2 and C3.

- **C2** carries CH_3 and H. C beats H → **CH_3** has priority.
- **C3** carries CH_3 and CH_2CH_3. Both attach through C: a tie. Go one atom further. CH_3 has (H, H, H); CH_2CH_3 has (**C**, H, H). C beats H at the first difference → **CH_2CH_3** has priority.
- **Z**: draw the CH_3 on C2 and the CH_2CH_3 on C3 on the **same side**.

Notice that in the Z isomer the two CH_3 groups are on **opposite** sides. Cis/trans by the methyls would give the wrong answer.

MARK-SCHEME LANGUAGE

- Z (Jan 2021): "Z isomer has the **highest priority groups** on the **same side** of the C=C"; priority by **atomic number**, not mass.
- Trans (Jun 2024): "the two (identical) groups / alkyl groups are on **opposite sides** of the C=C". "Molecule" for "group" does not score.
- A full name has the **E or Z**, the **position of the double bond**, the stem and **-ene** (Jan 2024): (E)-pent-2-ene, not "pentene" or "E-pentene".

Often asked as: draw the Z isomer of a named alkene and explain why it is Z; name a drawn isomer; MCQ on E/Z.

COMMON MISTAKE

Ranking groups by **mass** or by size instead of the **atomic number** of the first atom (WCH11 Jan 2021 ER).
Drawing the E isomer when Z was asked, or the **same** isomer twice in a different orientation (WCH11 Jan 2019, Oct 2019 ER).
Leaving out E/Z or the double-bond position in a name; "dimethylethene" for but-2-ene (WCH11 Jan 2024 ER).
In skeletal formulae, moving where the chain attaches when you swap groups (WCH11 Jan 2022 ER).

SPEC 5.4 Reactions of alkenes

WHAT THIS SPEC POINT ASKS

Describe these alkene reactions: (i) + hydrogen (nickel catalyst) → alkane; (ii) + halogen → dihalogenoalkane; (iii) + hydrogen halide → halogenoalkane; (iv) + steam (acid catalyst) → alcohol; (v) oxidation by acidified potassium manganate(VII) → diol.

5.4 The five reactions at a glance

Reagent	Conditions	Ethene gives	Type
H_2	**nickel** catalyst, about 150 °C (Pt or Pd also work)	ethane CH_3CH_3	addition / reduction (hydrogenation)
Br_2 (or Cl_2)	room temperature; liquid bromine, bromine in an organic solvent, or bromine water	1,2-dibromoethane CH_2BrCH_2Br	electrophilic addition
HBr (or HCl, HI)	room temperature, gaseous HBr	bromoethane CH_3CH_2Br	electrophilic addition
H_2O as **steam**	**phosphoric(V) acid (H_3PO_4)** catalyst; industrially about 300 °C and 60–70 atm	ethanol CH_3CH_2OH	addition (hydration)
$KMnO_4$ / H^+ (acidified potassium manganate(VII))	room temperature, shake	ethane-1,2-diol $HOCH_2CH_2OH$	oxidation

5.4 Equations

Reaction	Ethene	Propene
Hydrogenation	$CH_2=CH_2 + H_2 \rightarrow CH_3CH_3$	$CH_2=CHCH_3 + H_2 \rightarrow CH_3CH_2CH_3$ (propane)
Halogen	$CH_2=CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$	$CH_2=CHCH_3 + Br_2 \rightarrow CH_2BrCHBrCH_3$ (**1,2-dibromopropane**)
Hydrogen halide	$CH_2=CH_2 + HBr \rightarrow CH_3CH_2Br$	$CH_2=CHCH_3 + HBr \rightarrow CH_3CHBrCH_3$ (**2-bromopropane**, major) + $CH_3CH_2CH_2Br$ (1-bromopropane, minor)
Steam	$CH_2=CH_2 + H_2O \rightarrow CH_3CH_2OH$	$CH_2=CHCH_3 + H_2O \rightarrow CH_3CH(OH)CH_3$ (**propan-2-ol**, major) + $CH_3CH_2CH_2OH$ (propan-1-ol, minor)
$KMnO_4$ / H^+	$CH_2=CH_2 + [O] + H_2O \rightarrow HOCH_2CH_2OH$	$CH_2=CHCH_3 + [O] + H_2O \rightarrow HOCH_2CH(OH)CH_3$ (**propane-1,2-diol**)

- **[O]** stands for oxygen from the oxidising agent. Use it to balance oxidation equations.
- With $KMnO_4$ you see **purple → colourless**. The two –OH groups go on the **two carbons that were joined by the C=C** (adjacent carbons).
- Hydrogenation is used to harden vegetable oils into margarine (it removes some C=C bonds).
- For an **unsymmetrical** alkene (propene) with HBr or steam, two products are possible. Which one is major is explained in 5.6.

5.4 Thinking it through with a cyclic alkene

Cyclohexene behaves the same way. With H_2/Ni → cyclohexane; with Br_2 → 1,2-dibromocyclohexane; with HBr → bromocyclohexane; with steam/H_3PO_4 → cyclohexanol; with $KMnO_4/H^+$ → cyclohexane-1,2-diol. The ring stays; only the C=C changes.

WORKED EXAMPLE: HOW MANY C=C BONDS?

0.500 g of limonene, $C_{10}H_{16}$ (Mr 136.0), reacts completely with 176 cm^3 of hydrogen at room temperature and pressure (molar volume 24.0 $dm^3\ mol^{-1}$).
n(limonene) = 0.500 ÷ 136.0 = 0.003676 mol
n(H_2) = 0.176 ÷ 24.0 = 0.007333 mol
Ratio H_2 : limonene = 0.007333 ÷ 0.003676 = 1.99 ≈ **2**
One H_2 adds per C=C, so limonene has **2 C=C bonds**. (Its formula, $C_{10}H_{16}$, is 6 H short of $C_{10}H_{22}$: two C=C plus one ring.)

WORKED EXAMPLE: IODINE NUMBER

The iodine number is the mass of I_2 (in g) that adds to 100 g of a fat or oil. One I_2 adds per C=C.
Triolein (Mr 885.4) has 3 C=C, so 1 mol adds 3 mol I_2 = 3 × 253.8 = 761.4 g.
Iodine number = 761.4 × 100 ÷ 885.4 = **86.0**
Higher iodine number = more C=C = more unsaturated.

MARK-SCHEME LANGUAGE

- Hydrogenation catalyst: "**nickel**" (ALLOW Pd, Pt, Rh). IGNORE heat. Acids or zeolite are wrong; a catalyst listed with other substances is not awarded (Oct 2025).
- Hydration (Jan 2026, 2 marks): "**steam**" (ALLOW $H_2O(g)$, or water with a temperature above 100 °C; DO NOT AWARD water below 100 °C) **and** "**phosphoric(V) acid / H_3PO_4**" (ALLOW sulfuric acid, H^+). A wrong extra reagent such as $KMnO_4$ caps the answer at 1. Ignore pressure.
- $KMnO_4$: "**acidified potassium manganate(VII)**"; product a **diol with OH on adjacent carbons**.

Often asked as: give reagents and conditions; name or draw the product from propene, a branched alkene or cyclohexene; calculate the number of C=C bonds from a volume of H_2 or a mass of I_2/Br_2.

COMMON MISTAKE

Writing "phosphoric acid" without steam, or "water" without saying steam; giving "heat" or a temperature above 400 °C for hydration (WCH11 Jan 2020, Jan 2023 ER).
"K_2MnO_4" for potassium manganate(VII). It is **$KMnO_4$**; write the name if you're unsure (WCH11 Oct 2019 ER).
Putting both Br atoms or both OH groups on the **same** carbon; writing an alcohol instead of a diol; a 1,3-diol (WCH11 Jan 2022, Jan 2023 ER). Never bond Mn into the organic product.
"Butanol" with no position number (WCH11 Oct 2021 ER).
Forgetting to multiply by the number of C=C bonds in a moles calculation (WCH11 Jun 2022 ER).

BEYOND THE SPEC, BUT EXAMINED

Hydrogenation is **not** electrophilic addition (WCH11 Jan 2020).
Hydration of two different alkenes can give a set of positional isomers: pent-1-ene and pent-2-ene between them give pentan-1-ol, pentan-2-ol and pentan-3-ol (WCH11 Oct 2025).
Iodine-number and H_2-volume calculations to find the number of C=C bonds (WCH11 Jan 2026).

SPEC 5.5 Testing for C=C

WHAT THIS SPEC POINT ASKS

Know the test for a C=C double bond using bromine or bromine water.

5.5 The test

	Alkene (C=C present)	Alkane or cycloalkane (no C=C)
Shake with **bromine water** at room temperature	**orange → colourless**	**no change** (stays orange)
Shake with **acidified $KMnO_4$**	**purple → colourless**	no change (stays purple)

- The bromine **adds across the C=C** (5.4), so the coloured Br_2 is used up.
- Alkanes don't react with bromine in the dark: their substitution reaction needs **UV light** (4.17).
- Do it **without heating** and **in the dark** (or at least not in strong sunlight), so a positive result can only come from addition.
- Gases such as propene can be **bubbled through** bromine water or shaken in a stoppered tube with it.

SUGGESTED PRACTICAL: COMPARING AN ALKANE AND AN ALKENE

Use cyclohexane (an alkane) and cyclohexene (an alkene), a few drops of each.

1. **Bromine water**: shake. Cyclohexene decolourises it (orange → colourless). Cyclohexane: no change.
2. **Acidified $KMnO_4$**: shake. Cyclohexene: purple → colourless. Cyclohexane: stays purple.
3. **Combustion** (a few drops on a watch glass, in a fume cupboard): both burn. Cyclohexene gives a **smokier (sootier) flame** because it has a higher ratio of carbon to hydrogen.

Other suggested practicals use the same test: **cyclohexene** made from cyclohexanol (heated with concentrated phosphoric(V) acid, then distilled off) and **limonene** extracted from orange peel by steam distillation both decolourise bromine water.
Safety: both liquids are flammable (no flames nearby except the controlled combustion test); bromine water is harmful.

MARK-SCHEME LANGUAGE

- Bromine water test (Jun 2019): reagent "bromine water / bromine solution / $Br_2(aq)$"; observation "**yellow / orange / (red-)brown → colourless**". ALLOW "decolourises". IGNORE heat; adding an unnecessary condition such as UV **negated** the mark.
- Write "**colourless**", not "clear". Clear means see-through, and coloured solutions are clear too.
- A ring compound with no C=C: "**no change / stays orange**" (Oct 2025).
- Liquid bromine itself is **red-brown** (Oct 2020).

Often asked as: a test to distinguish propane from propene, with both observations; a test to show a compound from a reaction scheme is unsaturated.

COMMON MISTAKE

"Bromine water goes clear" or "turns red". It is orange to **colourless** (WCH11 Oct 2019, Jan 2020, Jan 2021 ER).
Writing about two layers when the substances tested were **gases** (propane, propene) (WCH11 Oct 2019 ER).
Describing the product ("1,2-dibromopropane forms") when the question asks what you **see** (WCH11 Jan 2021 ER).

SPEC 5.6 Electrophilic addition: mechanisms

WHAT THIS SPEC POINT ASKS

Describe, with curly-arrow diagrams and evidence, the electrophilic addition of (i) bromine and hydrogen bromide to ethene and (ii) hydrogen bromide to propene. Arrows start from a bond or a lone pair. Know the relative stability of primary, secondary and tertiary carbocations.

5.6 Why "electrophilic addition"?

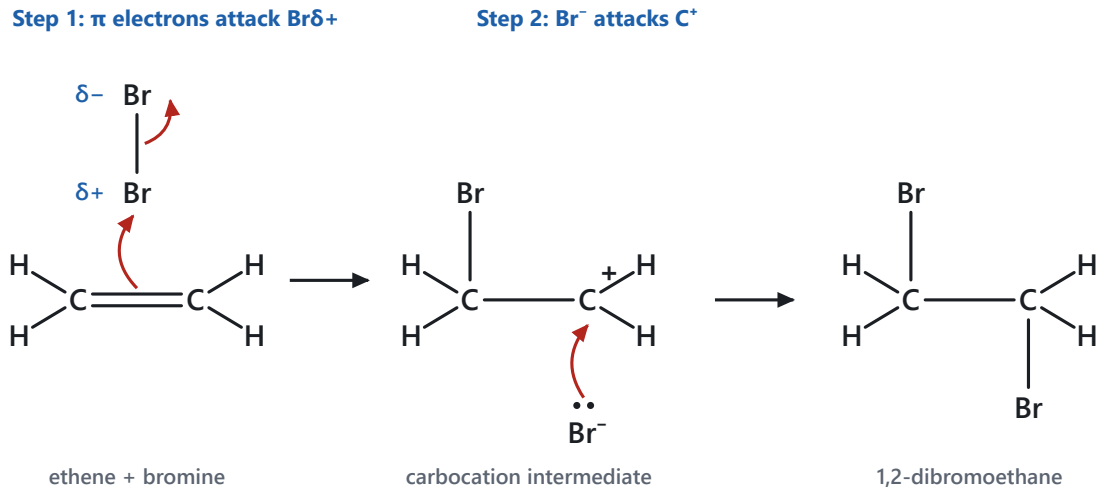
- The C=C is a region of **high electron density** (the π bond).
- An **electrophile** (electron-pair acceptor, 4.8) is attracted to it.
- The π electrons form a new bond to the electrophile, and two new single bonds end up where the double bond was: an **addition**.

5.6 Rules for every curly arrow

- A full curly arrow = movement of a **pair** of electrons.
- It **starts** at a **bond** (C=C, H–Br, Br–Br) or a **lone pair**. It **ends** at the atom that receives the electrons.

- Show **δ+ / δ−** on the electrophile, a **full charge** on the carbocation (+) and on the halide ion (−), and a **lone pair** on the halide ion.

5.6 Mechanism 1: bromine + ethene



Bromine + ethene. Step 1: two arrows (C=C to Brδ+; Br–Br bond to Brδ−). Step 2: lone pair on Br^- to C^+.

1. **The dipole is induced.** Br_2 is non-polar. As it approaches, the electron-rich C=C **repels** the electrons in the Br–Br bond, so the nearer Br becomes **δ+** and the further Br **δ−**.
2. **Arrow 1** starts in the **middle of the C=C** and points to the **Brδ+**. The π electrons form a new C–Br bond.
3. **Arrow 2** starts on the **Br–Br bond** and points to the **Brδ−**. The Br–Br bond breaks **heterolytically**, giving Br^-.
4. **Intermediate**: one carbon now has the Br; the other carbon has lost its share of the π electrons, has only three bonds and is a **carbocation** (C^+). Br^- is the other product of step 1.
5. **Arrow 3** starts at a **lone pair on Br⁻** and points to the **C⁺**. A new C–Br bond forms.
6. **Product**: CH_2BrCH_2Br, 1,2-dibromoethane.

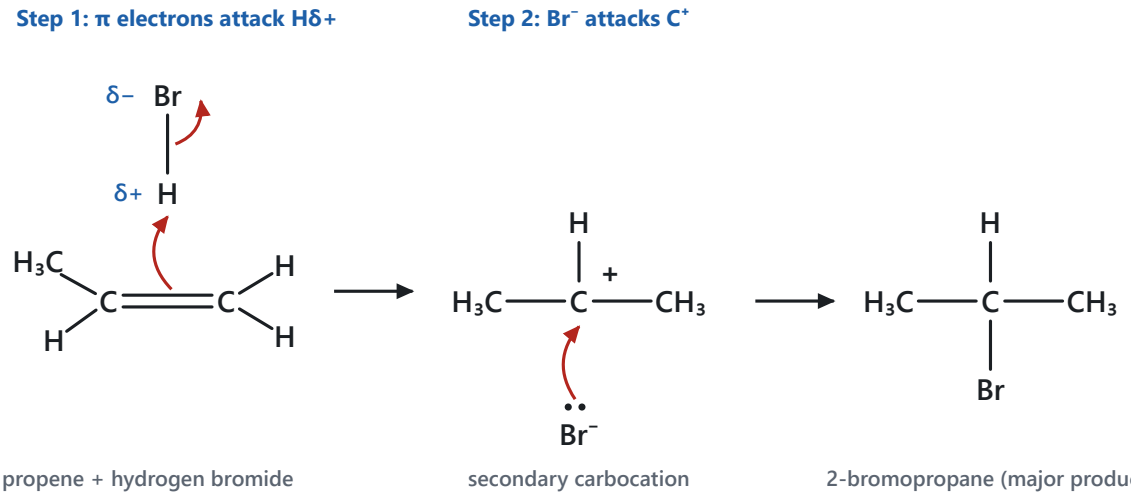
5.6 Mechanism 2: hydrogen bromide + ethene

HBr is **already polar**: Br is more electronegative, so it is **Hδ+–Brδ−**. The pattern is the same as for Br_2:

1. **Arrow 1**: from the **C=C** to the **Hδ+**.
2. **Arrow 2**: from the **H–Br bond** to the **Brδ−** (heterolytic fission, giving Br^-).
3. Intermediate: $CH_3CH_2^+$ (a primary carbocation) and Br^-.
4. **Arrow 3**: from a **lone pair on Br⁻** to the **C⁺**.
5. Product: CH_3CH_2Br, bromoethane.

Ethene is symmetrical, so it doesn't matter which carbon gets the H: there is only one product.

5.6 Mechanism 3: hydrogen bromide + propene



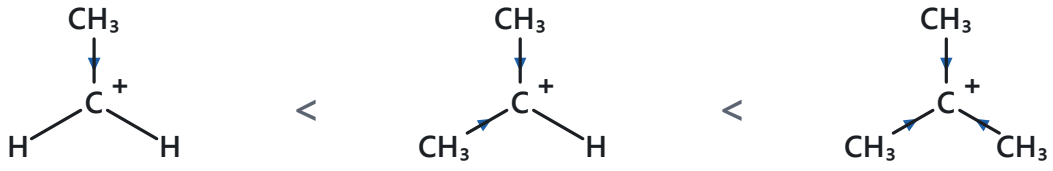
HBr + propene by the main route: H adds to the CH_2 end, giving the more stable secondary carbocation and so 2-bromopropane.

Propene is **unsymmetrical**, so the H can add to either carbon of the C=C:

H adds to	Carbocation formed	Type	Product
the **end** carbon (CH_2)	$CH_3–C^+H–CH_3$	**secondary** (2°): C^+ joined to **two** alkyl groups	**2-bromopropane** (**major**)
the **middle** carbon (CH)	$CH_3–CH_2–C^+H_2$	**primary** (1°): C^+ joined to **one** alkyl group	1-bromopropane (minor)

The arrows are drawn exactly as for ethene. Make sure your intermediate is the **secondary** carbocation if you are asked for the major product.

5.6 Why the secondary carbocation wins



Each alkyl group pushes electron density towards C^+ (small arrows). More alkyl groups spread the positive charge more, so the ion is more stable.

- **Alkyl groups are electron-releasing**: they push electron density towards the positively charged carbon (a positive inductive effect).
- This **spreads out (stabilises) the positive charge**. The more alkyl groups attached to C^+, the more stable the carbocation.
- Stability: **tertiary (3°) > secondary (2°) > primary (1°)**.
- The more stable carbocation forms **more easily** (lower activation energy), so most of the product comes from it. That product is the **major product**.

The same reasoning predicts any unsymmetrical case. 2-methylpropene + HBr goes mostly through the **tertiary** carbocation $(CH_3)_3C^+$, giving **2-bromo-2-methylpropane** as the major product.

KEY IDEA

Major product = the one formed from the **more stable carbocation intermediate**. Compare the **intermediates**, not the products.

5.6 Evidence for the mechanism

- **A carbocation intermediate is attacked by whatever nucleophile is around.** Ethene + bromine in the presence of **chloride ions** (NaCl solution) gives 1-bromo-2-chloroethane as well as 1,2-dibromoethane, but **no** 1,2-dichloroethane. So the first step always adds a Br atom (the Brδ+ end of Br_2), and the second step adds whichever negative ion reaches the C^+ first.
- With **bromine water**, water molecules also attack the carbocation, and the main product is **2-bromoethanol** ($BrCH_2CH_2OH$).
- With propene, HBr gives **mostly 2-bromopropane**. That matches the prediction from carbocation stability.

5.6 Unsymmetrical electrophiles

Work out which end is δ+ using electronegativity. The **δ+ atom** adds first (it is the electrophile); the **δ− atom** ends up on the carbocation carbon.

Electrophile	Dipole	With propene, major product
HBr	Hδ+–Brδ−	2-bromopropane, $CH_3CHBrCH_3$
BrCl	**Brδ+**–Clδ− (Cl more electronegative)	1-bromo-2-chloropropane, $CH_3CHClCH_2Br$
ICl	**Iδ+**–Clδ−	2-chloro-1-iodopropane, $CH_3CHClCH_2I$
H_2SO_4	Hδ+ on an O–H bond	propyl hydrogensulfate, $CH_3CH(OSO_3H)CH_3$

In each case the δ+ atom bonds to the **CH_2 end**, so the C^+ is on the middle carbon (secondary).

MARK-SCHEME LANGUAGE

- HBr mechanism (Jan 2023, 4 marks, six points): **dipole on H–Br**; arrow from the **H–Br bond to Br**; curly arrow from the **C=C to Hδ+**; **correct carbocation**; arrow from a **lone pair on Br^- to C^+**; **charge on bromide**. Marks go by how many of these points you include, so leave none out.
- Br_2 mechanism (Jan 2019): **induced dipole** on Br–Br; one arrow from C=C to Brδ+; one from the Br–Br bond to (or just beyond) Brδ−; intermediate; lone pair on Br^- with the arrow from it to C^+. IGNORE bond angles. Incorrect partial charges are penalised.
- Stability (Jan 2023, 2 marks): "the product is formed from the **secondary carbocation (intermediate)**" and "the **secondary carbocation is more stable** than the primary". Oct 2022: "the **tertiary carbocation** is more stable than the primary"; ALLOW "three methyl groups attached to the positive C". IGNORE Markovnikov's rule.
- BrCl: dipole Brδ+–Clδ− (Jan 2021); ICl: Iδ+ (Jun 2021).

Often asked as: the full mechanism for a named alkene (ethene, propene, 2-methylpropene, cyclohexene, a cyclic alkene); identify the carbocation intermediate; explain why one product is major; spot and correct errors in a given mechanism.

COMMON MISTAKE

Arrows starting from an **atom** or from a **charge** instead of from the bond or the lone pair; arrows pointing the wrong way. This is in every examiner report from 2019 to 2024.
No dipole on HBr or Br_2, or δ+ on the wrong atom (WCH11 Jan 2019, Oct 2019, Jan 2021 ER).
Writing Brδ− in step 2 instead of a full **Br^-** charge with a **lone pair** (WCH11 Oct 2020, Jan 2021, Oct 2022, Jan 2024 ER).
Using ethene when the question names a different alkene (WCH11 Jan 2019, Jan 2023, Oct 2023 ER). Practise with several alkenes.
Using HBr when bromine was asked, or the other way round (WCH11 Jan 2019 ER).
Drawing the primary carbocation when the major product comes from the secondary (WCH11 Jun 2022, Oct 2022 ER).
"2-bromopropane is a secondary carbocation", "secondary halogenoalkanes are more stable", or comparing the stability of the **products**: it is the **intermediate** carbocations you compare (WCH11 Jan 2023, Oct 2023, Oct 2024 ER).
Quoting Markovnikov's rule instead of explaining carbocation stability. The rule is not on the spec and earns nothing by itself (WCH11 Jan 2023 ER).

BEYOND THE SPEC, BUT EXAMINED

Unsymmetrical electrophiles: BrCl (WCH11 Jan 2021), ICl (WCH11 Jun 2021), and H_2SO_4 adding to an alkene. For H_2SO_4: arrow from C=C to the H of an O–H; arrow from that O–H bond to O; carbocation + HSO_4^-; lone pair on the O^- of HSO_4^- to C^+ (WCH11 Oct 2024, nine-point scheme).
Acid-catalysed hydration of propene: arrow from C=C to H^+; secondary carbocation; arrow from a **lone pair on the O of H_2O** to C^+; the O then loses H^+, giving **propan-2-ol** and **regenerating H^+** (the catalyst) (WCH11 Jan 2021).
How the **π electrons induce a dipole** in Cl_2 or Br_2 (WCH11 Jan 2025).
Correcting a wrong mechanism: reverse the dipole on Br_2; start the arrow at the π bond; give Br^- a lone pair and charge (WCH11 Oct 2019).

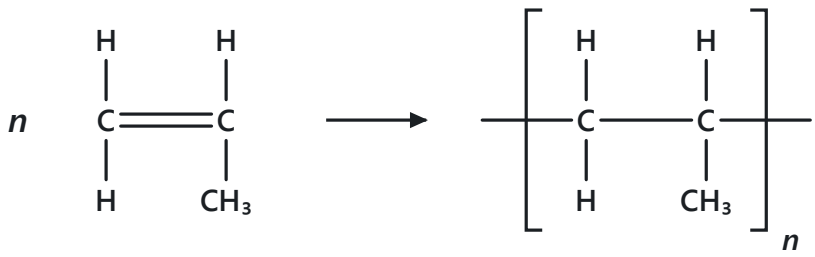
SPEC 5.7 Addition polymerisation

WHAT THIS SPEC POINT ASKS

Describe the addition polymerisation of alkenes. Draw the repeat unit from the monomer, and the monomer from the repeat unit.

5.7 What happens

- **Monomers**: small molecules with a **C=C**.
- Under suitable conditions (high pressure and an initiator, or a catalyst) the **π bond in each monomer breaks** and the monomers join into a long saturated chain: the **polymer**.
- The polymer is the **only product** (atom economy 100%).
- The polymer chain has **no C=C** left. It is saturated and very unreactive, like an alkane.



propene (monomer) poly(propene), shown as its repeat unit

Polymerisation of propene. The repeat unit has two main-chain carbons, the CH_3 as a side group, extension bonds through the brackets, and n outside.

5.7 Monomer → repeat unit

1. Draw the monomer with the **C=C horizontal** and the other four groups **above and below** (not along the line).
2. Change the C=C to a **C–C single bond**.
3. Add an **extension bond** to the left of the first carbon and to the right of the second.
4. Put **square brackets** across the extension bonds and write ***n*** at the bottom right.
5. For the equation, write ***n*** in front of the monomer too.

$n\ CH_2{=}CHCH_3 \rightarrow -[CH_2{-}CH(CH_3)]-_n$

The repeat unit's main chain is always just two carbons (from a monomer with one C=C). Everything else hangs off as side groups.

5.7 Repeat unit → monomer

1. Take the **two main-chain carbons** inside the brackets.
2. Remove the extension bonds and the brackets.
3. Put a **C=C** between those two carbons. Keep the side groups.

5.7 Common addition polymers

Monomer	Polymer	Repeat unit
ethene $CH_2{=}CH_2$	poly(ethene)	$-[CH_2{-}CH_2]-$
propene $CH_2{=}CHCH_3$	poly(propene)	$-[CH_2{-}CH(CH_3)]-$
chloroethene $CH_2{=}CHCl$	poly(chloroethene), PVC	$-[CH_2{-}CHCl]-$
tetrafluoroethene $CF_2{=}CF_2$	poly(tetrafluoroethene), PTFE	$-[CF_2{-}CF_2]-$
phenylethene $CH_2{=}CHC_6H_5$	poly(phenylethene), polystyrene	$-[CH_2{-}CH(C_6H_5)]-$
but-1-ene $CH_2{=}CHCH_2CH_3$	poly(but-1-ene)	$-[CH_2{-}CH(C_2H_5)]-$
but-2-ene $CH_3CH{=}CHCH_3$	poly(but-2-ene)	$-[CH(CH_3){-}CH(CH_3)]-$
methyl 2-methylpropenoate $CH_2{=}C(CH_3)COOCH_3$	Perspex	$-[CH_2{-}C(CH_3)(COOCH_3)]-$

WORKED EXAMPLE: NUMBER OF REPEAT UNITS

A sample of poly(propene) has an average Mr of 210 000.
Repeat unit C_3H_6: Mr = (3 × 12.0) + (6 × 1.0) = 42.0
Number of repeat units = 210 000 ÷ 42.0 = **5000**

EXAM TIP

"Draw **two** repeat units" means four main-chain carbons in one continuous chain, with extension bonds at each end. Two separate bracketed units, or one, or three, don't score (WCH11 Jan 2019).

MARK-SCHEME LANGUAGE

- Repeat unit (Jan 2026): "diagram showing **extension bonds**"; ALLOW CH_3 written as a group; DO NOT AWARD **missing H** or a skeletal formula.
- Equation (Jun 2019): ***n*** before the monomer **and** after the polymer brackets; "the equation must be balanced".
- Polymer is saturated: **no C=C** in the repeat unit (Oct 2021 ER).
- Addition v condensation (Oct 2020): poly(lactic acid) is a **condensation** polymer that gives off a small molecule (water); propene polymerisation gives **no** other product (addition). ALLOW "higher atom economy".

Often asked as: draw the repeat unit (propene, but-1-ene, chloroethene, tetrafluoroethene); draw two repeat units; deduce the monomer from a polymer section; write the equation with *n*.

COMMON MISTAKE

Three carbons in the main chain (e.g. putting the CH_3 of propene into the backbone); leaving out the CH_3; keeping the C=C in the repeat unit (WCH11 Jan 2021, Oct 2021 ER).
Drawing poly(ethene) or poly(but-2-ene) when but-1-ene was asked (WCH11 Oct 2021 ER).
Missing *n* on one side; extension bonds that don't pass through the brackets (WCH11 Oct 2020, Jun 2019, Jan 2023 ER).
Saying propene polymerisation "produces hydrogen". Addition polymerisation produces **nothing** else (WCH11 Oct 2020 ER).
Drawing trifluoroethene for tetrafluoroethene: tetra = **four** F (WCH11 Jun 2024).

BEYOND THE SPEC, BUT EXAMINED

Monomers with two C=C bonds (dienes). Buta-1,3-diene, $CH_2{=}CH{-}CH{=}CH_2$, polymerises to a repeat unit with **four** main-chain carbons and a C=C in the middle: $-[CH_2{-}CH{=}CH{-}CH_2]-$. Working backwards, a repeat unit with a C=C inside the chain came from a monomer with two C=C bonds (WCH11 Oct 2023, Oct 2022).
Mr of a repeat unit and the number of repeat units in a chain (WCH11 Oct 2023).
Why polymers suit some uses: low density, don't corrode, don't release metal ions into water (WCH11 Jan 2025).

SPEC 5.8 Dealing with polymer waste

WHAT THIS SPEC POINT ASKS

Understand how chemists reduce the problems of polymer disposal by (i) developing biodegradable polymers and (ii) removing toxic waste gases produced when polymers are burned.

5.8 The problem

- Addition polymers have **strong, non-polar C–C and C–H bonds** and no reactive functional group, so microorganisms' enzymes can't break them down. They are **non-biodegradable** and last for hundreds of

years.
- **Landfill**: takes up land; plastics persist; litter harms wildlife.
- **Incineration**: releases energy (which can generate electricity) and takes little space, but makes CO_2 and **toxic gases**. Burning **PVC** releases **hydrogen chloride**, HCl. Incomplete burning also makes CO and soot.

5.8 What chemists do

(i) Develop biodegradable polymers

- A **biodegradable** polymer is broken down by **microorganisms** (bacteria, fungi) into harmless products such as CO_2, water and biomass.
- Examples: **poly(lactic acid), PLA**, made from plant starch (corn, sugar cane). It is a **condensation** polymer with ester links that can be hydrolysed. Starch-based plastics and compostable bags.
- Benefits: less waste to landfill; less litter and harm to wildlife; no need to incinerate; often made from **renewable** plant material rather than crude oil.
- Limits: needs the right conditions (moisture, warmth, microbes) to break down; uses farmland; may still release CO_2 or methane as it decomposes.

(ii) Remove toxic gases from incinerator waste

- The waste gases are passed through **scrubbers** containing a **base**, which **neutralises** acidic gases such as HCl and SO_2.
  - $CaO(s) + 2HCl(g) \rightarrow CaCl_2(s) + H_2O(l)$
  - $Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$
  - $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
- A **large surface area** (spraying a slurry of the base, or a fine powder) speeds up the reaction.
- HCl is also very soluble, so water sprays help to remove it.

Other approaches

- **Recycling**: sorting, melting and remoulding thermoplastics (saves crude oil and energy).
- **Feedstock recycling**: cracking waste polymers back into small molecules (monomers, fuels) to make new chemicals.
- **Energy recovery**: burning to generate electricity, with the gases cleaned.

MARK-SCHEME LANGUAGE

- Biodegradable (Oct 2020), three of: takes **less time** to break down; **reduces waste to landfill**; no need to incinerate; reduces litter / harm to wildlife; breaks down to **non-harmful products**; made from **renewable** sources / conserves crude oil. DO NOT AWARD "no waste", "no pollution". "Biodegradable" means **broken down by microorganisms**; "breaks down naturally" alone did not score.
- Incineration (Oct 2021, Jan 2023, Jun 2024): advantages: **energy released** to generate electricity or heat; less space than landfill; pollutants can be trapped. Disadvantages: **toxic / corrosive gases** (name **HCl** from PVC); particulates; not all polymers can be recycled.
- Removing HCl (Jun 2019, Oct 2023): **neutralise with a base**, e.g. **calcium hydroxide** or calcium oxide; a large surface area increases the rate. "Scrubbers" alone isn't enough.
- Chemists limit the problem by (Oct 2022): developing **biodegradable** polymers; **removing toxic waste gases** from incineration; converting waste polymers into useful feedstock.

Often asked as: advantages and disadvantages of incineration and landfill; what biodegradable means and why it helps; how HCl is removed from incinerator gases, with an equation.

COMMON MISTAKE

Describing what the public should do (reuse, recycle bins) when the question asks what **chemists** do (WCH11 Oct 2022 ER).
Saying incineration "uses" a lot of energy: it **releases** energy (WCH11 Oct 2021 ER).
Confusing biodegradable with recyclable. Non-biodegradable polymers can be recycled too (WCH11 Oct 2020 ER).
Mentioning CO_2 and global warming when the question says not to (WCH11 Oct 2021 ER).
Suggesting a **catalytic converter** to remove HCl (WCH11 Jun 2019 ER). Use a base.

BEYOND THE SPEC, BUT EXAMINED

Poly(lactic acid) as a condensation polymer compared with poly(propene) (WCH11 Oct 2020).
Landfill gas: anaerobic decay makes methane and CO_2 (WCH11 Jun 2022).